WORKSHOP · GEOMUNDUS 2026

Flood Mapping from Satellite Imagery with ML4Floods

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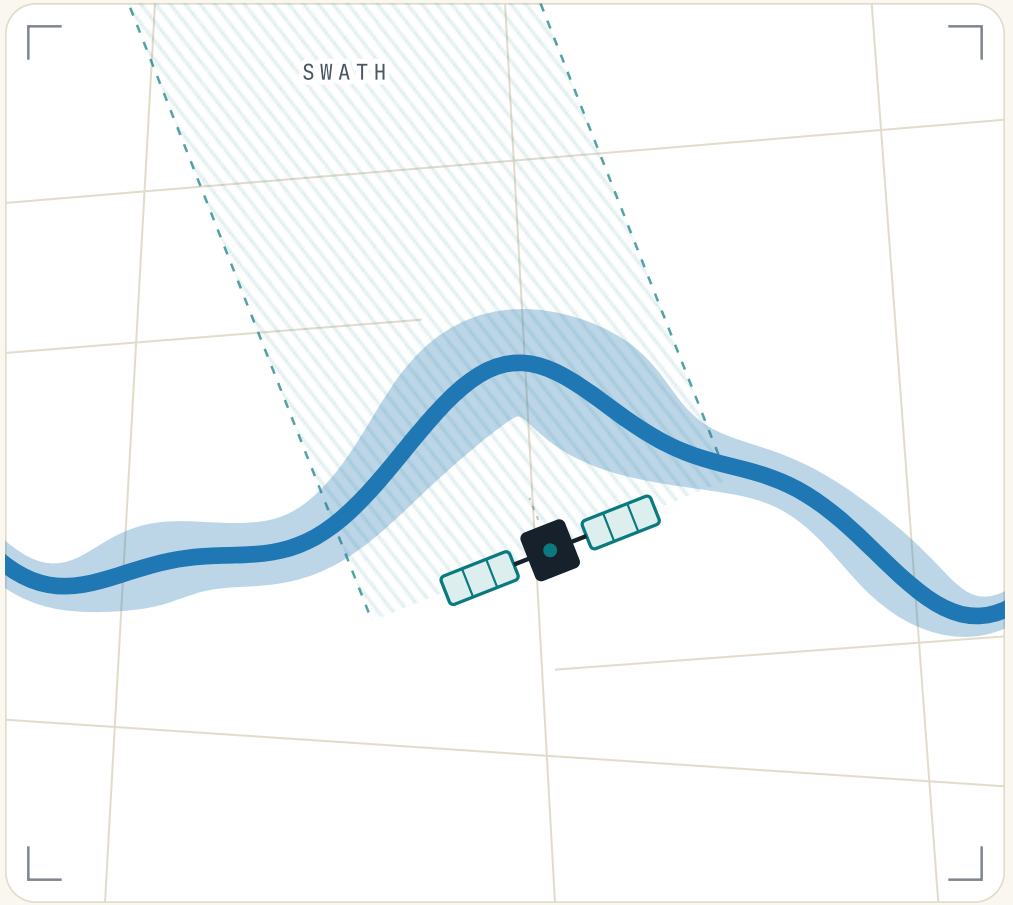

Image and Signal Processing group (ISP)

Universitat de València · isp.uv.es

AI for remote sensing, Earth & climate, and the brain

Geospatial foundation models · Causal inference · Dynamical systems · Physics-aware ML · Explainable AI · **Extreme events** · Uncertainty · Visual neuroscience · Generative models

Team

remote sensing + AI within ISP: floods · methane · clouds · super-resolution

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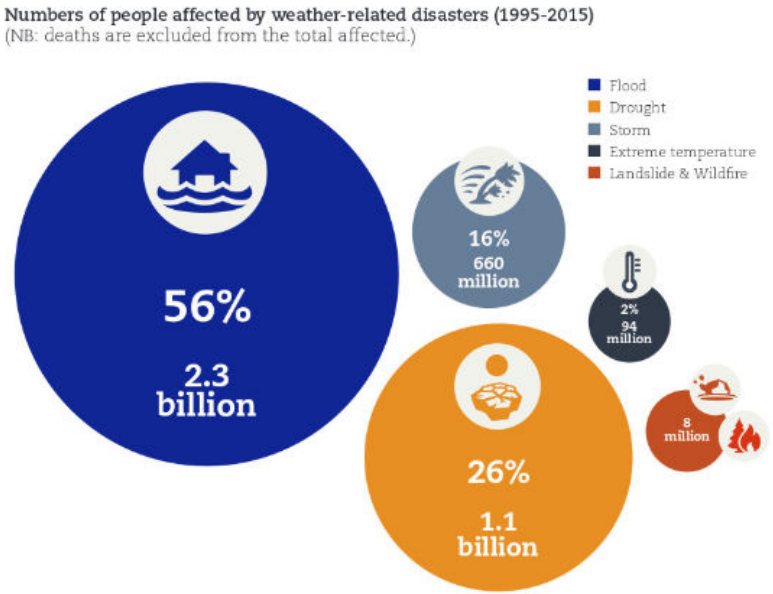
Simon Donike

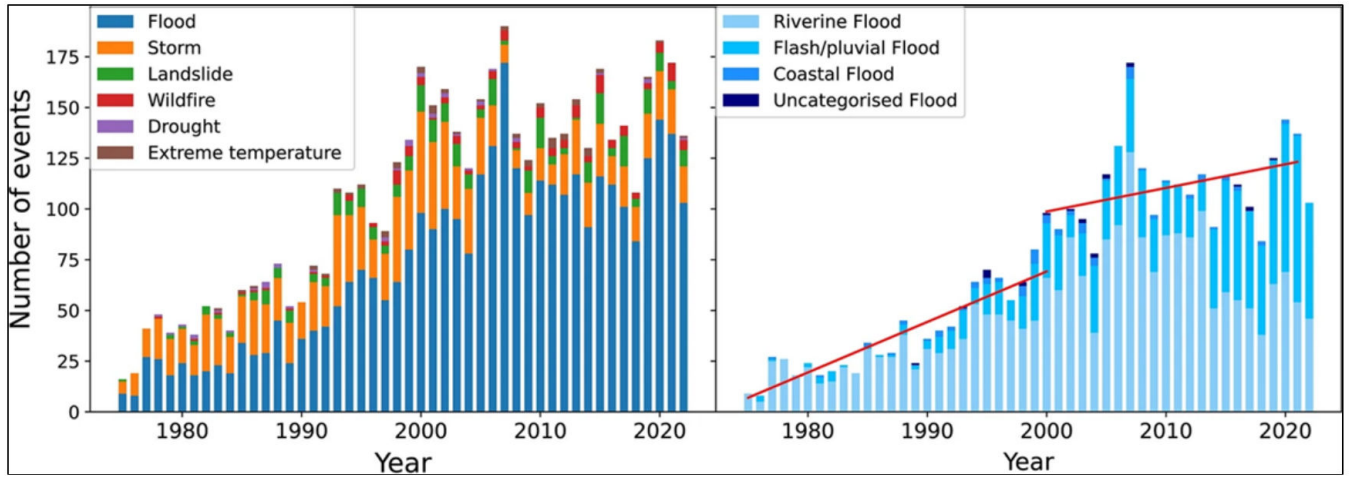
Julio Contreras

Enrique Portalés-Julià

Floods: the most widespread disaster, and increasing



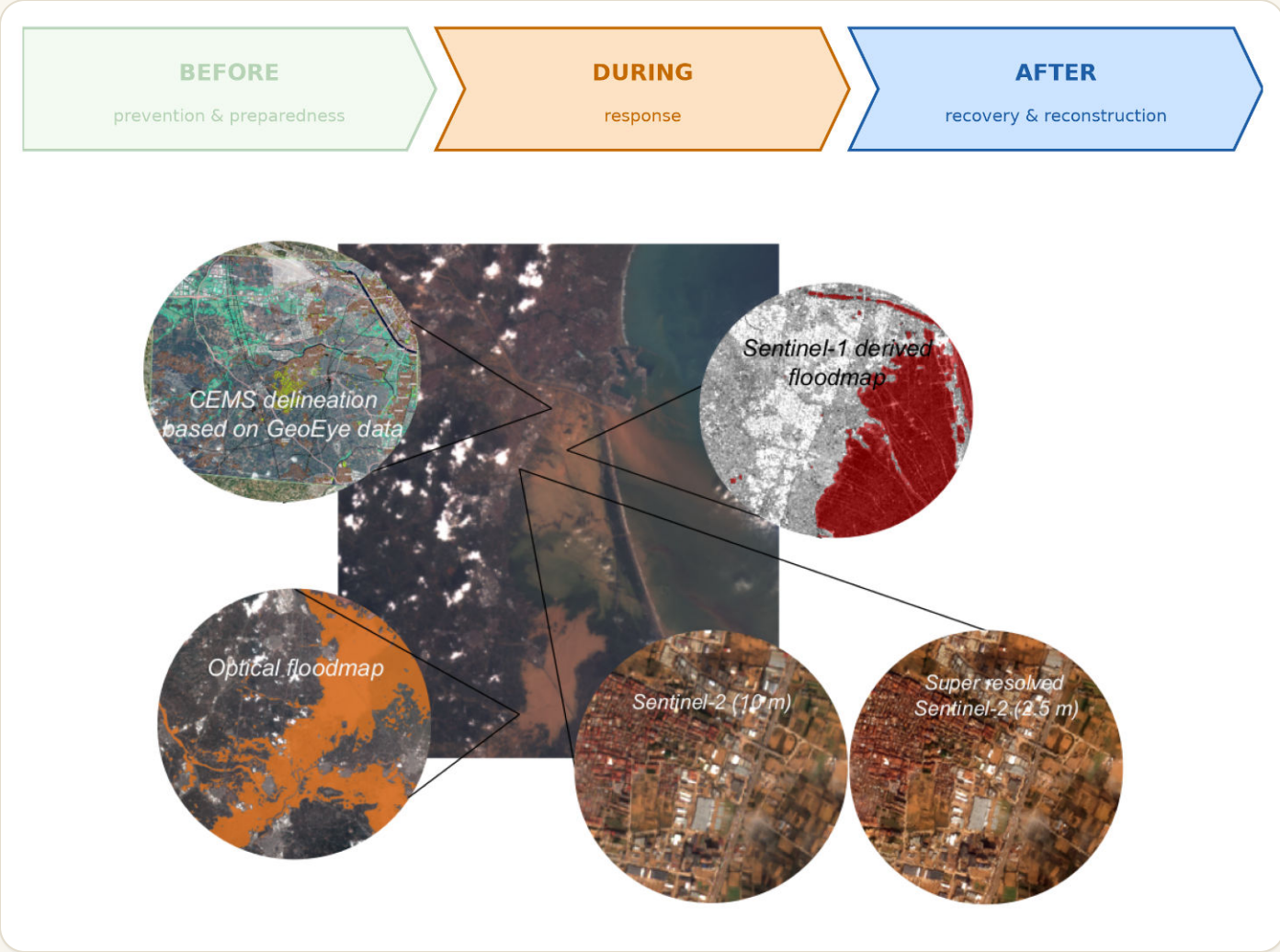
People affected by weather-related disasters, 1995-2015: floods 56 %. CRED & UNISDR (2015)



Jonkman et al. (2024)

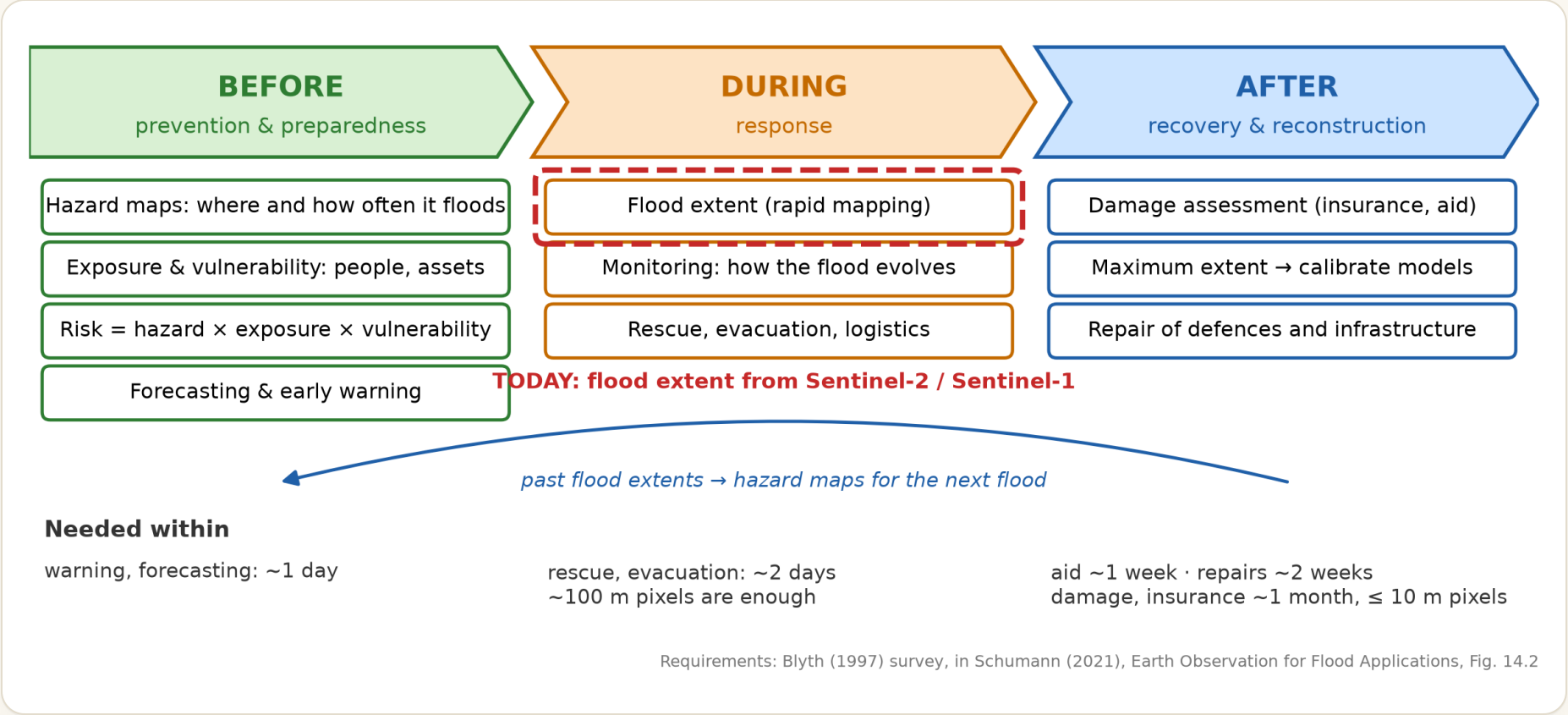
Number of disaster events per year, 1975-2022. NASA ARSET (2026), Floods Part 1 (Jonkman et al., 2024; EM-DAT)

Valencia DANA, October 2024



Why remote sensing for floods?

- Large areas, repeated observations, free archives (Copernicus, Landsat) · operational: Copernicus EMS, GFM, UNOSAT



Today

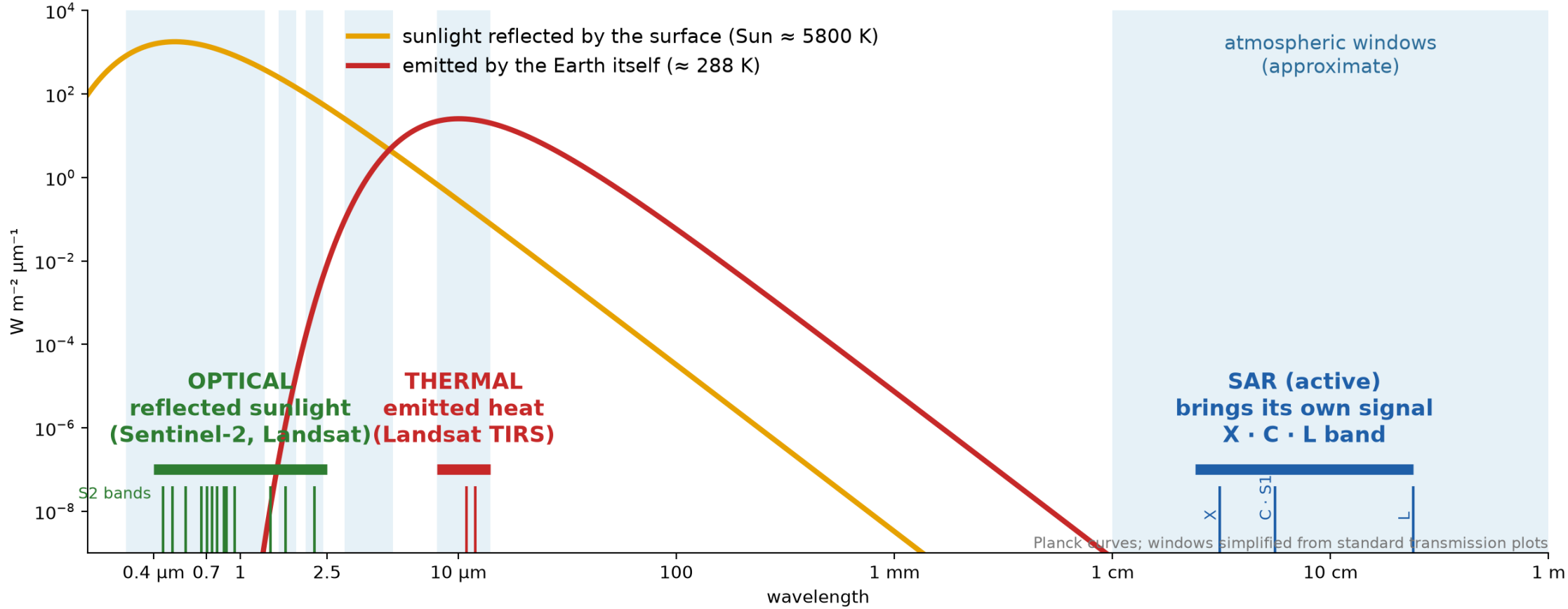
- Remote sensing fundamentals: optical + SAR
- Hands-on 1 — classical flood mapping
- Break
- Deep learning for flood mapping: WorldFloods & ml4floods
- Hands-on 2 — the ml4floods pipeline
- From data to action + wrap-up

github.com/kipoju/geomundus-ml4floods

01 Optical remote sensing

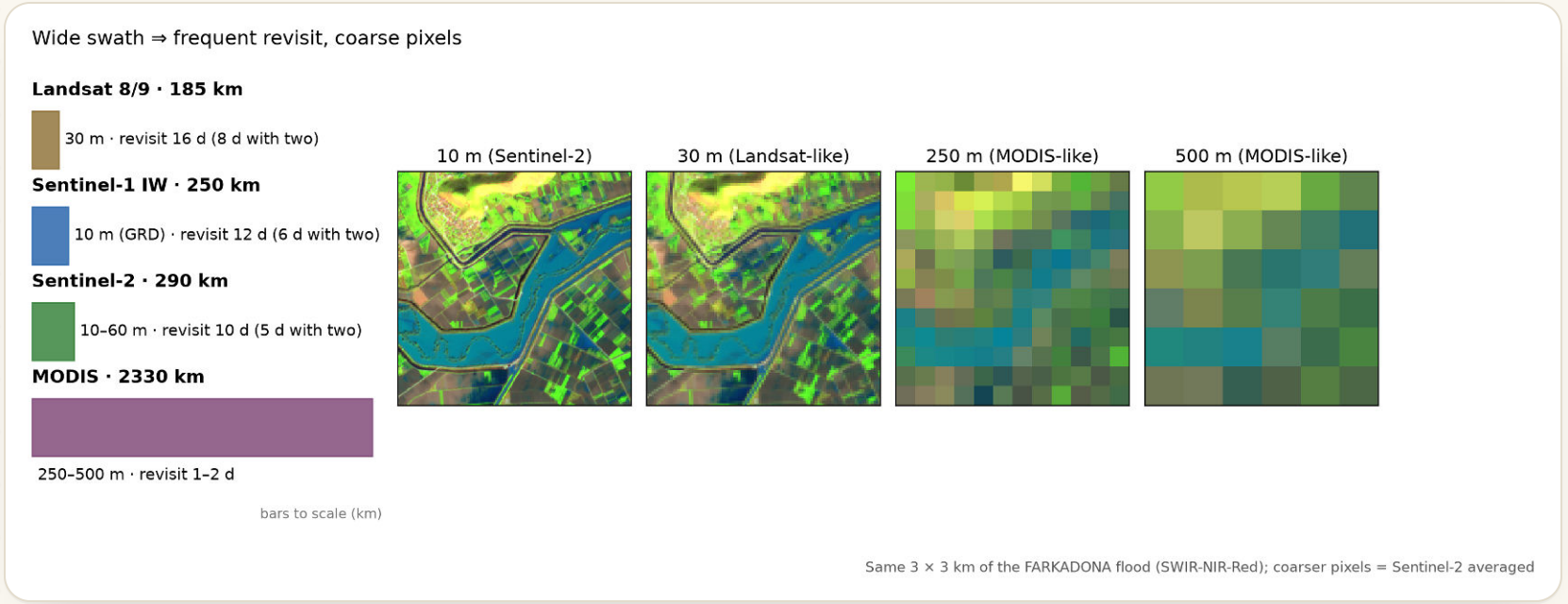
Spectral signatures, water indices and thresholds

Sunlight, heat and radar: the electromagnetic spectrum



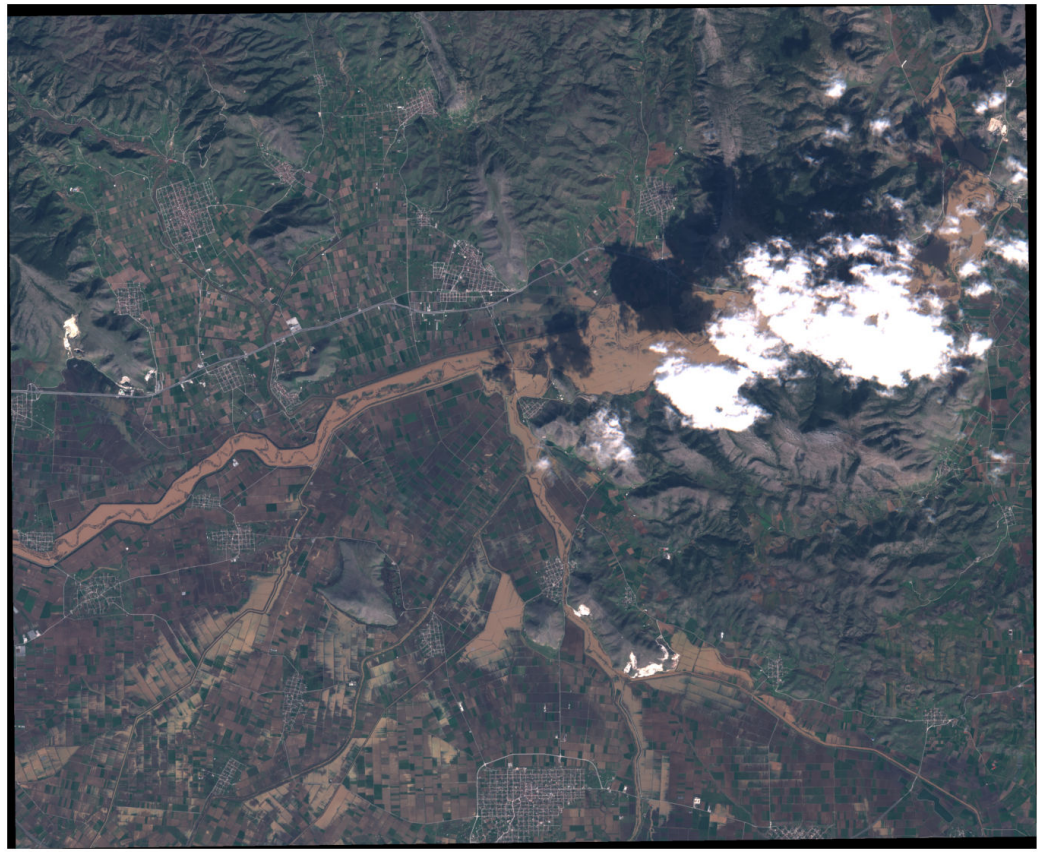
Optical missions for flood mapping

	SENTINEL-2	LANDSAT 8/9	MODIS	PLANETSCOPE
Resolution	10-60 m	30 m	250-500 m	~3 m
Revisit	5 days	8 days (both)	1-2 days	daily
SWIR	yes	yes	yes	no
Access	free	free	free	commercial

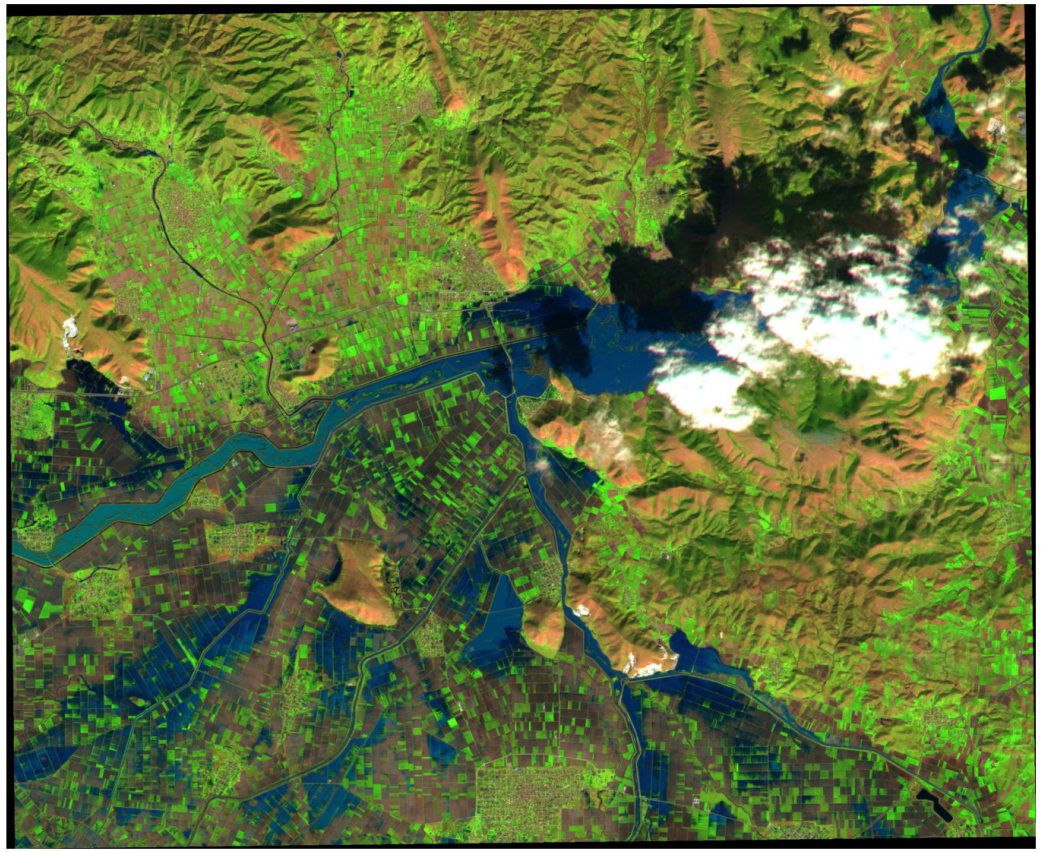


True colour vs false colour

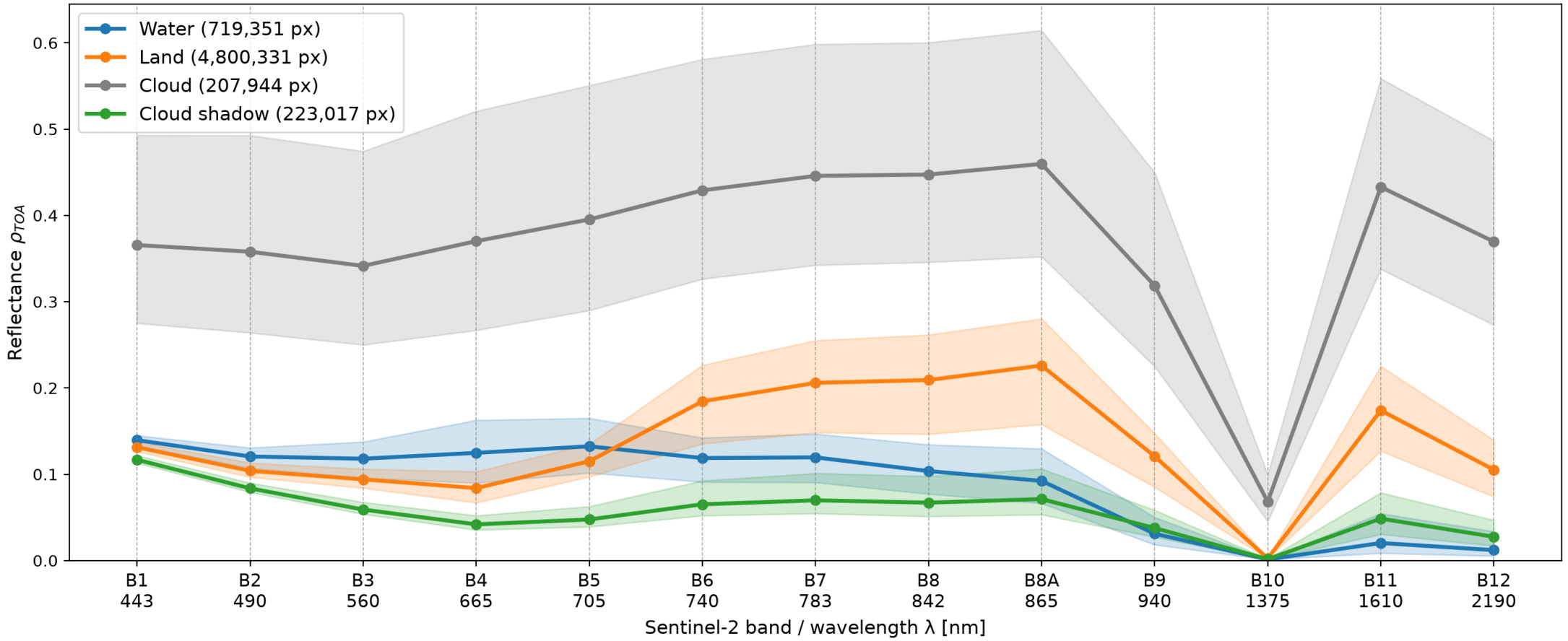
True colour (R, G, B) = (B4, B3, B2)



False colour (SWIR, NIR, Red) = (B11, B8, B4)



Spectral signatures of water, land, clouds and shadows



Water indices

WATER → INDEX > 0

NDWI · MCFEETERS 1996

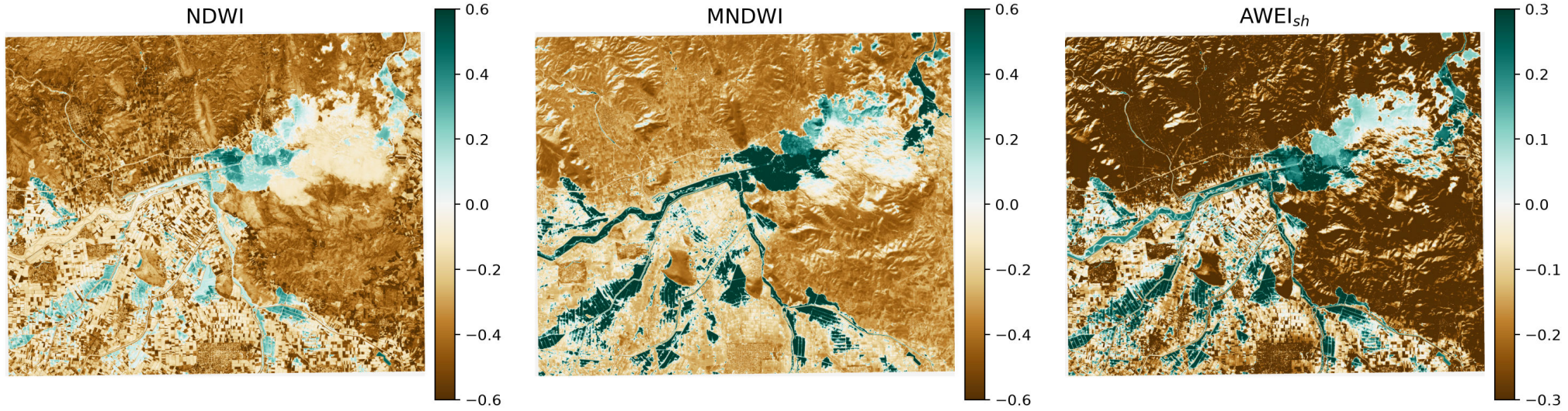
$$\mathrm{NDWI} = \frac{G - \mathrm{NIR}}{G + \mathrm{NIR}}$$

MNDWI · XU 2006

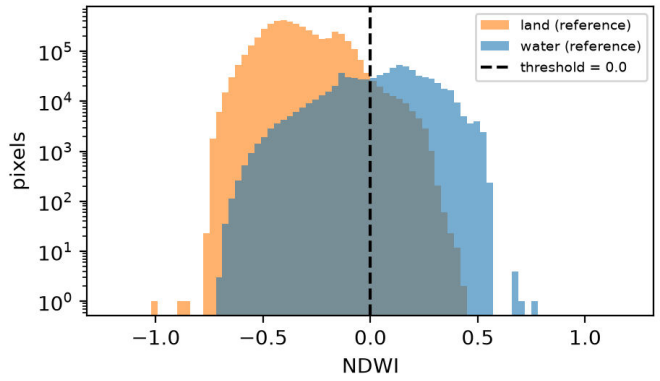
$$\mathrm{MNDWI} = \frac{G - \mathrm{SWIR}_1}{G + \mathrm{SWIR}_1}$$

$\mathrm{AWEI}_{\mathrm{SH}}$ · FEYISA ET AL. 2014

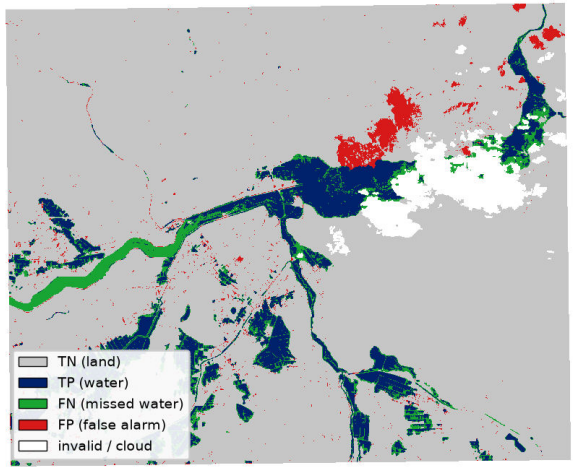
$$B + 2.5\,G - 1.5\,(\mathrm{NIR} + \mathrm{SWIR}_1) - 0.25\,\mathrm{SWIR}_2$$

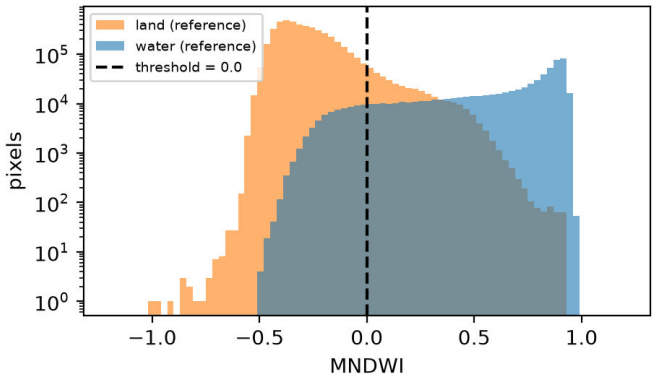


From index to flood map (threshold = 0)

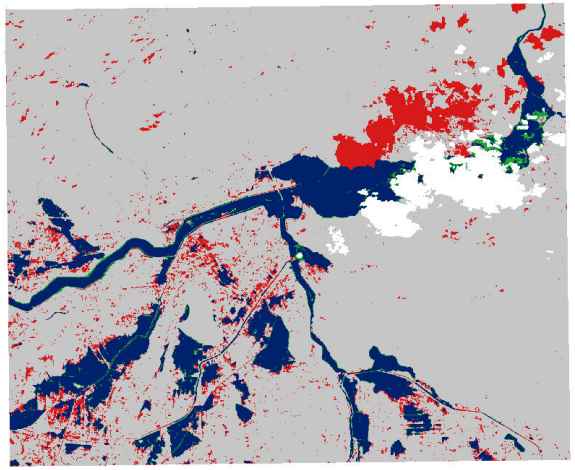


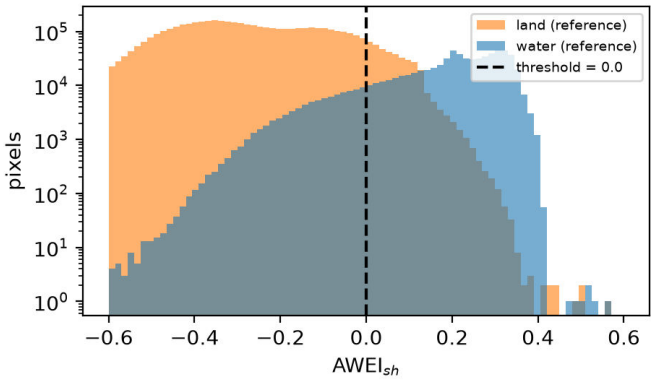
NDWI > 0 — P 0.81 · R 0.65 · IoU 0.56



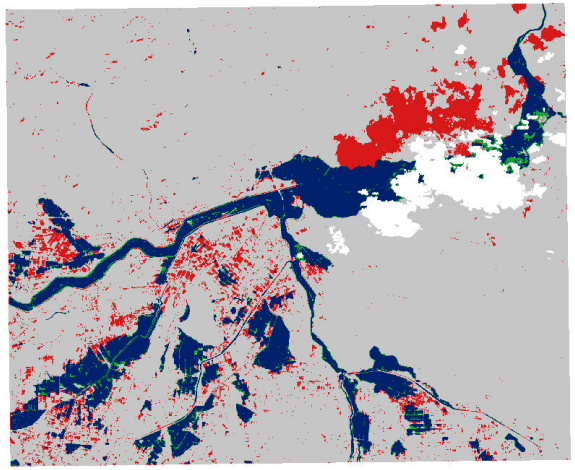


MNDWI > 0 — P 0.64 · R 0.91 · IoU 0.60

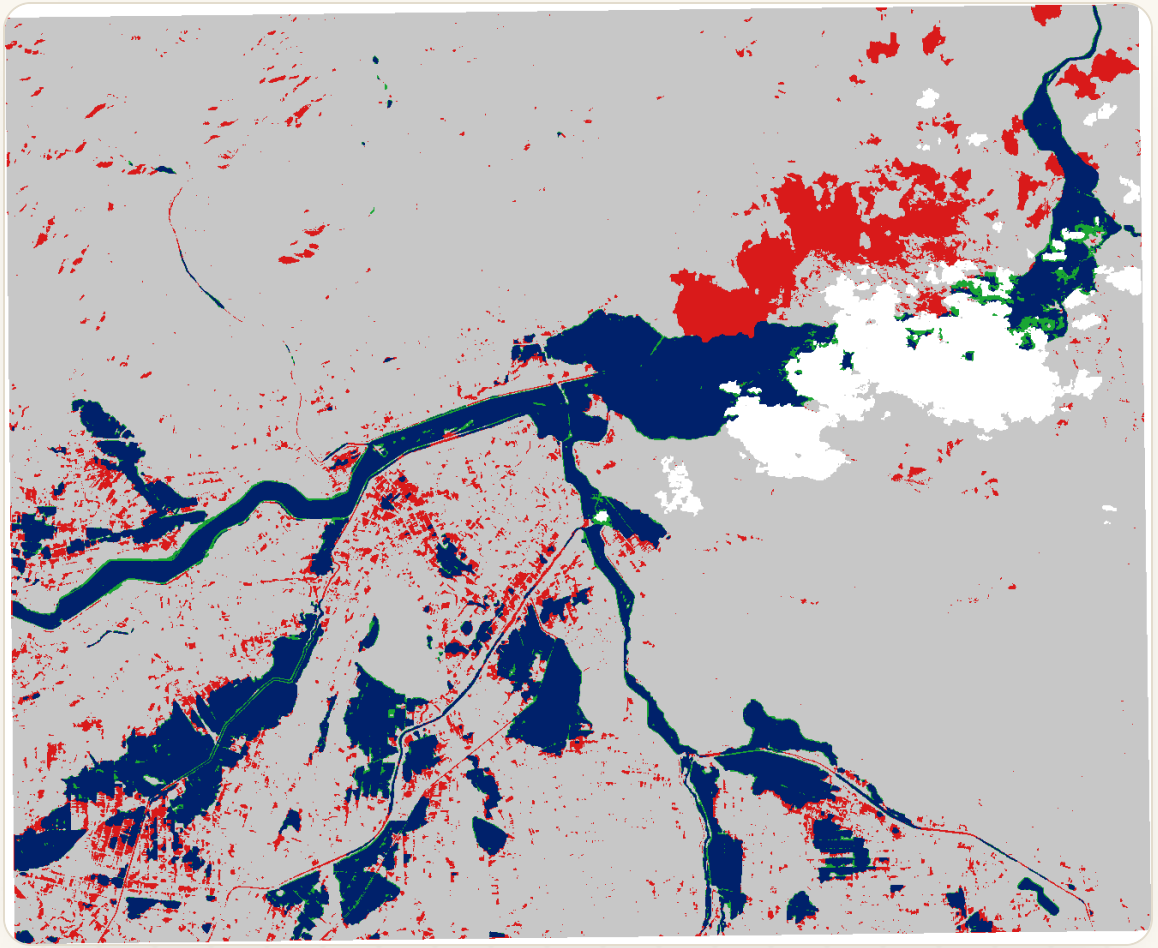




AWEI$_{sh}$ > 0 — P 0.62 · R 0.87 · IoU 0.57



Try it: find your own threshold



EMSR271 · FARKADONA · SENTINEL-2 · VIEW: CONFUSION

MNDWI THRESHOLD τ **0.00**

-0.5 WATER ABOVE τ 0.5

water land REFERENCE

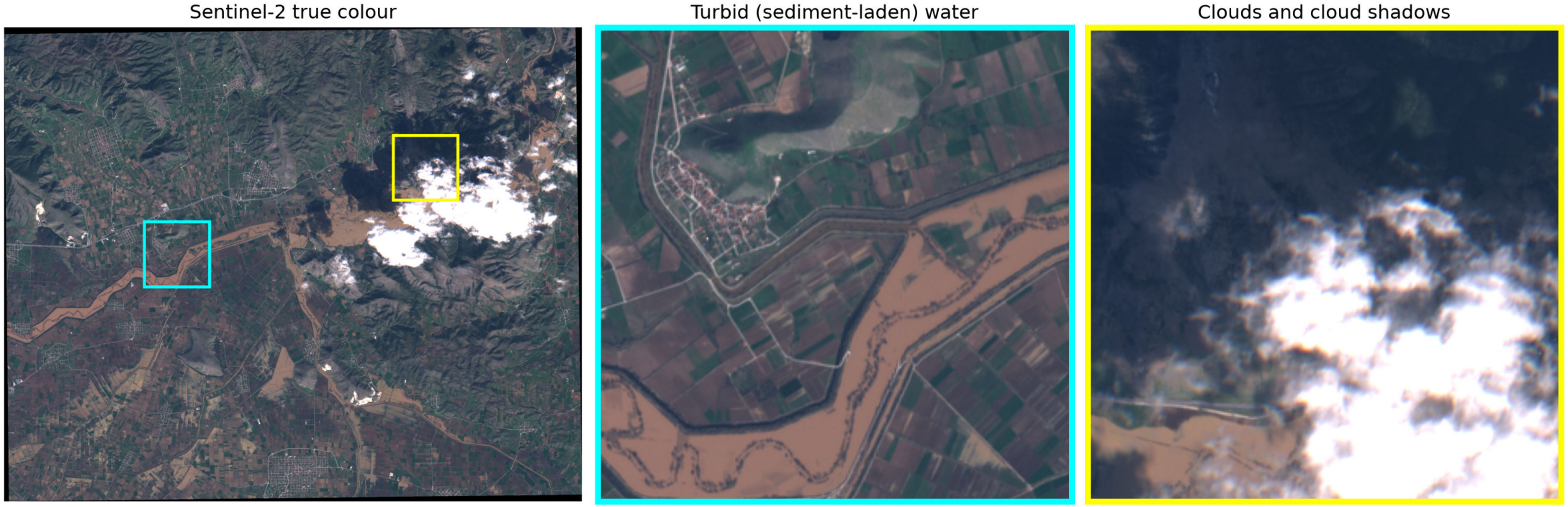
PRECISION	RECALL	IOU
0.64	0.91	0.60

CONFUSION | MNDWI | RGB | SWIR

TP FN FP TN cloud / invalid

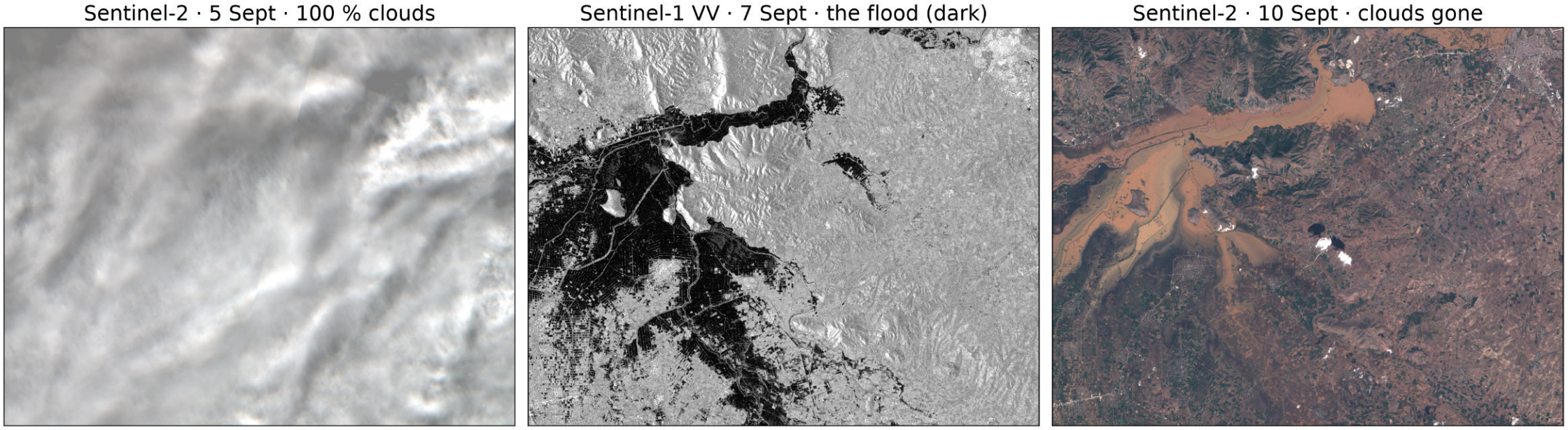
Limits of optical flood mapping

- Clouds hide the flood · cloud shadows look like water · turbid, sediment-laden water
- Flooded vegetation and urban floods · the best threshold changes from scene to scene



Clouds: the biggest limit of optical

- Floods come with storms: while the water is out, the sky is often still covered. Sentinel-2 saw **nothing**; a radar saw the whole flood through the clouds



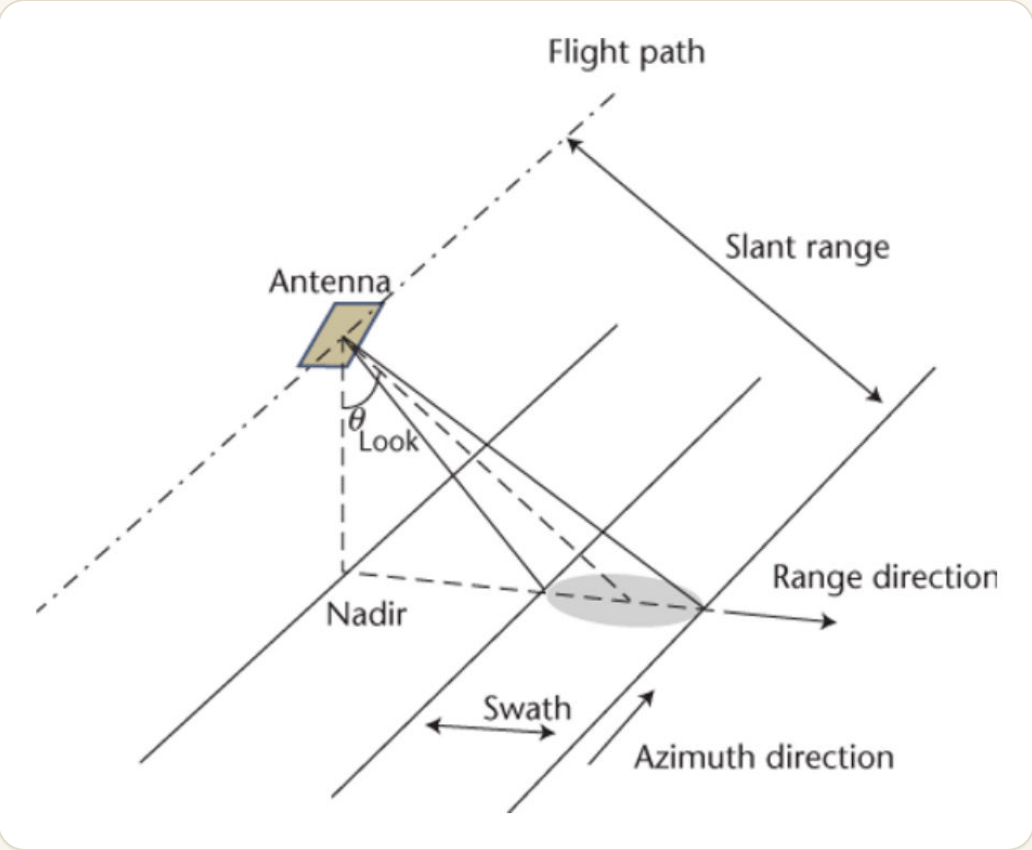
02

Synthetic Aperture Radar (SAR)

How the image is formed · what a pixel measures · what makes it dark or bright

Radar: a pulse and its echo

- **Active**: it sends its own microwave pulse → works day and night, through clouds
- It looks **sideways**: looking straight down, two points left and right of the track would be at the same distance
- Each pixel = **how much of the pulse comes back** (the *backscatter*)
- Where the pixel goes: **range** (across the track) from the echo delay; **azimuth** (along the track) from the satellite moving on
- **Incidence angle** θ: how slanted the pulse hits the ground (Sentinel-1: 29–46°)



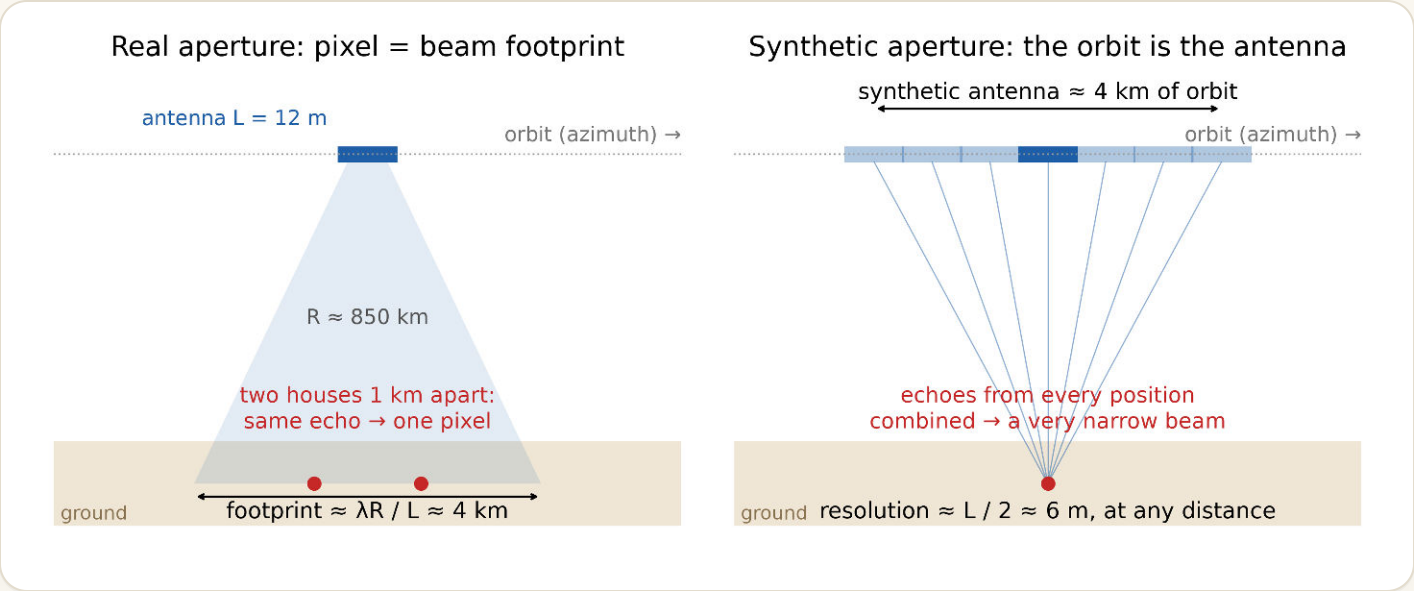
SAR imaging geometry. Credit: NASA (via Portalés-Julià, 2025)

Why ”synthetic aperture”?

- **Without SAR**, the pixel is the beam footprint: $\delta_{\text{az}} = \lambda R / L$. For 5 m pixels from Sentinel-1 (λ = 5.6 cm, $R \approx$ 850 km) we would need an antenna of

$$L = \frac{\lambda R}{\delta_{\text{az}}} = \frac{0.056\ \text{m} \times 850\ \text{km}}{5\ \text{m}} \approx \mathbf{9.5}\ \text{km}$$

- **With SAR**, the 12 m antenna records each point along ~4 km of orbit and the echoes are combined (synthetic antenna): azimuth $\delta_{\text{az}} \approx L/2 \approx 6$ m, at any distance; range $\delta_r = c/2B$, set by the pulse bandwidth B



Sentinel-1

	SENTINEL-1 (COPERNICUS)
Band	C-band, 5.4 GHz (λ ≈ 5.6 cm)
Mode over land	Interferometric Wide swath (IW): 250 km, incidence 29–46°
Product	GRD: geocoded, 10 m pixels, VV + VH, calibrated σ^0
Repeat	12 days per satellite, 6 days with two (S1B lost Dec 2021, S1C launched Dec 2024)
Access	free and open: backbone of operational flood monitoring (e.g. CEMS GFM)
Our scene	S1A descending, 28 Feb 2018 04:39 UTC (≈ 5 h before the Sentinel-2 image)

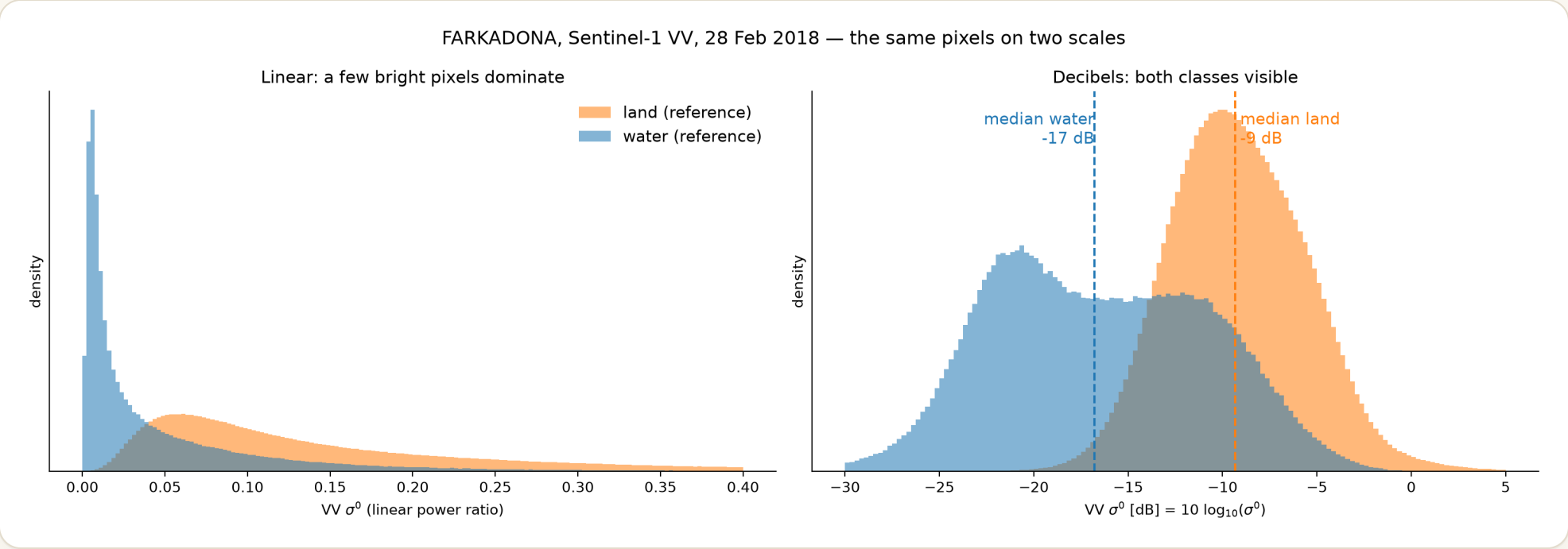
The radar equation: how much comes back

$$P_{\text{received}} = \frac{P_{\text{sent}}\, G^2\, \lambda^2}{(4\pi)^3\, R^4} \cdot A \cdot \sigma^0$$

TERM	WHAT IT IS	WHO SETS IT
P_{sent}	power of the pulse	the radar
G	antenna gain: how much it focuses the beam (squared: it sends **and** receives)	the radar
λ	wavelength (Sentinel-1: 5.6 cm)	the radar
R^4	distance: the pulse spreads on the way down (R^2), the echo on the way back (R^2)	the orbit
A	ground area of the pixel	the geometry
σ^0	**backscatter coefficient**: fraction the ground sends back to the radar, per m²	**the ground**

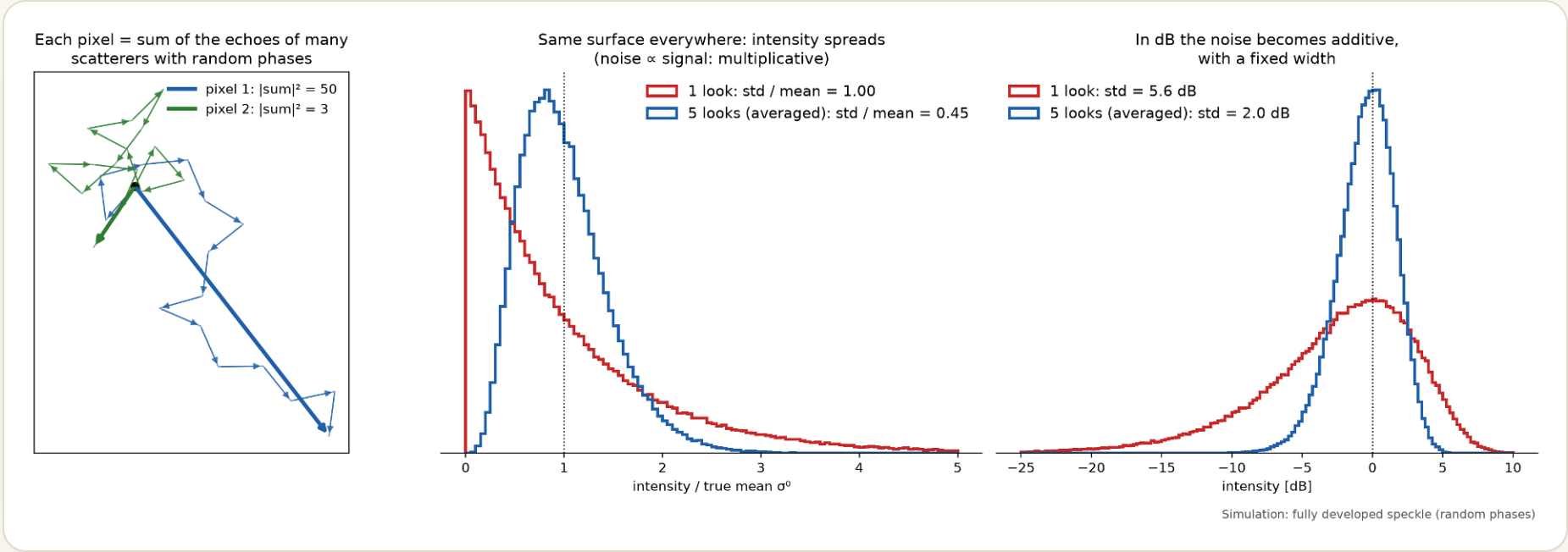
σ^0 in decibels

- σ^0 spans orders of magnitude → we use **decibels**: $\sigma^0_{\mathrm{dB}} = 10\,\log_{10}\sigma^0$ (-20 dB = 1 %, -10 dB = 10 %, 0 dB = 100 %)
- In linear units everything is crammed near 0; in dB, water and land become two humps



Speckle: why a SAR image is grainy

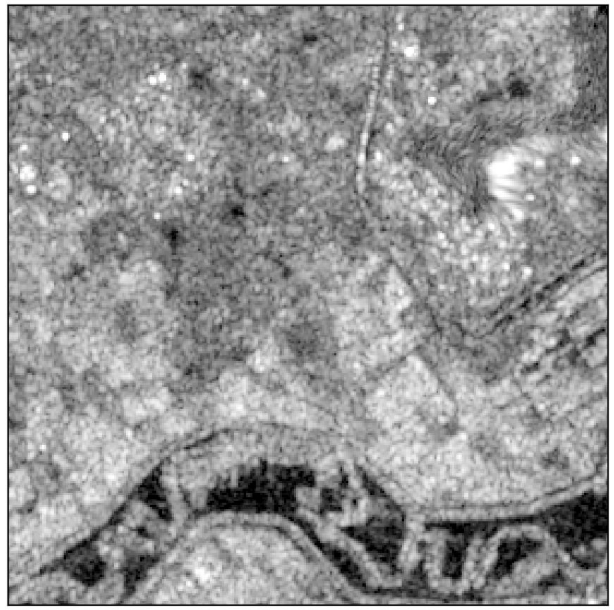
- A pixel is much larger than λ: its value is the **sum of the echoes of many small scatterers**, each with a random phase → they interfere, adding up in one pixel and cancelling in the next
- A perfectly uniform field gives a spread of values as large as the mean. **Averaging N looks** (pixels or dates) shrinks it ($\mathrm{std}/\mathrm{mean} = 1/\sqrt{N}$), at the cost of resolution



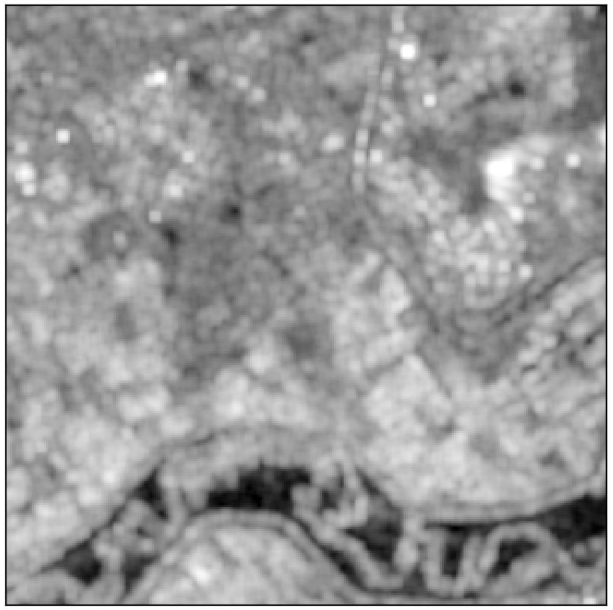
Speckle on our flood

- Averaging 5 × 5 pixels smooths the texture, but water and land **still overlap** → the rest is physics (part C), not noise

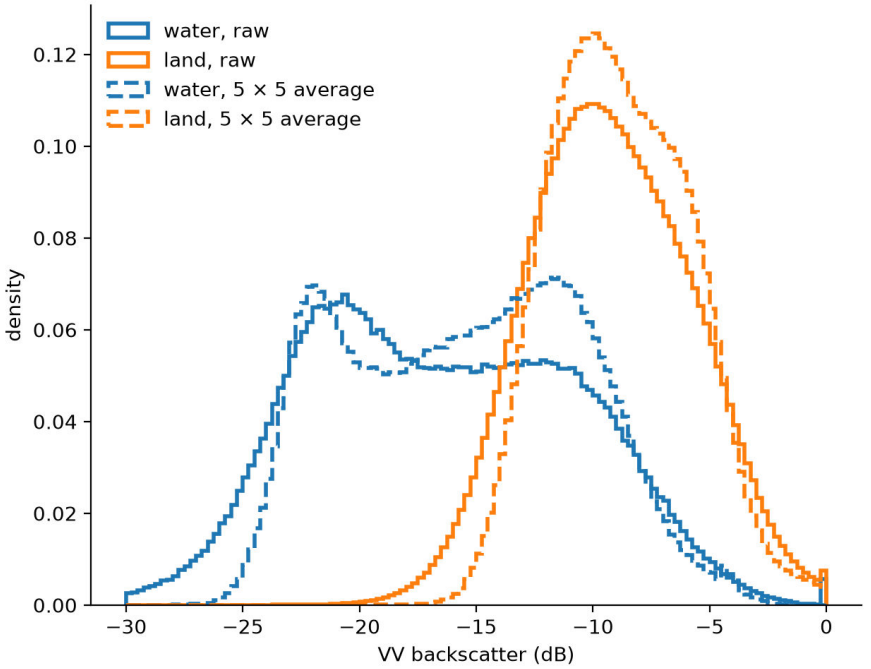
VV, raw (10 m pixels)



VV, 5 × 5 average

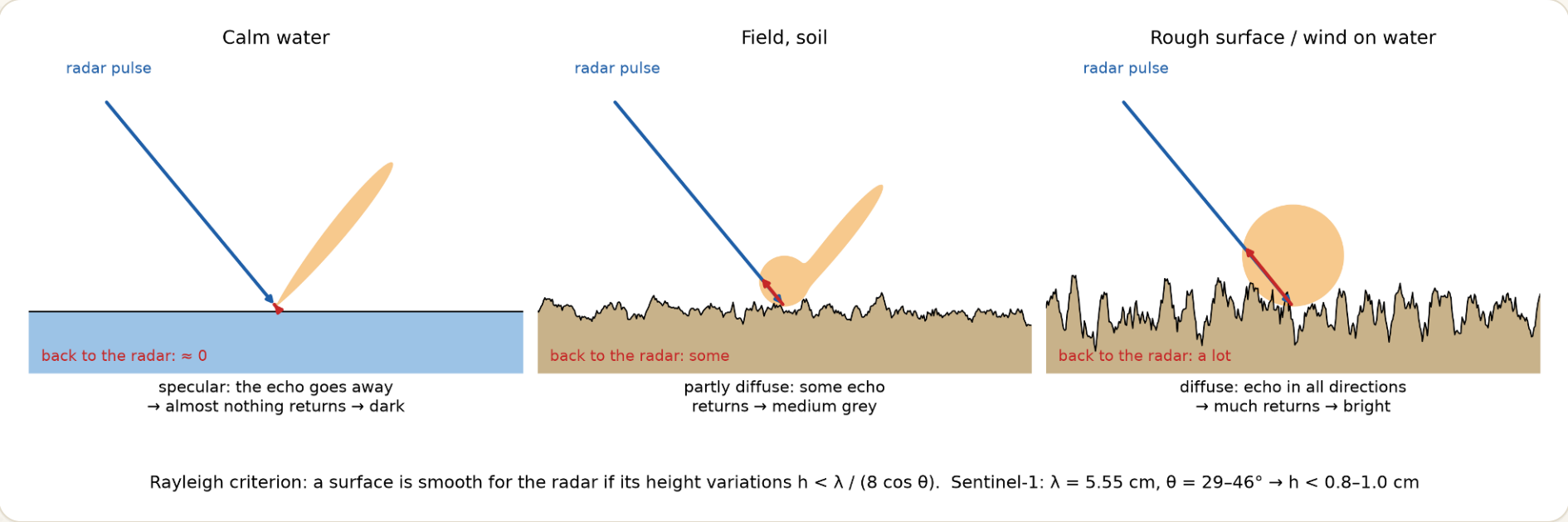


Whole FARKADONA scene, by reference class



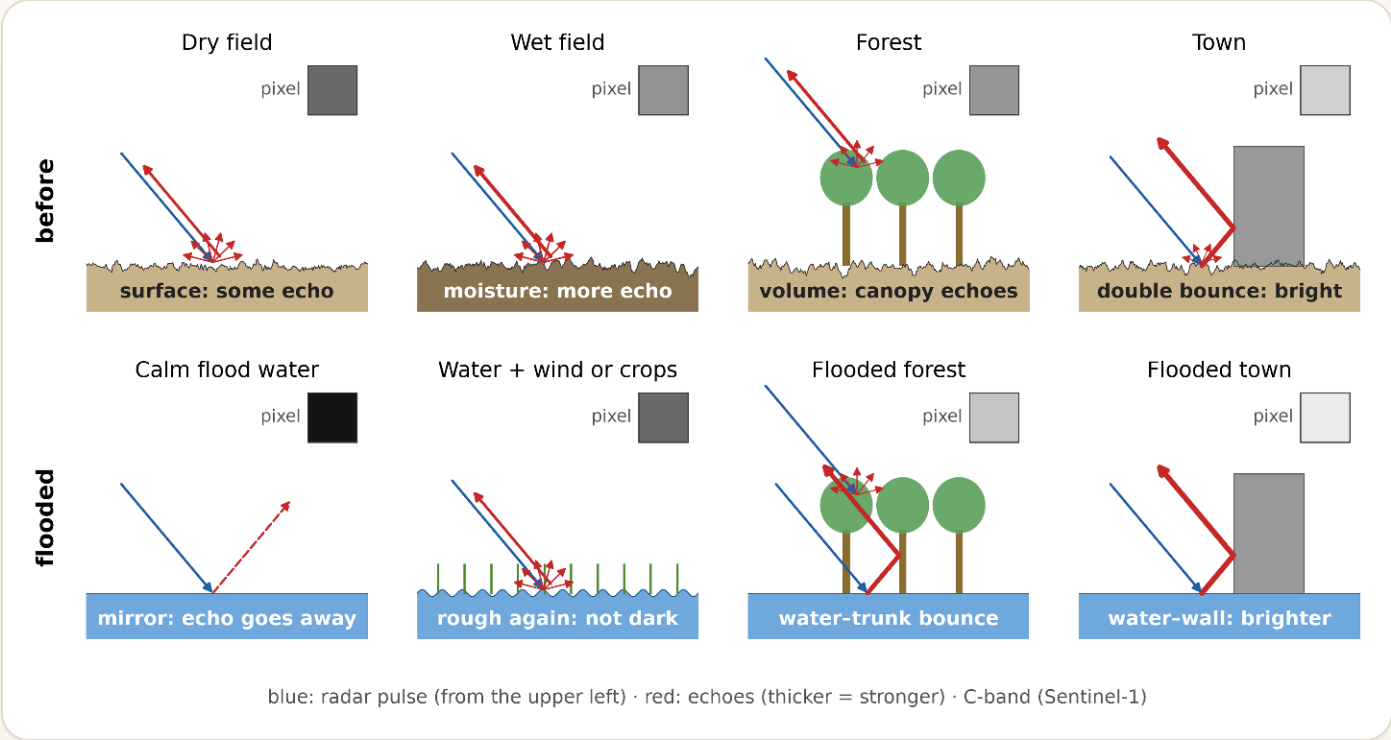
Why calm water is dark: roughness

- **Smooth** surface → reflects like a mirror, the echo goes **away** from a radar that looks sideways → dark
- **Rough** surface → scatters in all directions, part comes **back** → bright
- "Rough" is relative to the wavelength: for Sentinel-1, bumps larger than **~1 cm**. Calm water is smooth; a ploughed field or wind-blown water is rough



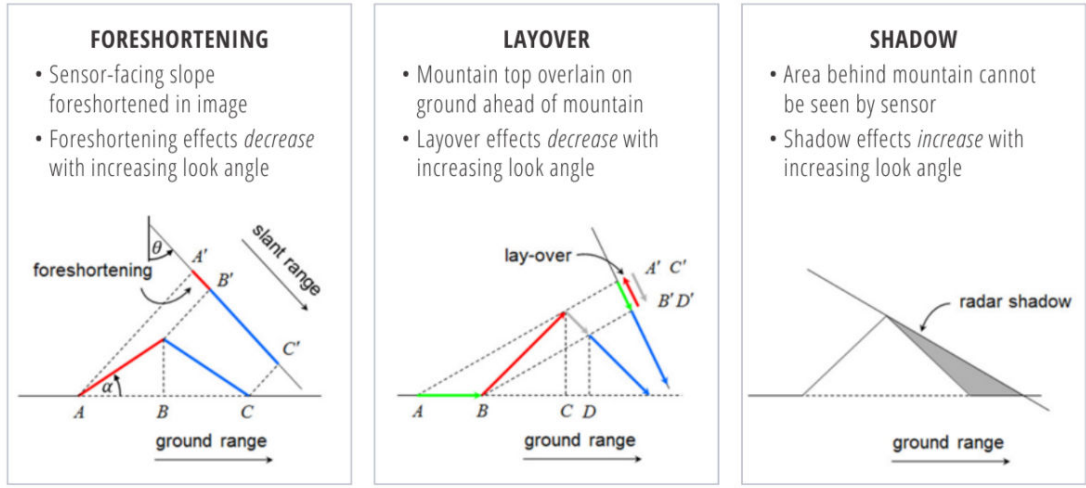
What sends the echo back

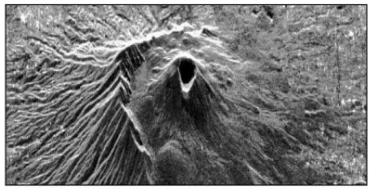
- **Surface**: roughness (previous slide) and **moisture**: wet soil reflects more → brighter · **Volume**: echoes bounce inside the canopy · **Double bounce**: ground + wall or trunk act as a corner → straight back, very bright
- When it floods, open fields go **dark**, but flooded forests and towns go **brighter**, and wind or emergent crops make the water rough again



Slopes: bright in front, dark behind

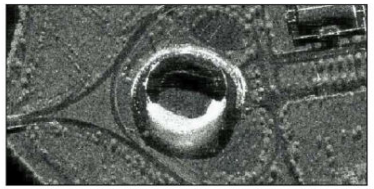
- Pixels are placed by **distance to the radar**: facing slopes come out **squashed and bright**, and behind a hill nothing is lit → **radar shadow, as dark as water**





Vesuvius (ERS-1): slopes facing the radar squashed and bright

Mountains: peaks laid over the valleys in front of them

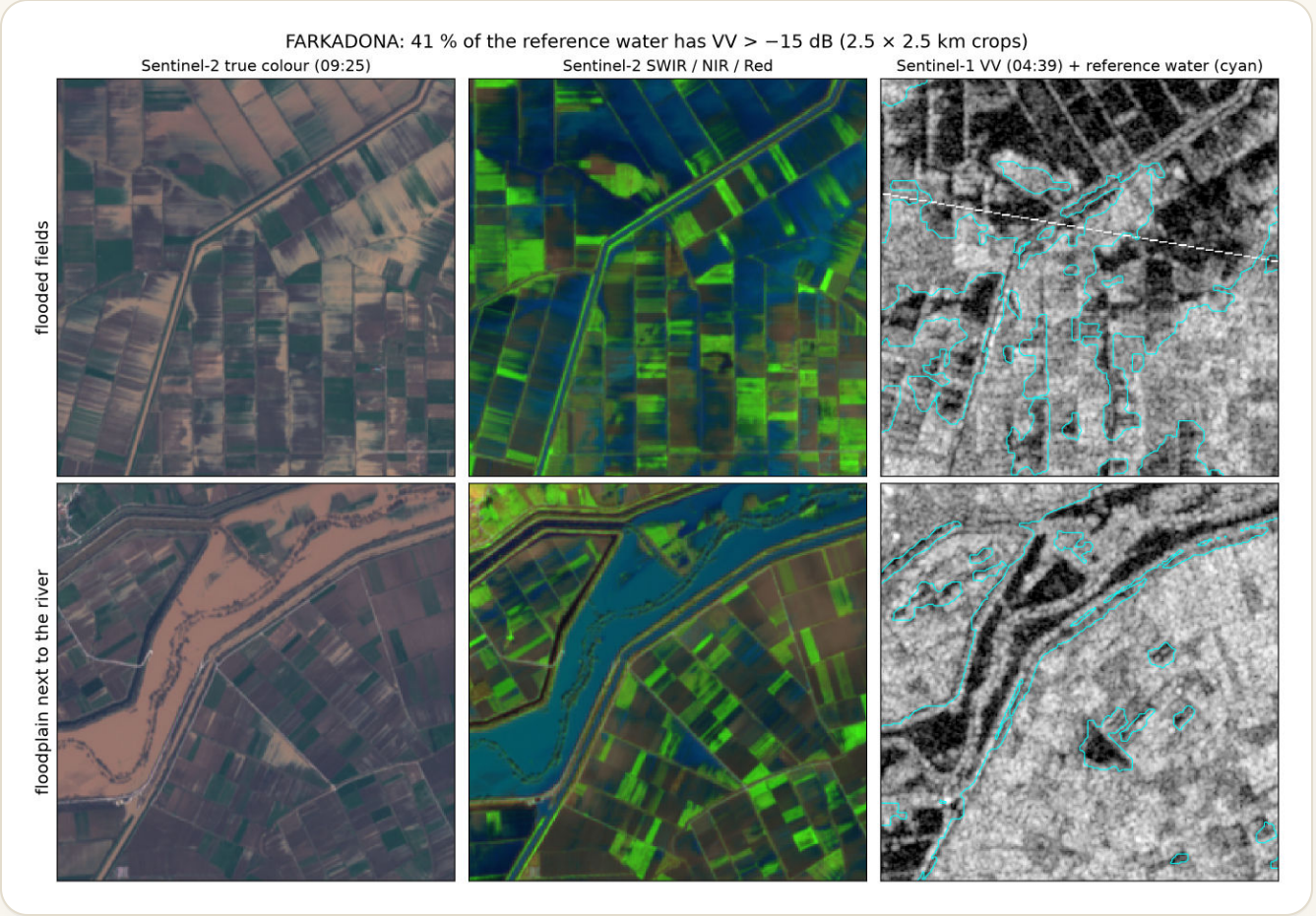


RFK Stadium: no echo behind the walls → dark, like calm water

Geometry: Flores-Anderson et al. (2019), The SAR Handbook, Fig. 2.4

Bright water in our flood

- **41 %** of the reference water of FARKADONA has VV > −15 dB
- Shallow water over crops and stubble, wind… or water that arrived after the radar pass (04:39 vs 09:25)
- Optical sees it, SAR does not → no single threshold will catch it: we need **context** (deep learning) or **both sensors**



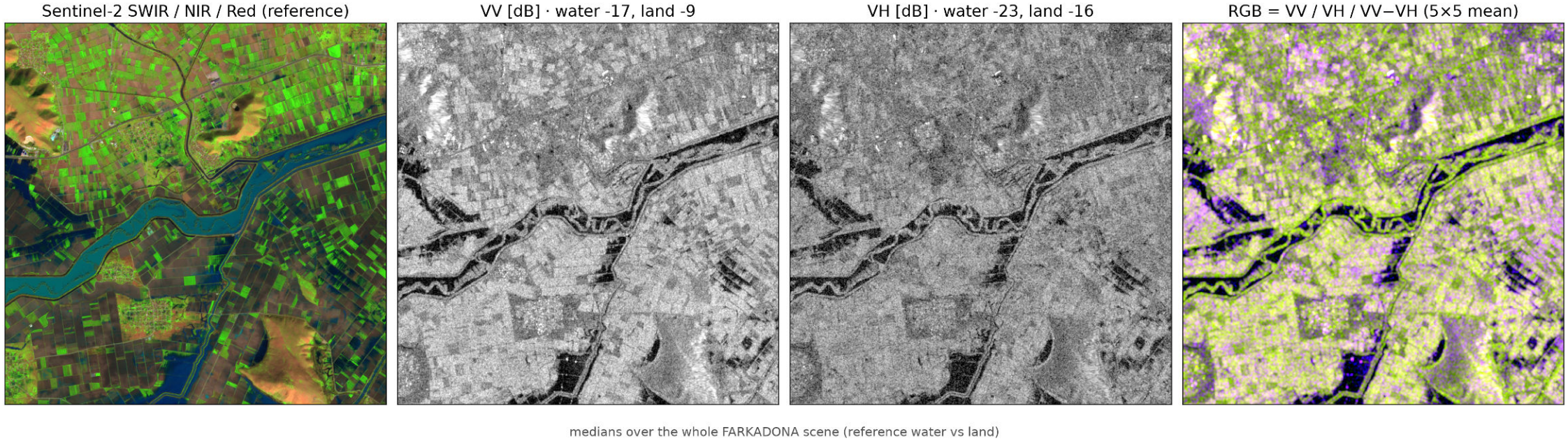
A cheat sheet of σ^0

Σ^0 (C-BAND, VV)	WHAT IS USUALLY THERE	WHY
above 0 dB	towns, slopes facing the radar	double bounce, foreshortening
−10 to 0 dB	forests, very rough surfaces	volume (random) scattering
−20 to −10 dB	crops and fields, wind-roughened water	roughness and soil moisture
below −20 dB	**calm water**, roads, very dry sand, radar shadow	almost no power returns to the radar

- Our flood: median **water −17 dB**, **land −9 dB**
- A rough guide, not a rule: it changes with **band** and **angle**
- Calm water shares its row with roads, sand and shadows → a threshold will confuse them

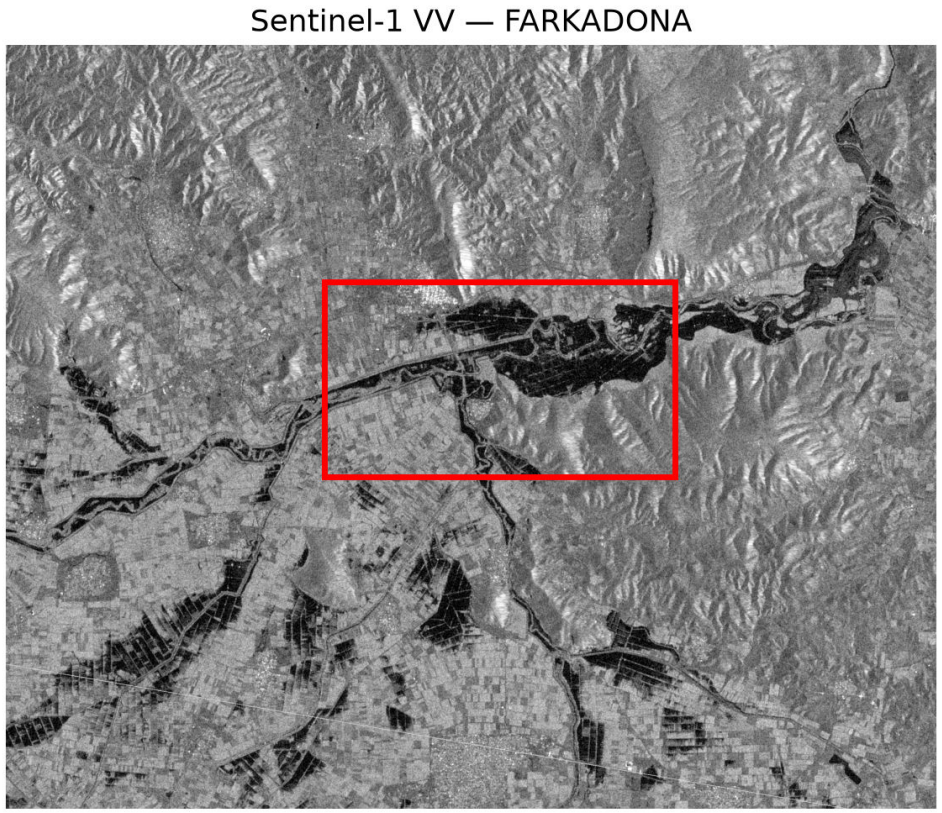
Polarisation: VV and VH

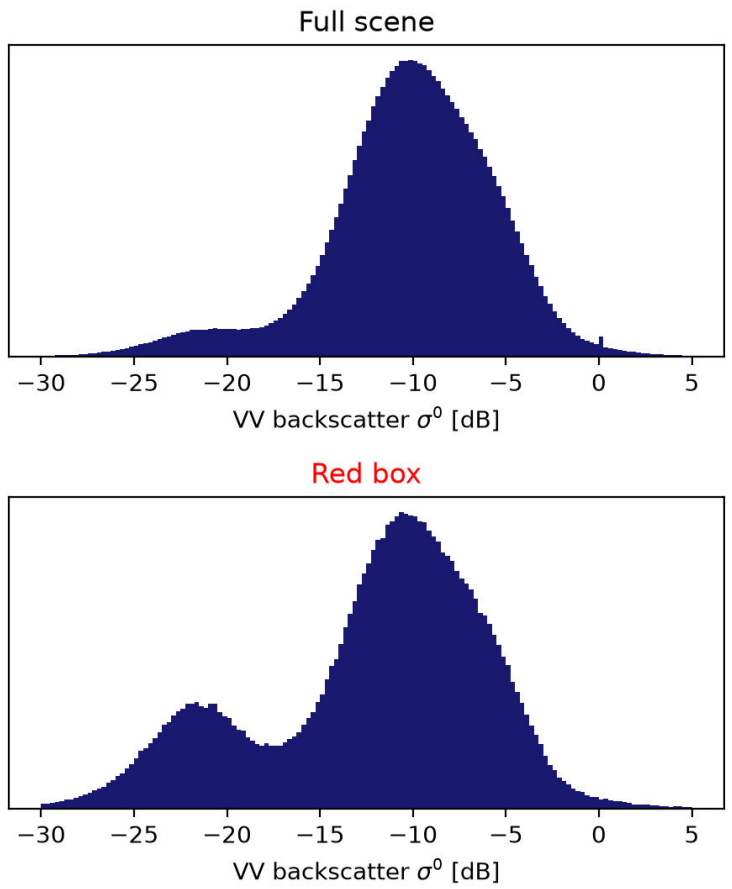
- **VV** (sent and received vertically): follows **surface roughness** → the best water/land contrast. The one we threshold next
- **VH** (sent vertically, received horizontally): only what **depolarises** the wave, mostly **volume** (vegetation) → weaker, vegetation stands out, water very dark too
- **RGB** (VV, VH, VV−VH): white where both agree (slopes facing the radar), green = vegetation, pink/purple = VV only (bare soil, double bounce in villages), black = water in both



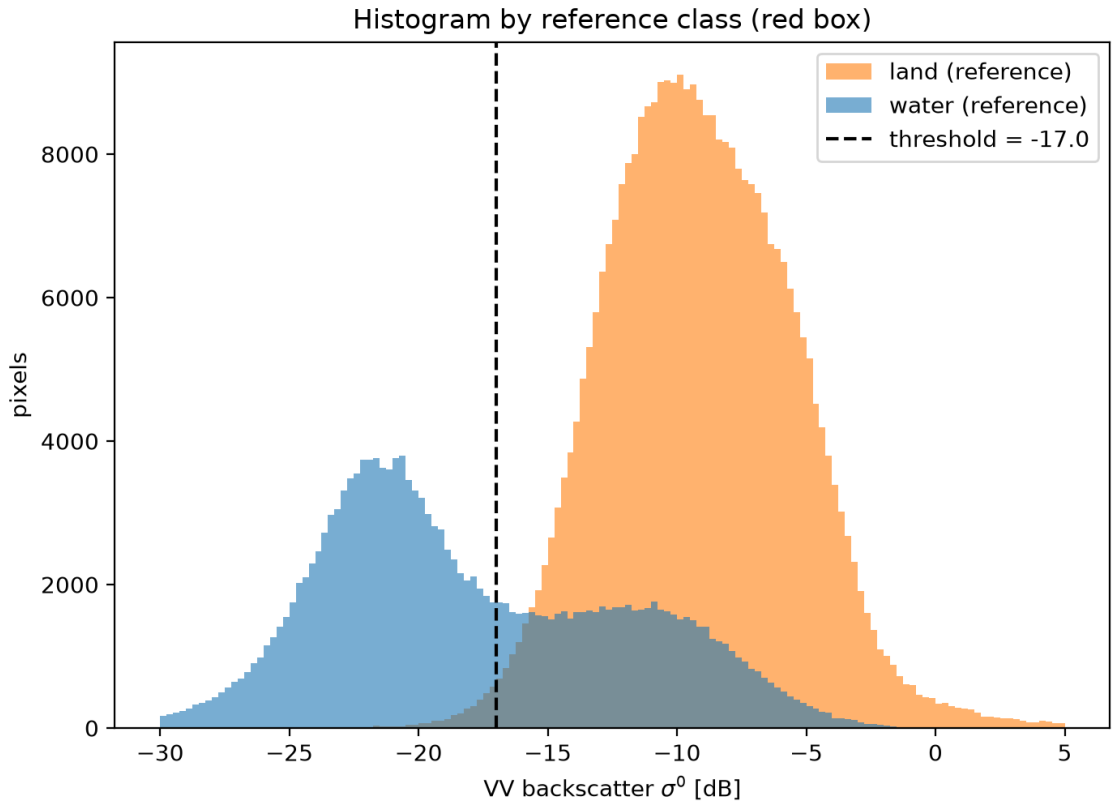
medians over the whole FARKADONA scene (reference water vs land)

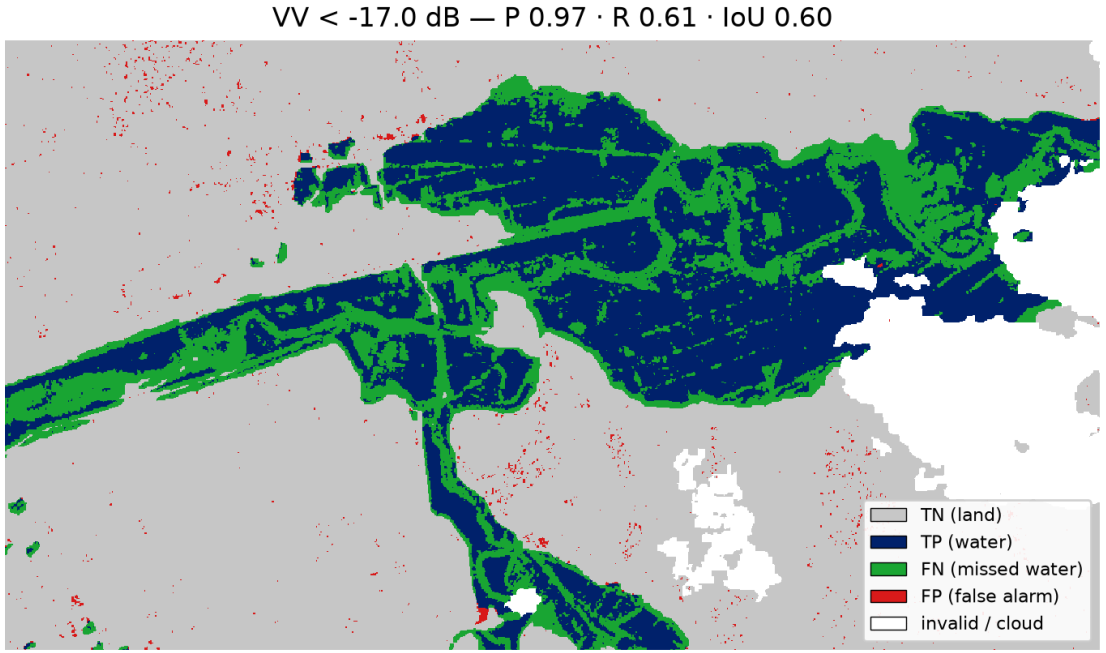
A flood seen by SAR: histograms





A threshold by eye: classes overlap





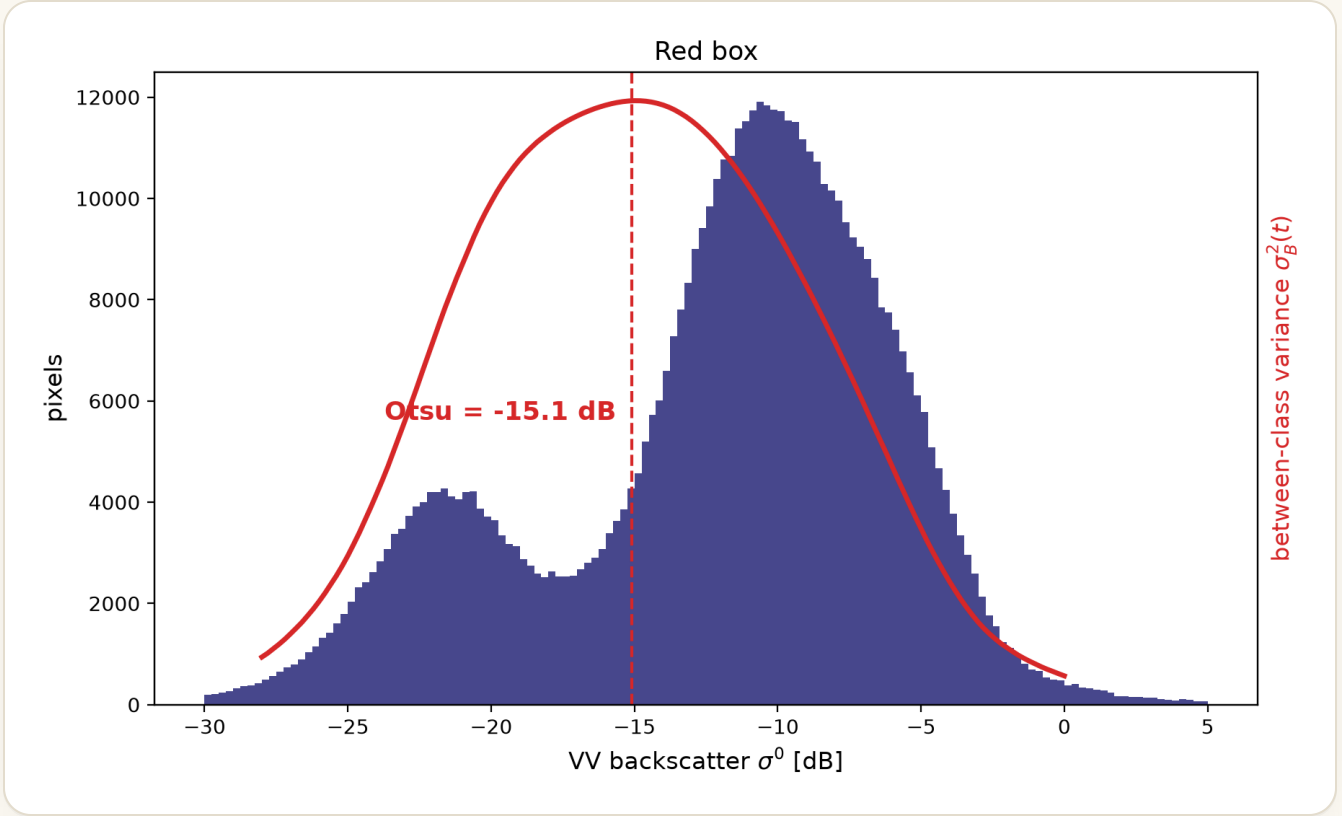
Otsu thresholding

- Try every threshold t: pixels $< t$ are *water*, $\geq t$ are *land*
- Keep the t that maximises the **between-class variance**

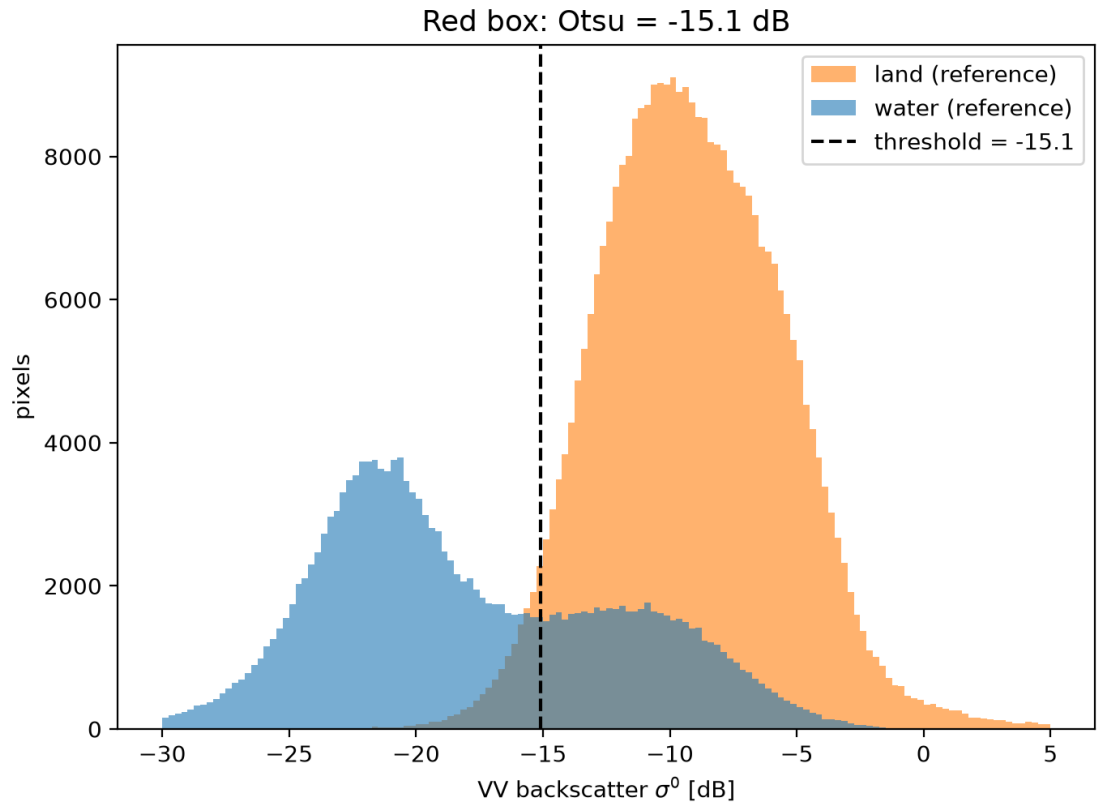
$$\sigma_B^2(t) = p_{\mathrm{w}}(t)\, p_{\mathrm{l}}(t)\, \big(\mu_{\mathrm{w}}(t) - \mu_{\mathrm{l}}(t)\big)^2$$

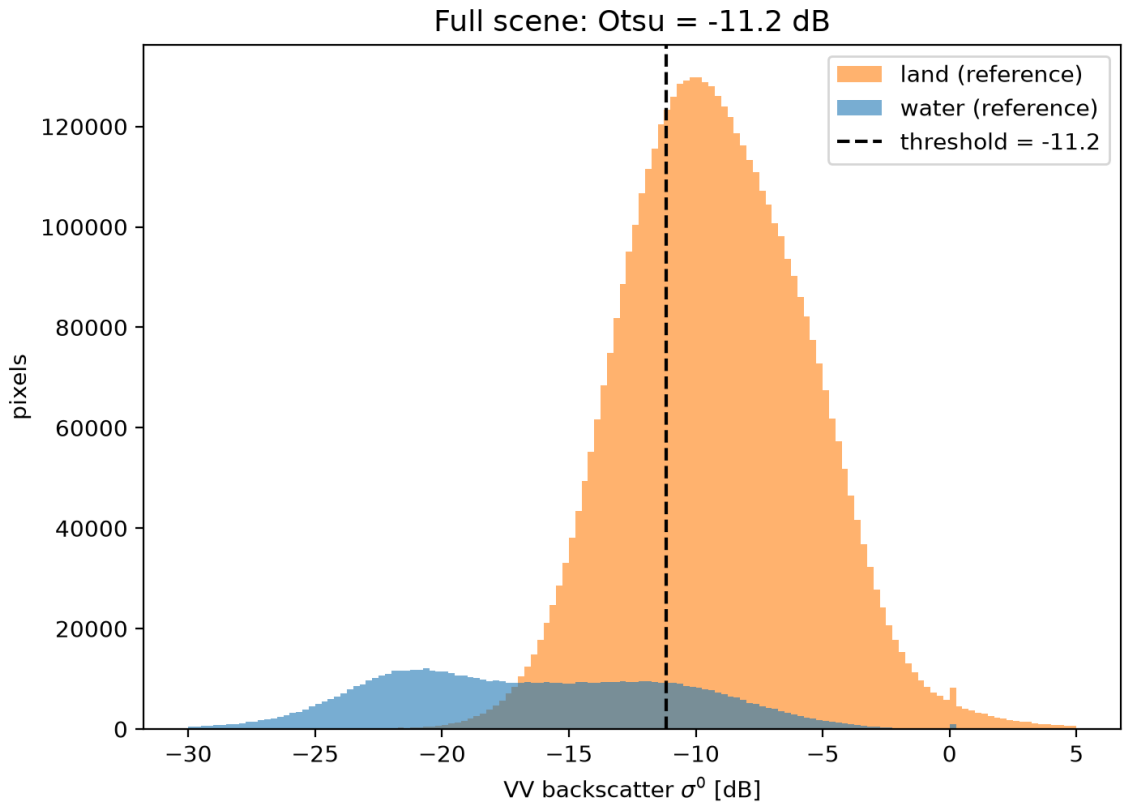
- Works when the histogram is **bimodal**

OTSU (1979)



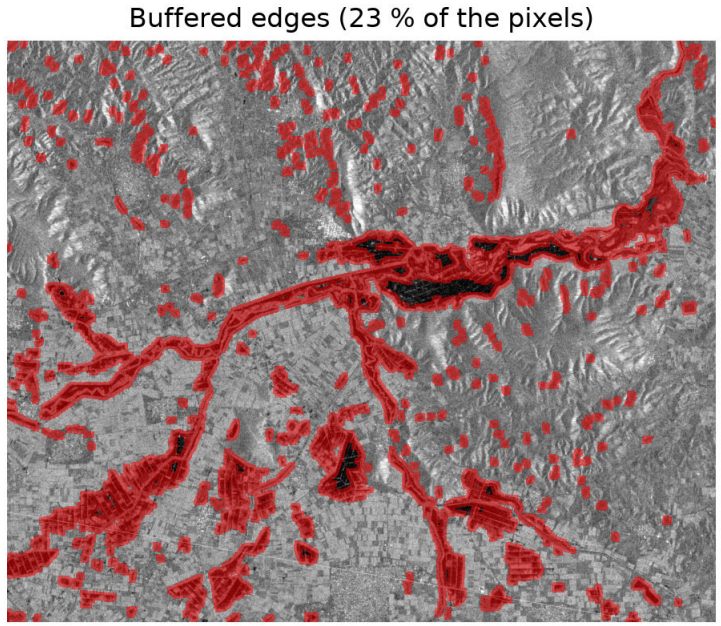
Why Otsu fails on the full scene

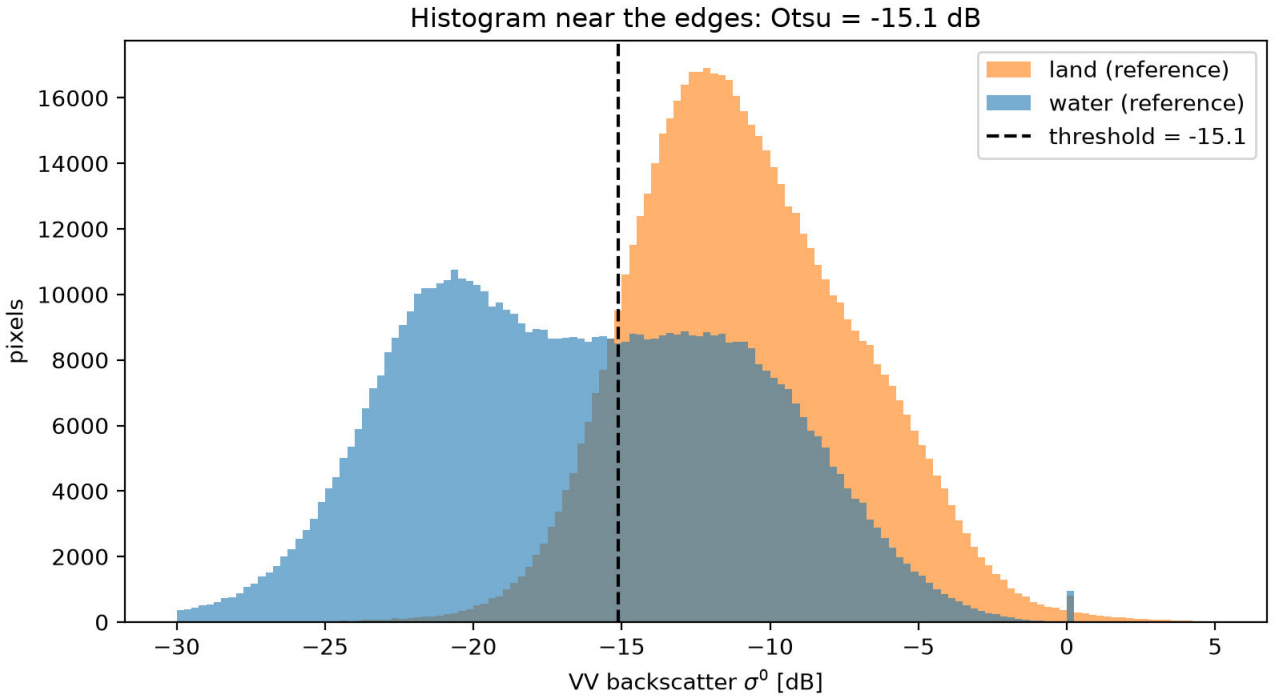




Edge Otsu

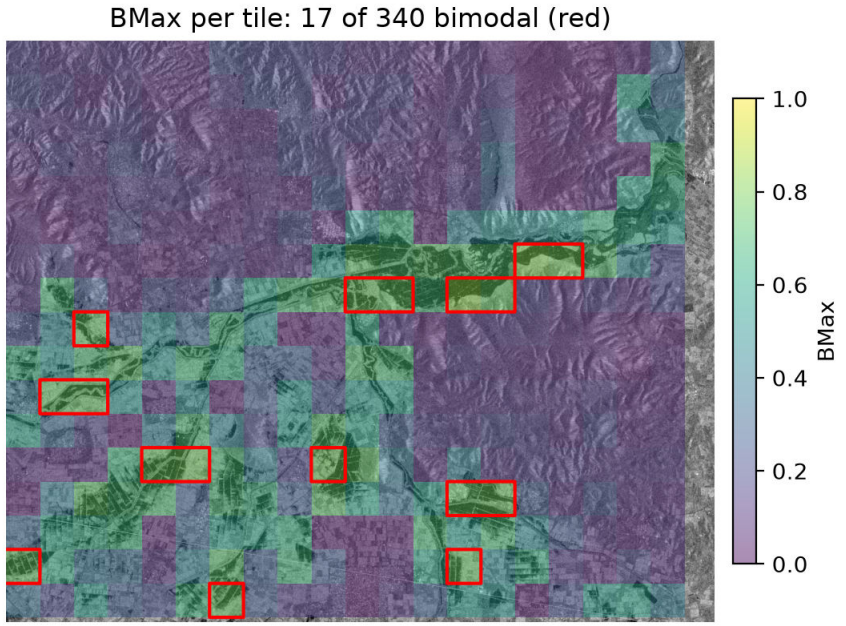
- Rough water mask → Canny edges → 100 m buffer → Otsu only near water/land edges (Markert et al., 2020)

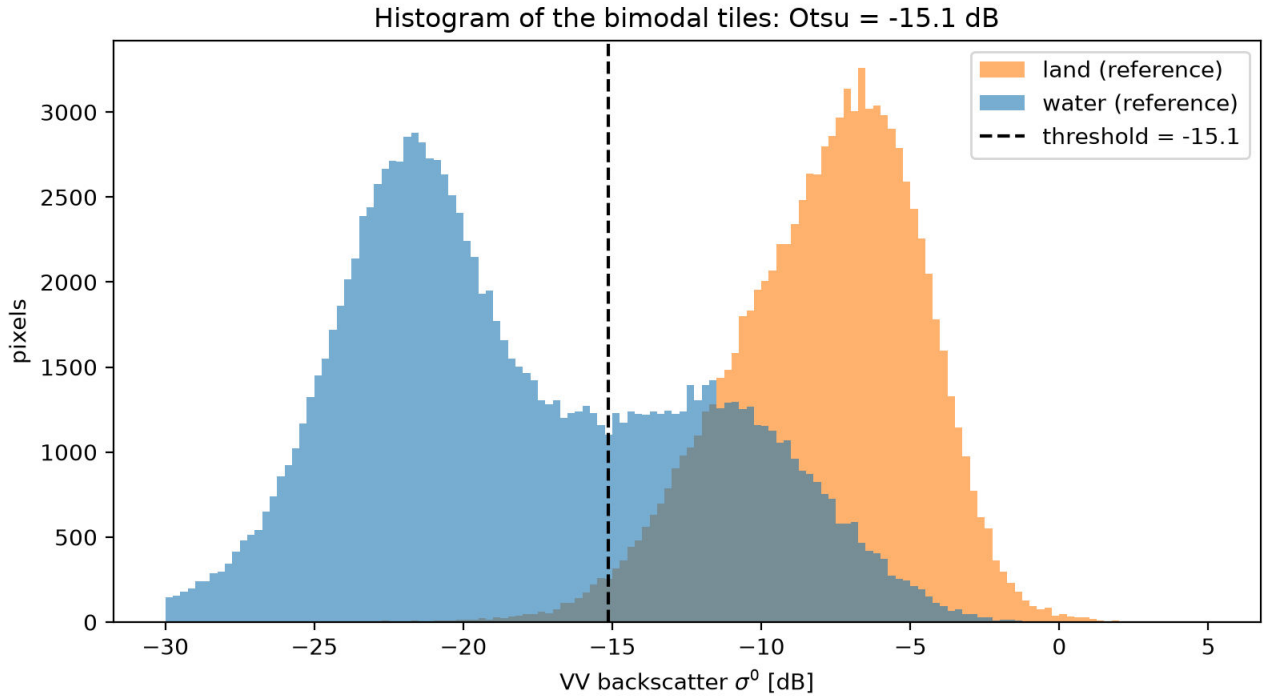




BMax Otsu

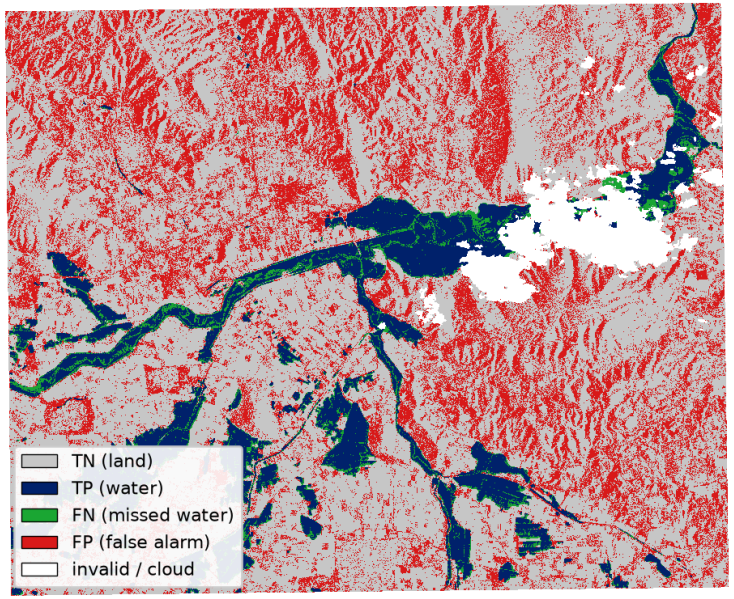
- Split in tiles → keep bimodal tiles (BMax > 0.75) → Otsu on them (Cao et al., 2019; Markert et al., 2020)



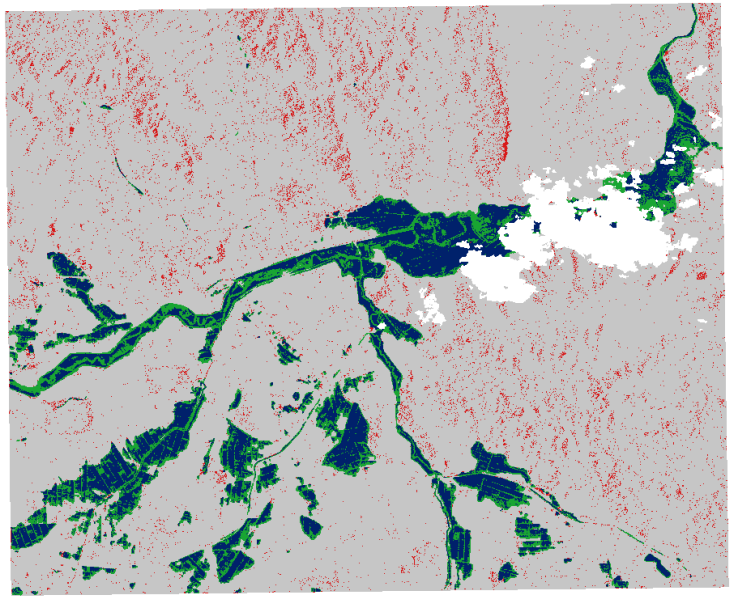


Comparison

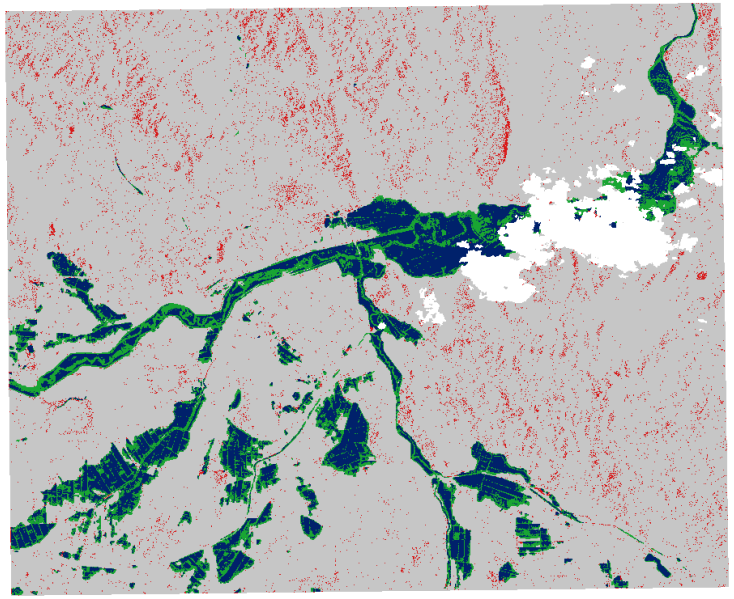
Otsu (full scene) (-11.2 dB)
P 0.28 · R 0.79 · IoU 0.26



Edge Otsu (-15.1 dB)
P 0.68 · R 0.59 · IoU 0.46

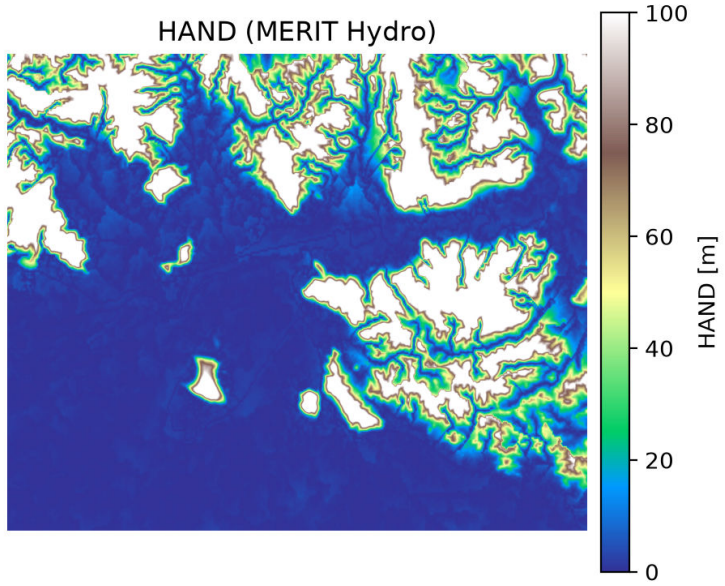


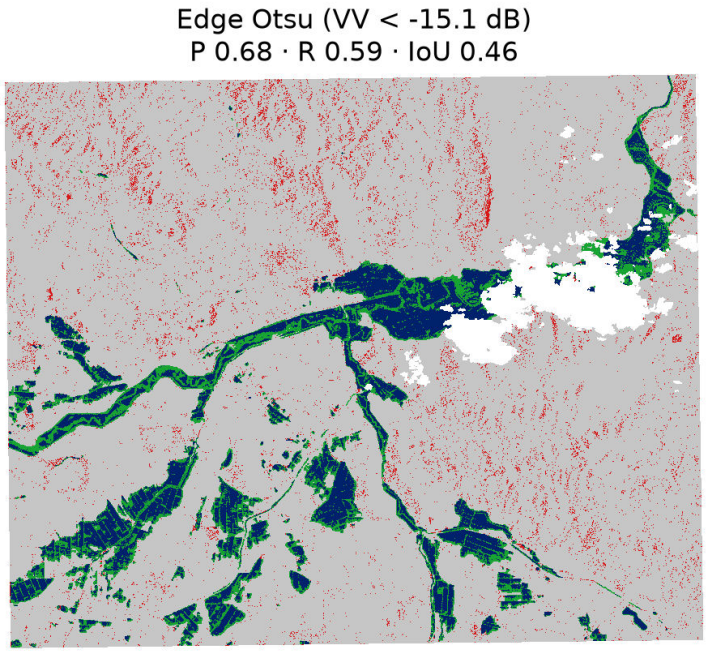
BMax Otsu (-15.1 dB)
P 0.69 · R 0.59 · IoU 0.46

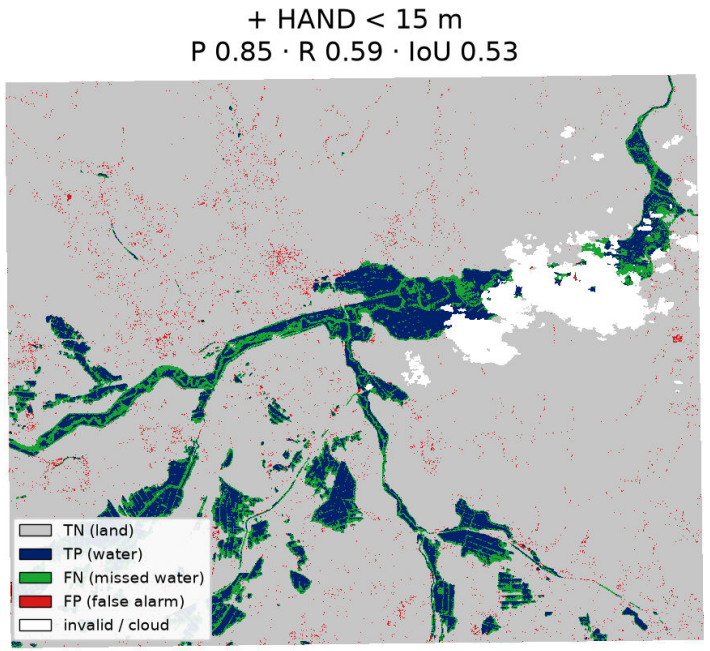


Adding terrain: HAND

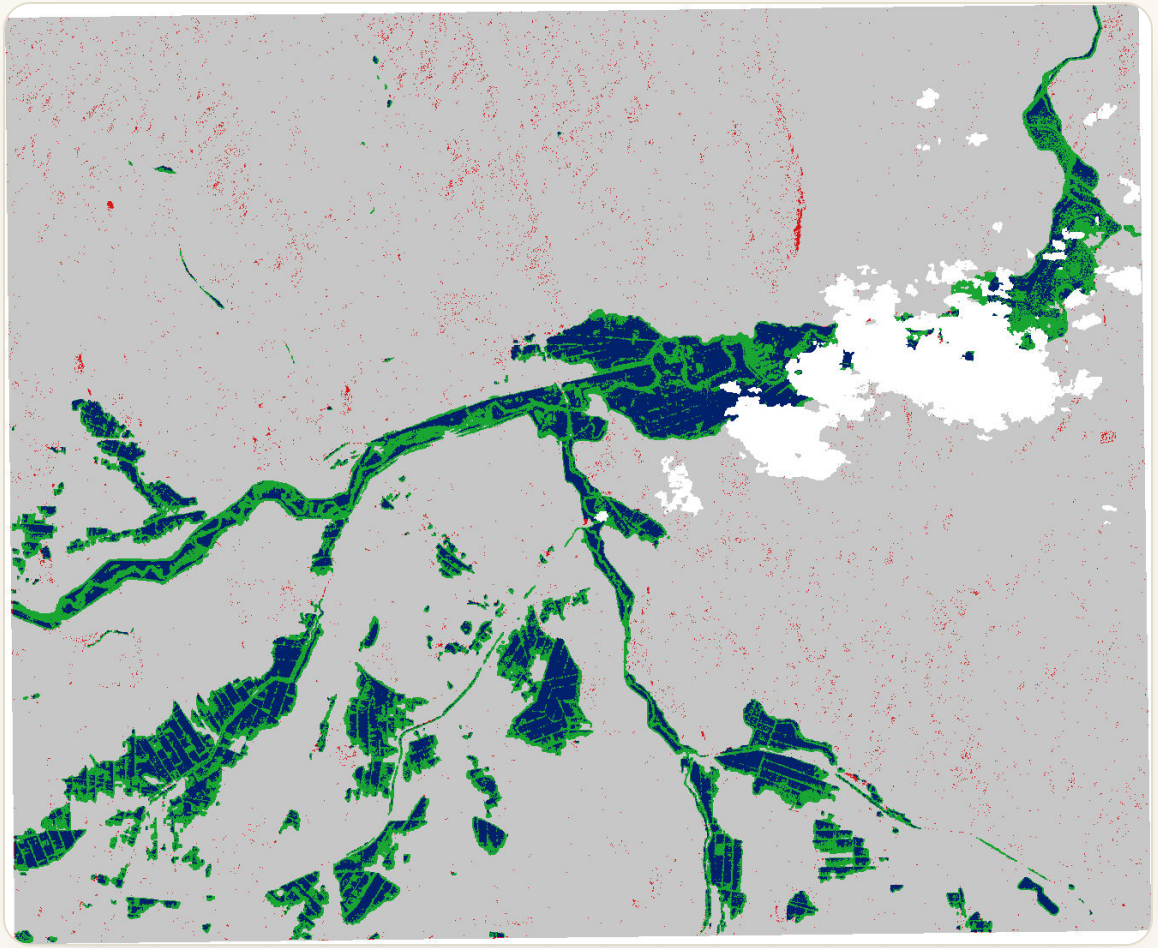
- Height Above Nearest Drainage (MERIT Hydro): remove water detected > 15 m above the river







Try it: VV threshold, Otsu and HAND

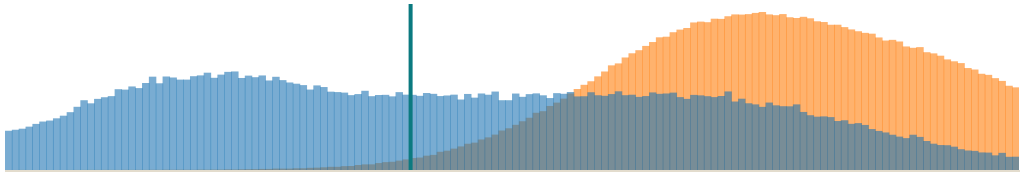


EMSR271 · FARKADONA · SENTINEL-2 · VIEW: CONFUSION

VV THRESHOLD τ

-17.0 dB

-25 WATER BELOW τ -5



water land REFERENCE

Otsu (whole scene) HAND < 15 m

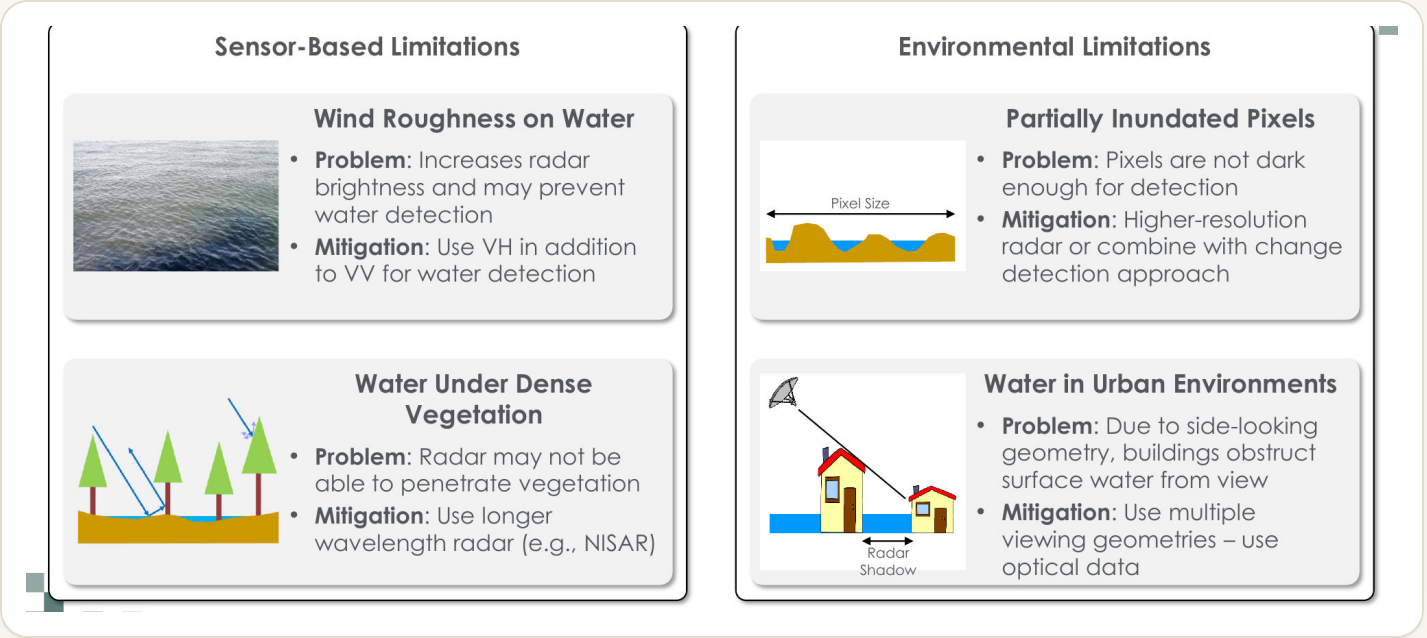
PRECISION	RECALL	IOU
0.89	0.49	0.46

CONFUSION VV RGB SWIR

TP FN FP TN cloud / invalid

Limits of rule-based methods

- Manual or scene-dependent thresholds
- Per-pixel decisions: no spatial context
- Clouds (optical) / smooth surfaces and shadows (SAR)
- → learn from data: deep learning



NASA ARSET (2023), SAR for Floods, Part 3

03

Hands-on 1

Classical flood mapping — notebooks 01a (optical) and 01b (SAR)

Break

10 min

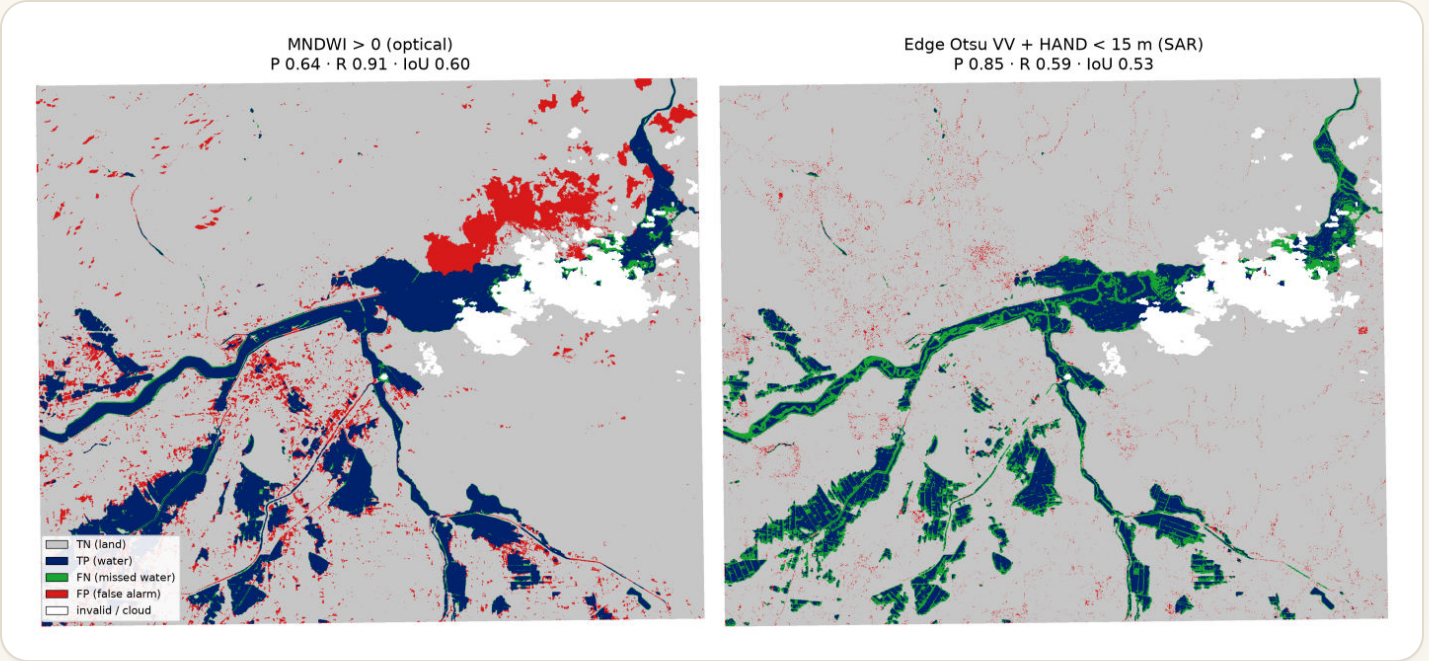
04

Deep learning for flood mapping

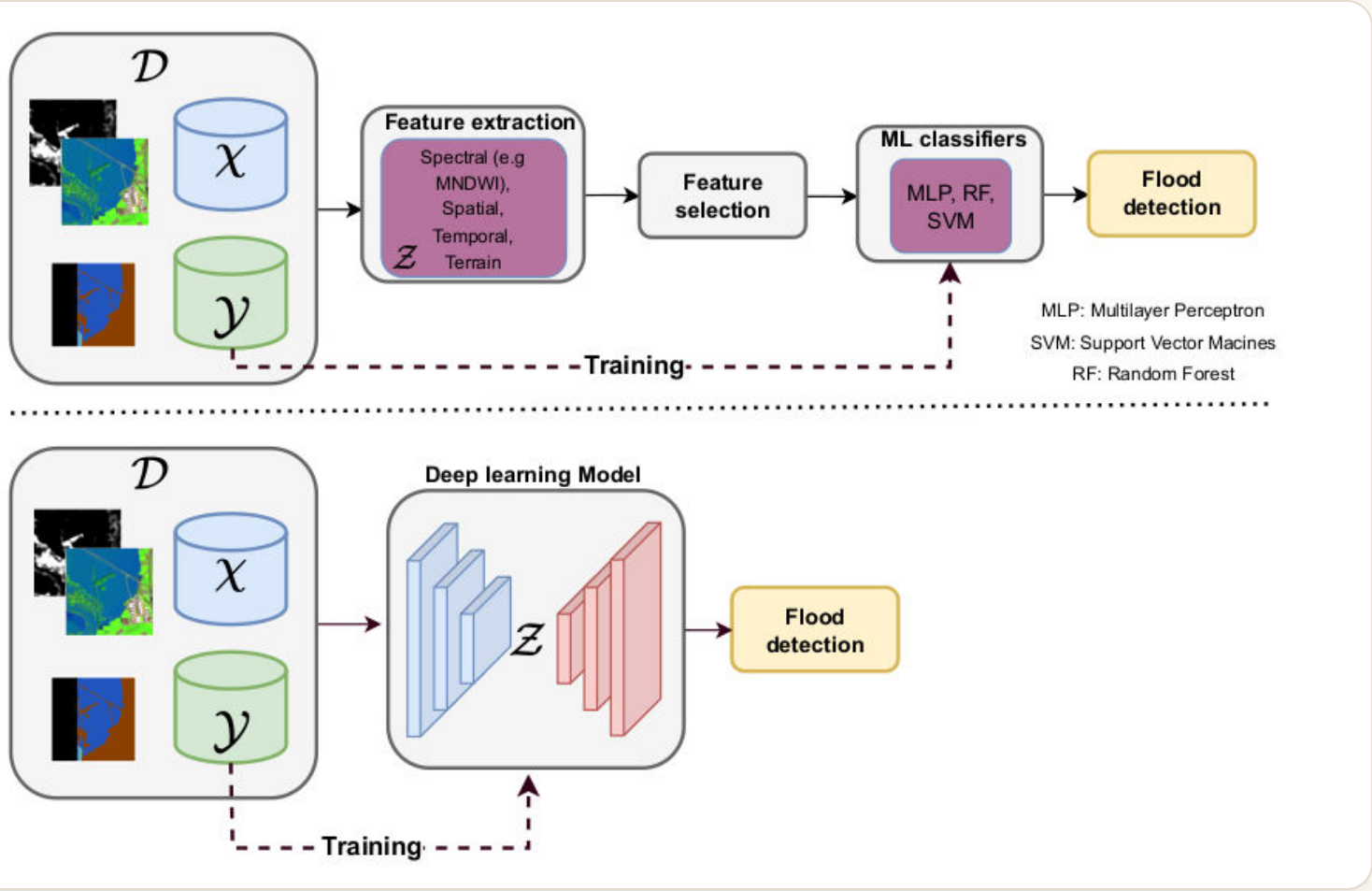
WorldFloods & ml4floods

Recap: where rules break

- Thresholds change from scene to scene
- Pixel by pixel: no spatial context
- Clouds and shadows (optical); smooth fields and slopes (SAR)
- More data does not make them better



Classical ML vs deep learning approaches



> For both approaches, success depends on **data quality**.

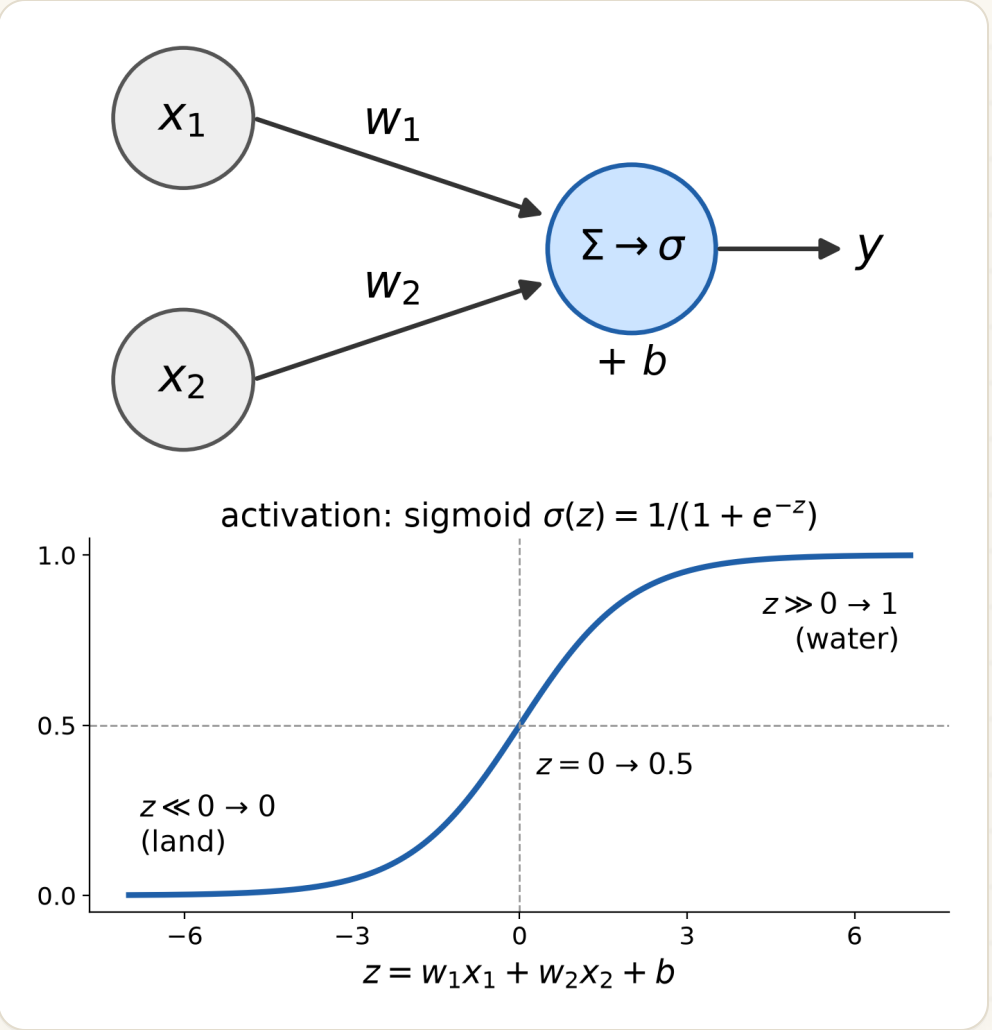
The artificial neuron

- Inputs x_1, x_2 — e.g. two bands of a pixel
- Weighted sum + bias: $z = w_1 x_1 + w_2 x_2 + b$
- An **activation** squashes z between 0 and 1 — the **sigmoid**:

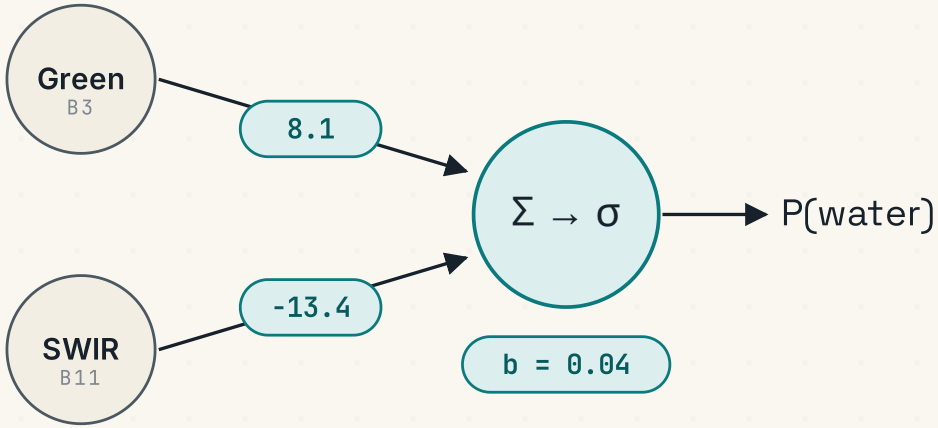
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$P(\text{water}) = \sigma\left(w_1 x_1 + w_2 x_2 + b\right)$$

- **Learning = finding w and b**



A threshold is already a neuron



$$P(\text{water}) = \sigma\left(w_1\, G + w_2\, \text{SWIR} + b\right)$$

BY HAND

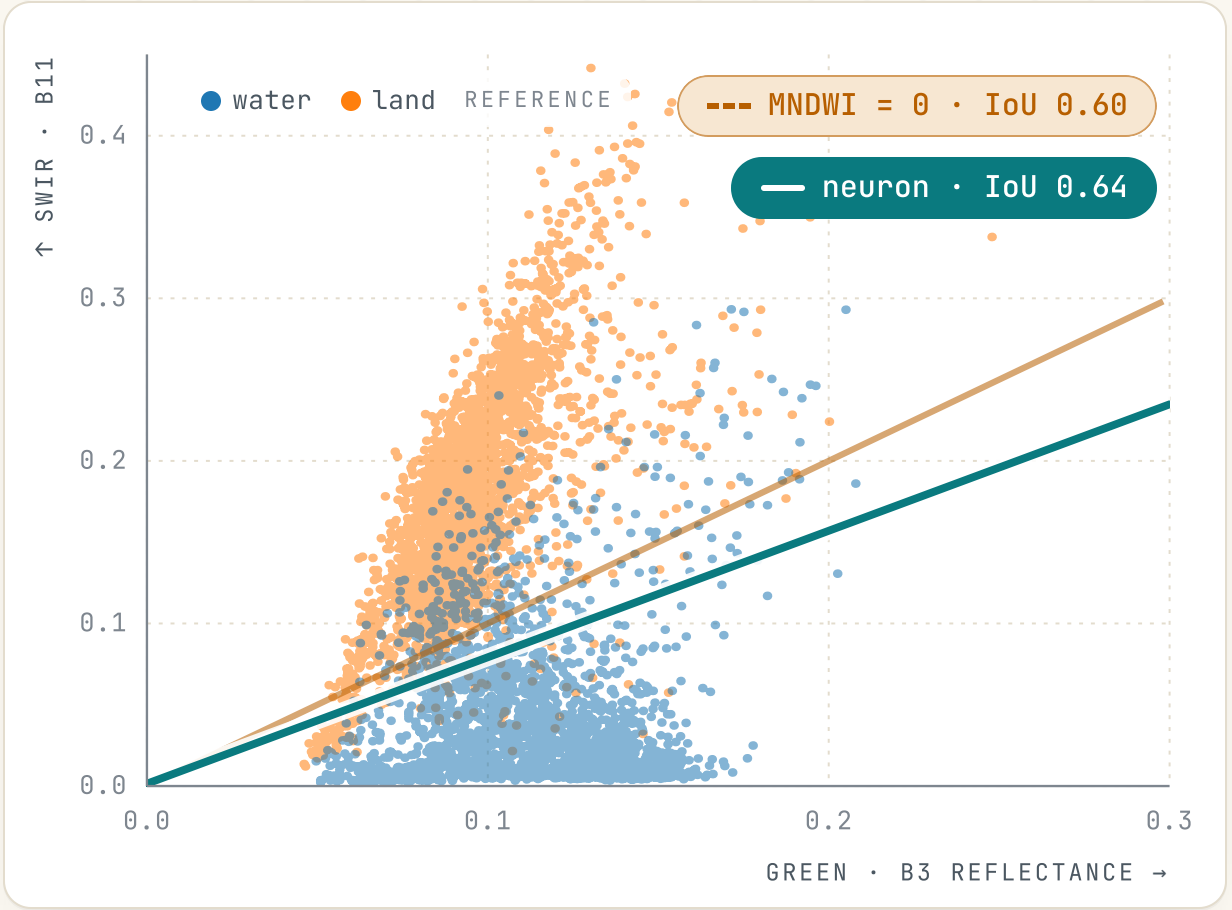
$$\text{MNDWI} > 0 \iff G - \text{SWIR} > 0$$

so `w = (1, -1), b = 0`

LEARNED

Logistic regression on `160 000` labelled px from the `8` train events

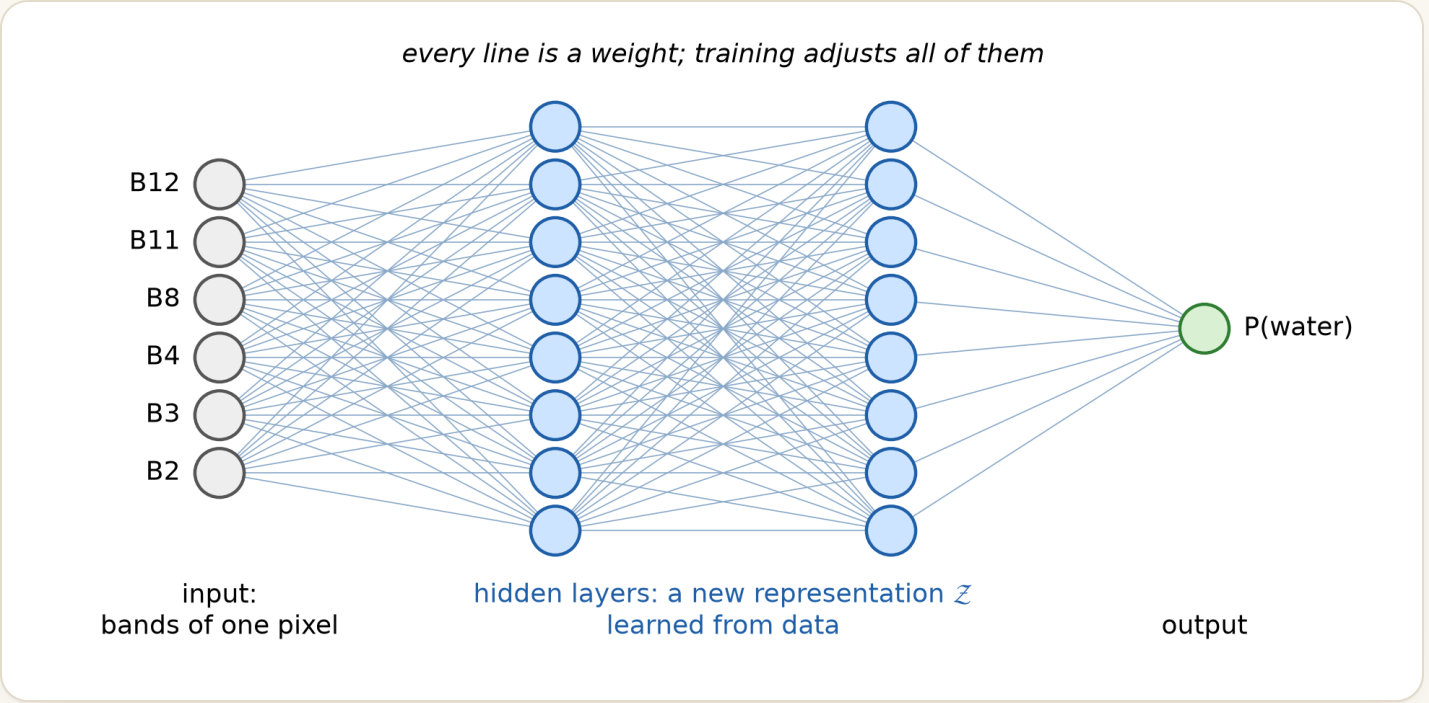
`w = (15.5, -26.5), b = 0.08`



FARKADONA (TEST EVENT) · 2 500 PX PER CLASS · SENTINEL-2

The multilayer perceptron (MLP)

- One neuron = one straight line
- Many neurons in layers → curved decision boundaries
- Every line is a weight: all of them are learned from labelled examples



How a network learns — and generalises

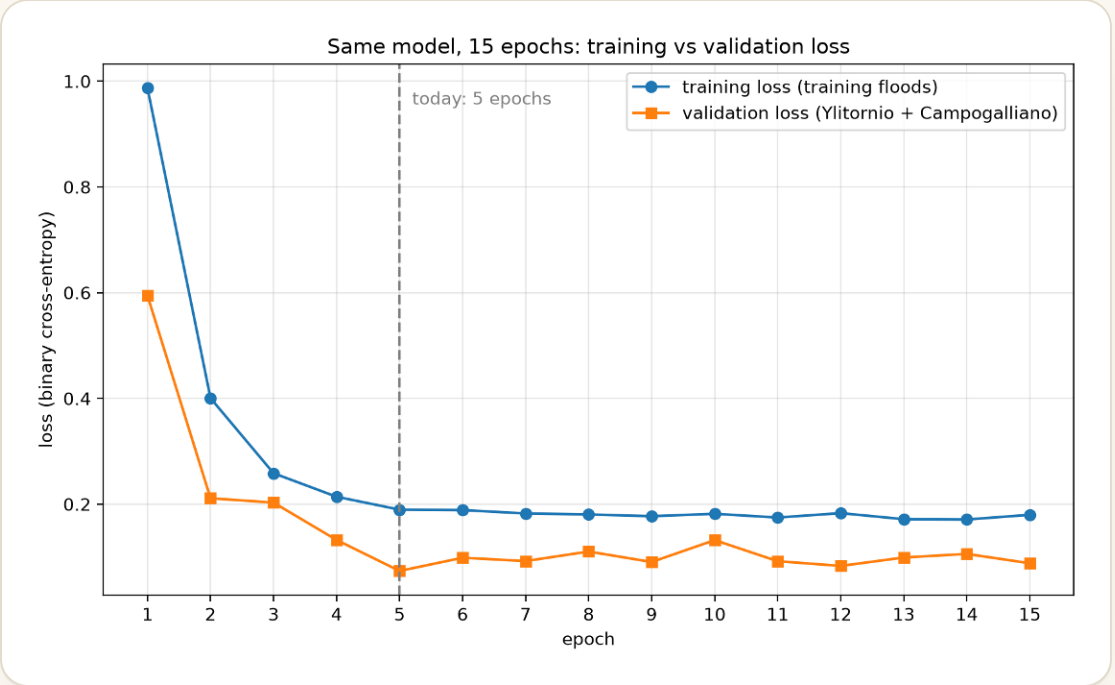
- **Loss**: how wrong $P(\text{water})$ is against the labels y (binary cross-entropy)

$$\mathcal{L} = -\big[\, y \log p + (1-y) \log(1-p) \,\big]$$

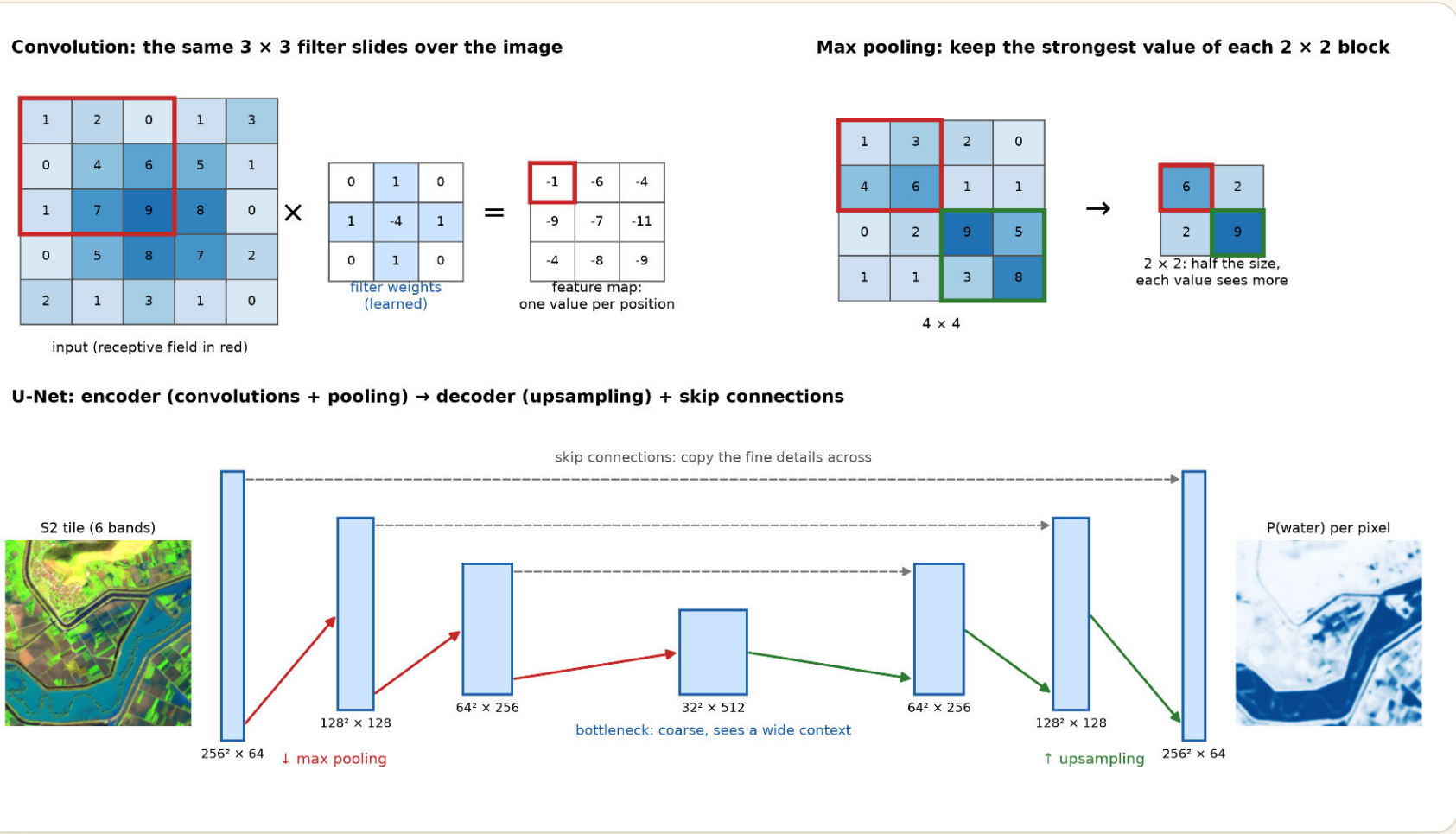
- **Gradient descent**: every weight a little downhill

$$w \leftarrow w - \eta \, \frac{\partial \mathcal{L}}{\partial w}$$

- Backpropagation is automatic: `loss.backward()` gives every $\partial \mathcal{L} / \partial w$ (PyTorch, TensorFlow, JAX)
- One batch of tiles at a time → memory does not grow with the dataset
- Training loss always goes down; **validation loss** (an event never trained on) tells us if the model generalises



From CNNs to U-Net



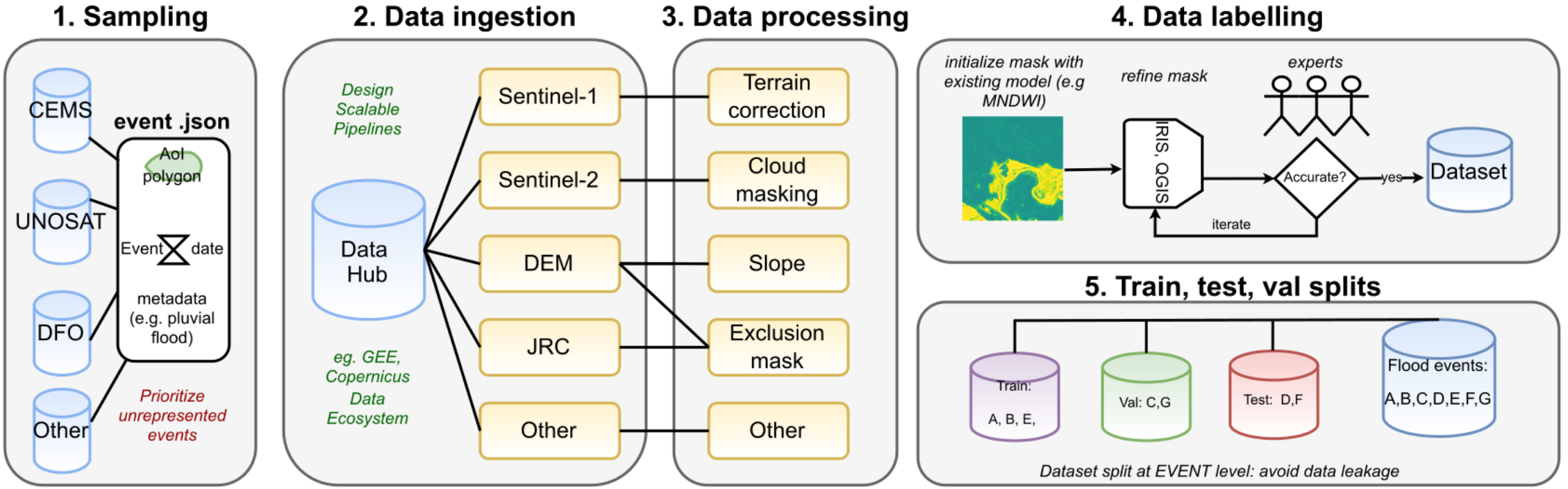
Towards global flood mapping

- WorldFloods: flood maps from Copernicus EMS & UNOSAT + Sentinel-2 (Mateo-García et al., 2021, Scientific Reports)
- ml4floods: open-source toolkit, from flood maps to trained models
- WorldFloods v2 + cloud-aware models (Portalés-Julià et al., 2023, Scientific Reports)
- One optical model for floods anywhere in the world

github.com/spaceml-org/ml4floods

Mateo-Garcia, G., Veitch-Michaelis, J., Smith, L. et al. (2021). Towards global flood mapping onboard low cost satellites with machine learning. *Scientific Reports*, 11.
Portalés-Julià, E., Mateo-García, G., Purcell, C. and Gómez-Chova, L. (2023). Global flood extent segmentation in optical satellite images. *Scientific Reports*, 13, 20316.

Flood detection datasets



Building a flood detection dataset: sampling, ingestion, processing, labelling and an event-level split. Portalés-Julià (2025), PhD thesis

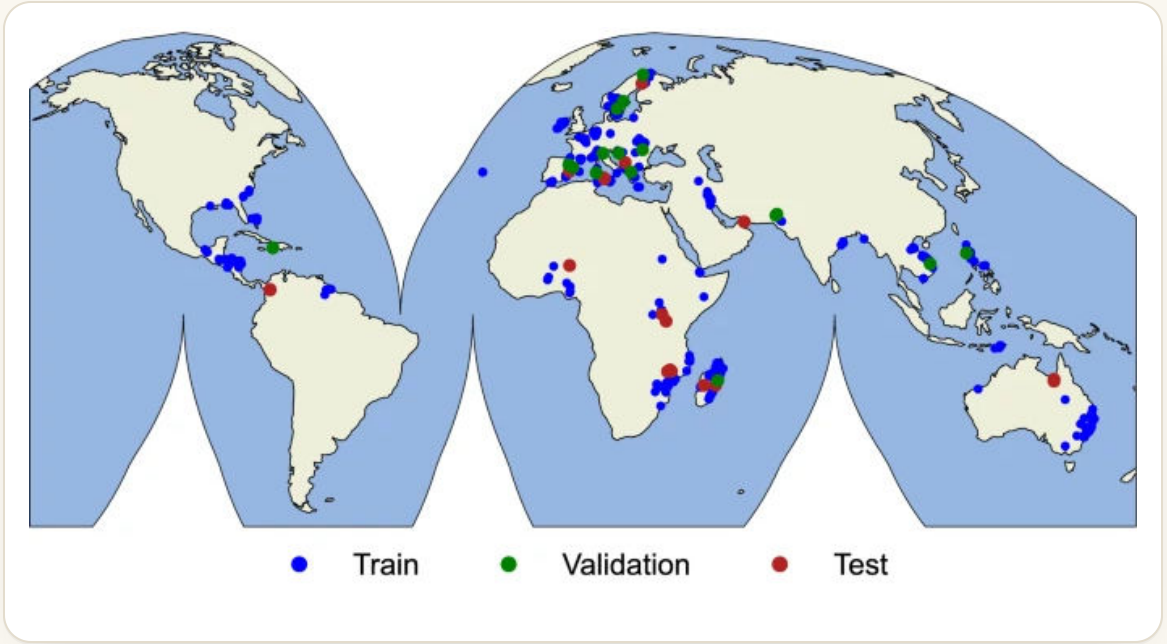
WorldFloods Dataset (2023)

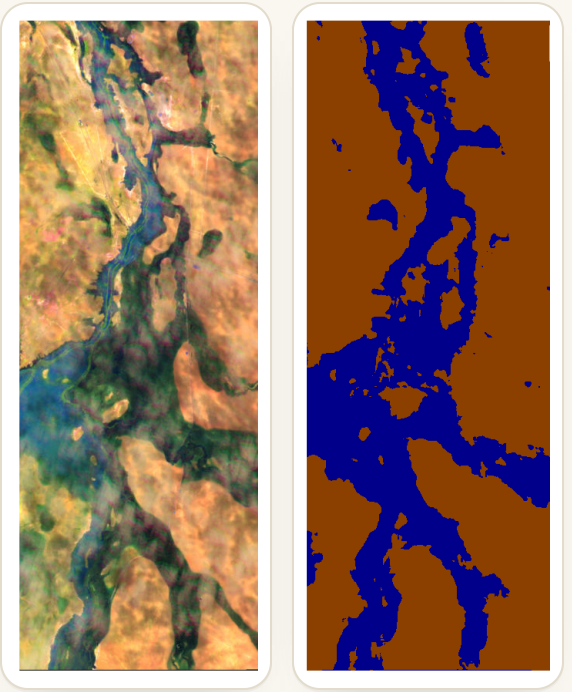
144
FLOODS

509
FLOOD MAPS

73 809
PATCHES · 256 × 256 PX

309 GB
TOTAL SIZE





Mateo-García, G. et al. Towards global flood mapping onboard low cost satellites with machine learning. Scientific Reports (2021) · Portalés-Julià, E. et al. (2023), Scientific Reports

Getting WorldFloods v2

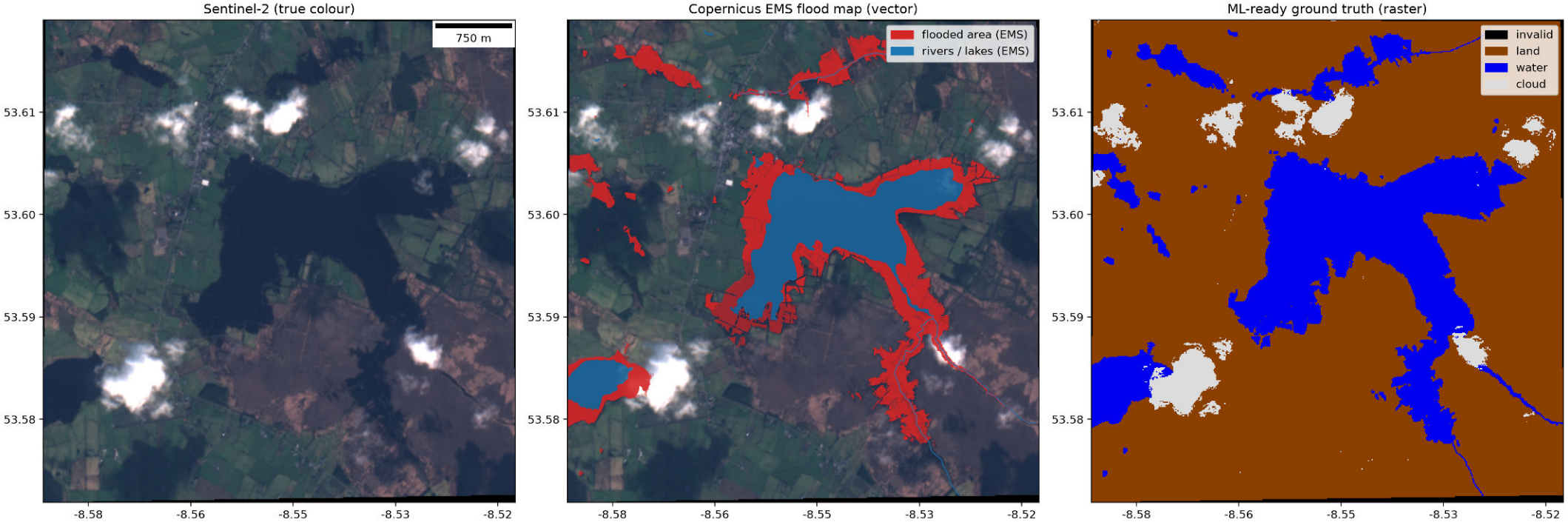
- `huggingface.co/datasets/isp-uv-es/WorldFloodsv2`
- ~102 GB in total (almost all Sentinel-2) → today we use **11 small events** (~0.7 GB)
- One file per flood map and product: `S2/` (13 bands, GeoTIFF), `gt/` (2 bands: cloud, water), `PERMANENTWATERJRC/`, `floodmaps/` (GeoJSON), `meta/` (JSON)
- Download single events with `huggingface_hub` (`allow_patterns`) or plain HTTPS — already in the layout that ml4floods reads

SPLIT	FLOOD MAPS	SENTINEL-2
train	475	92.7 GB
val	16	2.5 GB
test	18	3.1 GB

WorldFloods v2 on HuggingFace

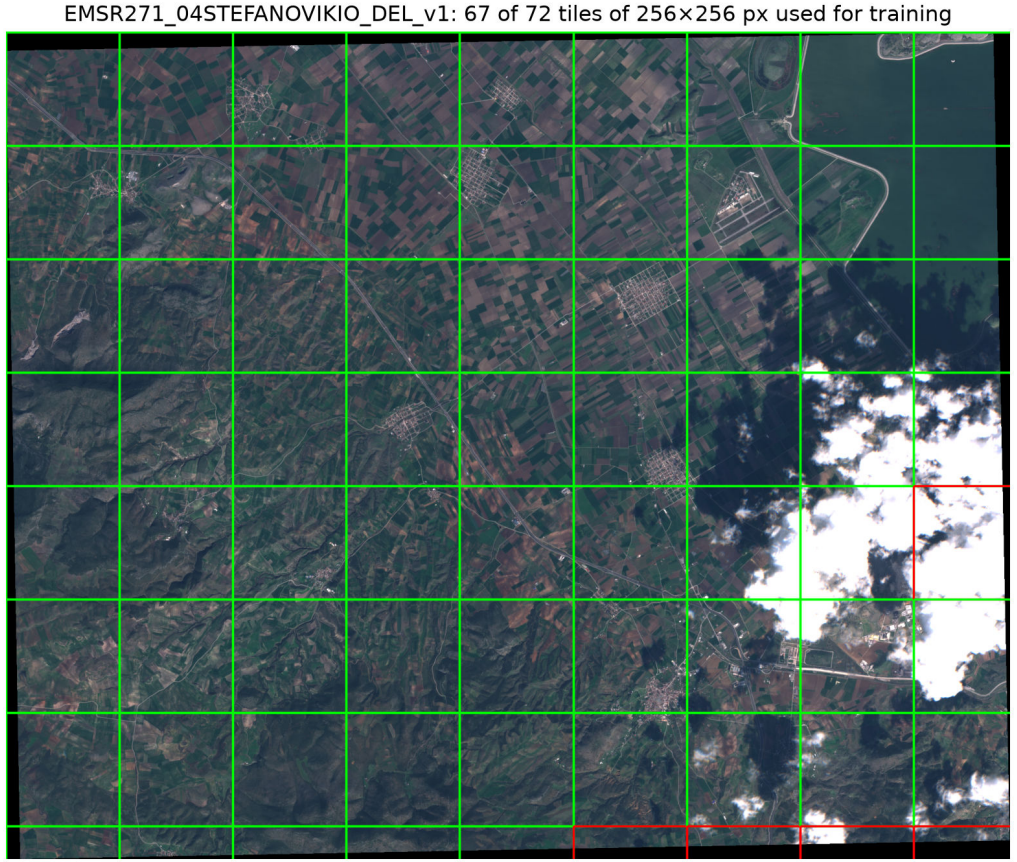
From someone else's flood map to ML-ready data

- Copernicus EMS / UNOSAT: vector maps drawn by analysts → Sentinel-2 of the same day → rasterised labels
- This is how WorldFloods was built (ml4floods.data)



Training on large scenes: tiling

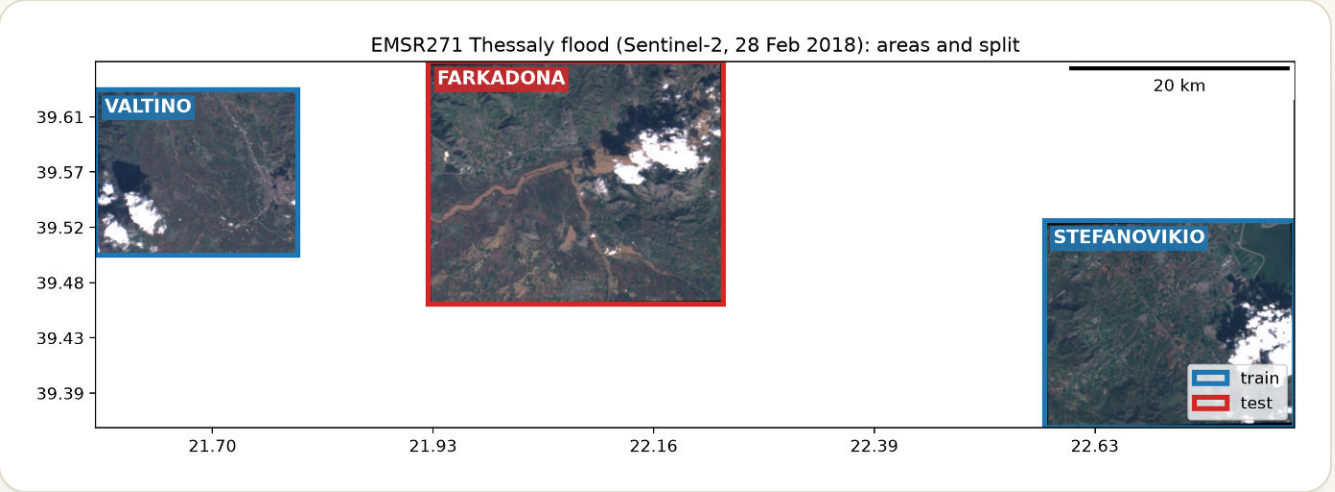
- Scenes of thousands of pixels → tiles of 256 × 256 px
- Tiles mostly cloudy or without data are dropped
- Inference: the whole scene tile by tile, stitched back



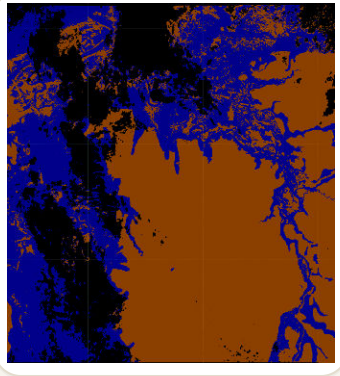
Data leakage: split by flood event

- Never split random tiles of the same event
- Train: VALTINO + STEFANOVIKIO (Thessaly) + 6 small floods from WorldFloods v2
- Validation: Ylitornio (Finland) + Campogalliano (Italy), clean references · Test: FARKADONA
- Choosing settings (e.g. the threshold) on the test event is leakage too

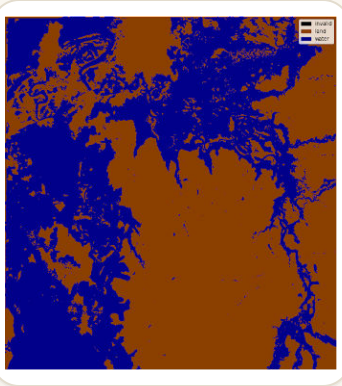
> DATA LEAKAGE
>
> Information from the evaluation data reaches the model (or our choices) during training → the scores look better than they will be on a **new flood**. Neighbouring tiles of the same event look alike, so a random split leaks.



Evaluation metrics



Ground truth



Prediction (water channel)

BINARY CONFUSION MATRIX · EXCLUDING BRIGHT CLOUDS

	REF. WATER	REF. LAND
PRED. WATER	TP	FP
PRED. LAND	FN	TN

$$\text{IoU} = \frac{TP}{TP + FN + FP}$$

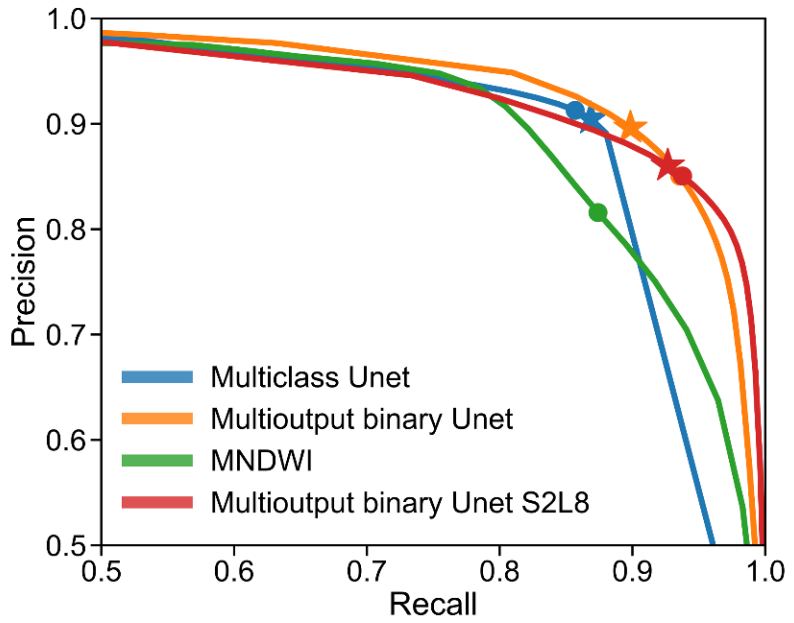
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

Performance metrics

ARCHITECTURE	MODEL VERSION	RECALL	PRECISION	IOU
U-Net	Multiclass	92.23	**91.51**	**85.29**
U-Net	Multioutput binary	94.19 ± 3.61	86.94 ± 2.50	82.45 ± 2.24
U-Net S2-to-L8	Multioutput binary	**96.35** ± 0.37	86.18 ± 1.22	83.49 ± 0.88
U-Net RGBNIR	Multioutput binary	93.50 ± 1.43	84.33 ± 2.57	79.86 ± 1.74
SimpleCNN	Multioutput binary	96.27	73.17	71.01
MNDWI	–	85.87	80.56	70.45
linear	Multioutput binary	87.06	73.27	64.62

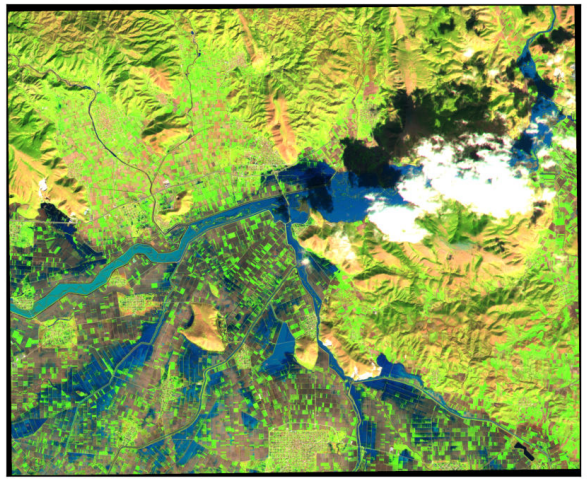
MEAN PER FLOOD · FIXED WATER THRESHOLD AT 0.5 PROBABILITY

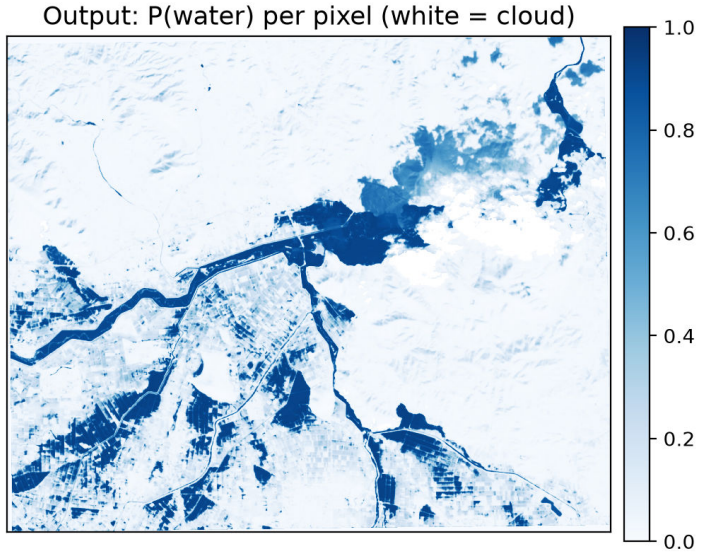


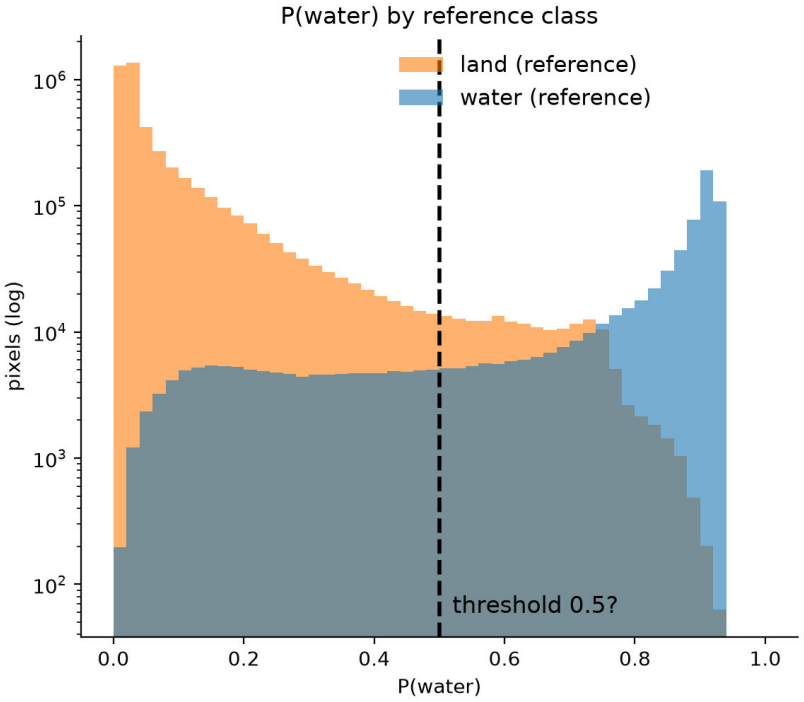
Varying the water threshold generates the precision-recall curves

What does the network output?

- Not a yes/no: a probability of water for every pixel (the sigmoid of the last neuron)
- Most pixels are sure (near 0 or near 1); the doubtful ones are in between → we must choose a threshold again

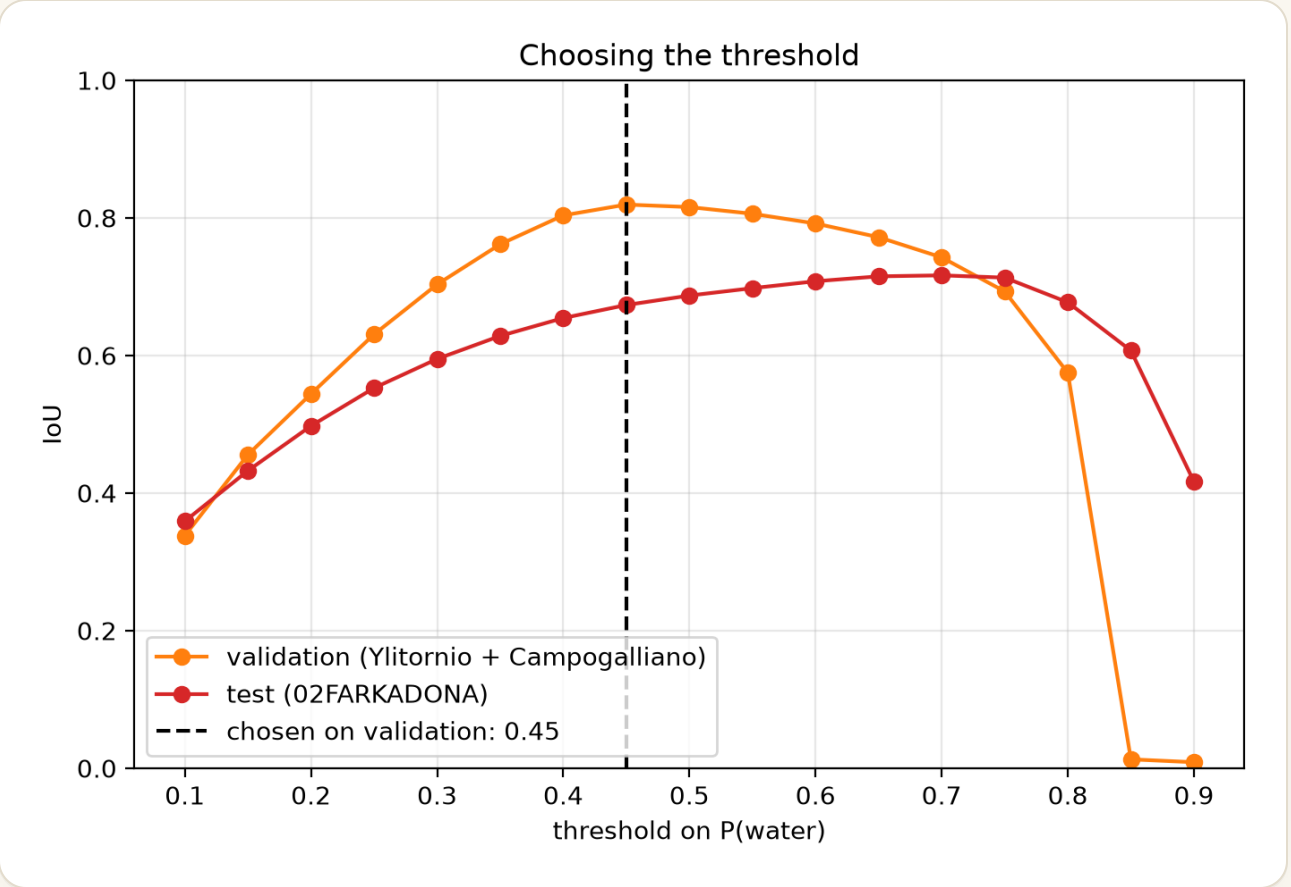






The threshold is a decision

- The model gives probabilities; the map depends on the threshold we choose
- Choose it on the validation floods, then apply it to the test event (choosing on test = leakage)
- Rescue teams prefer recall; damage assessment prefers precision



05

Hands-on 2

The ml4floods pipeline: train your own model, then use the global one

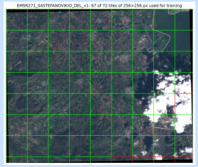
Roadmap

Notebook 02a — train your own model

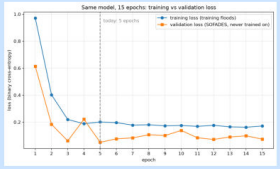
`ml4floods.data · ml4floods.models`

1. CEMS flood map → ML-ready labels

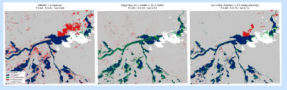
2. Split by event + 256 × 256 tiles



3. Trainer.fit + learning curves



4. Evaluate vs MNDWI / SAR



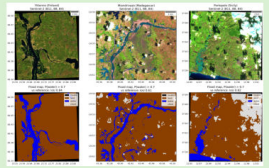
→

Notebook 02b — use the global model

`ml4floods.scripts.inference`

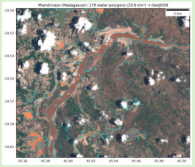
1. Weights from HuggingFace

2. Three new floods



3. Choose the threshold

4. Polygons → GeoJSON for a GIS

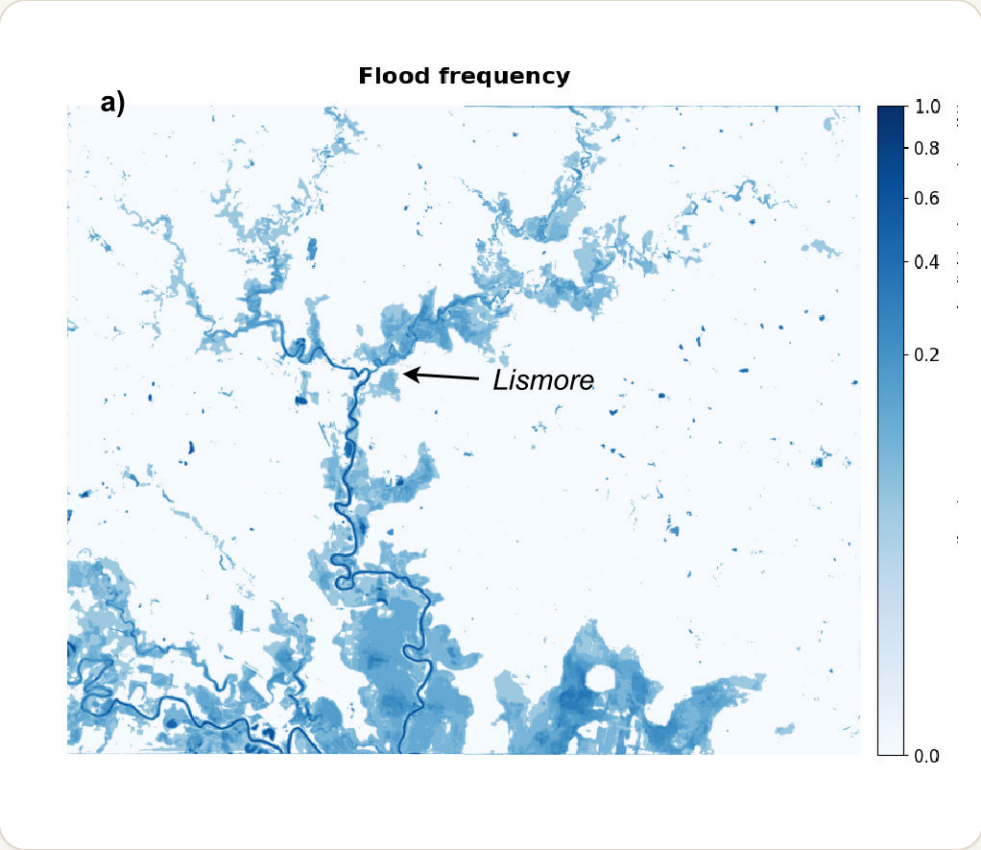


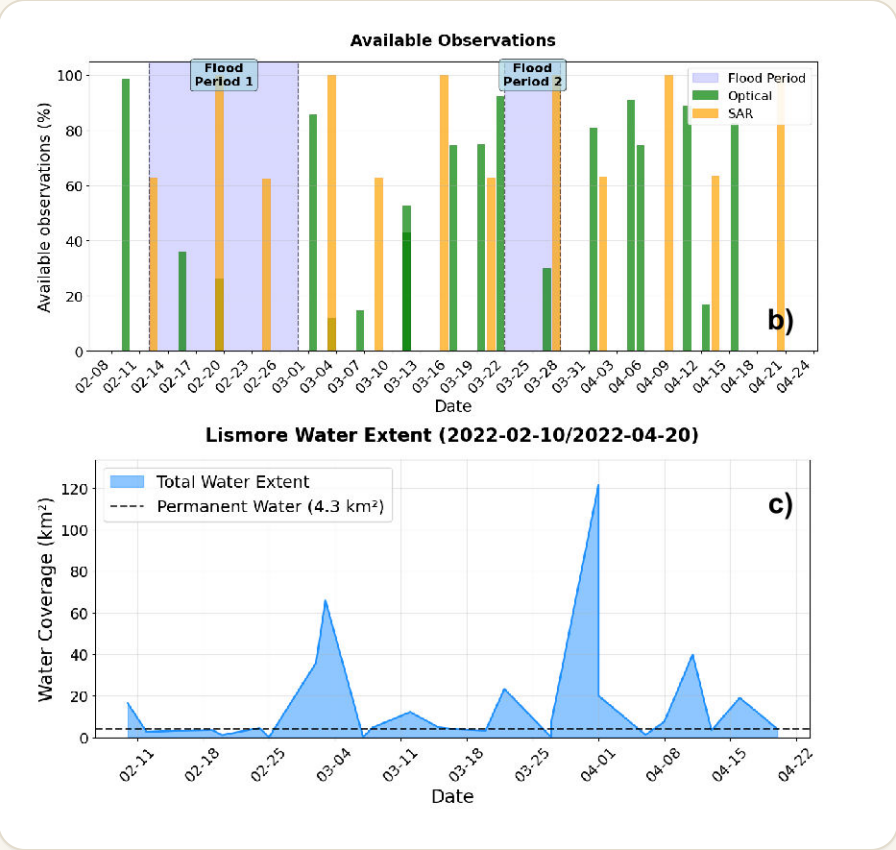
06

From data to action

Large-scale flood mapping

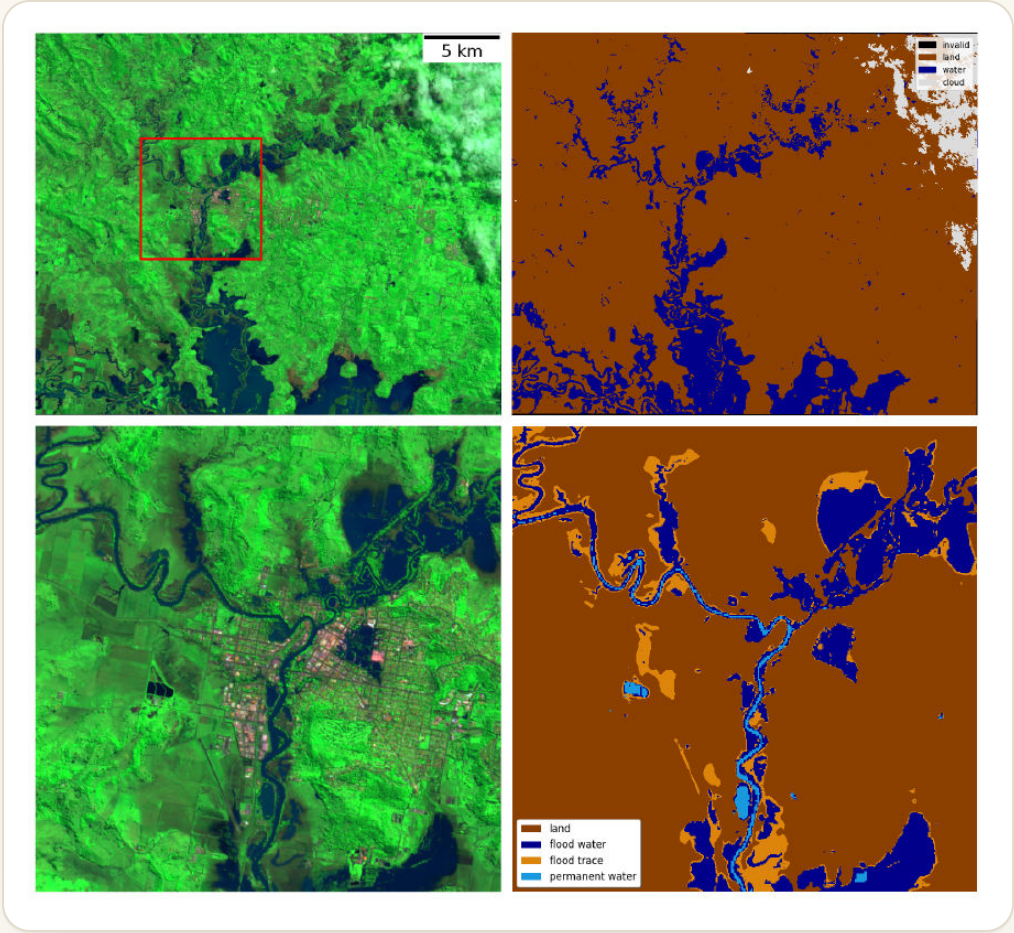
2022 Lismore floods





LISMORE WATER DYNAMICS · AUSTRALIA FLOODS 2022 · PORTALÉS-JULIÀ (2025), PHD THESIS

2022 Lismore floods: after the peak

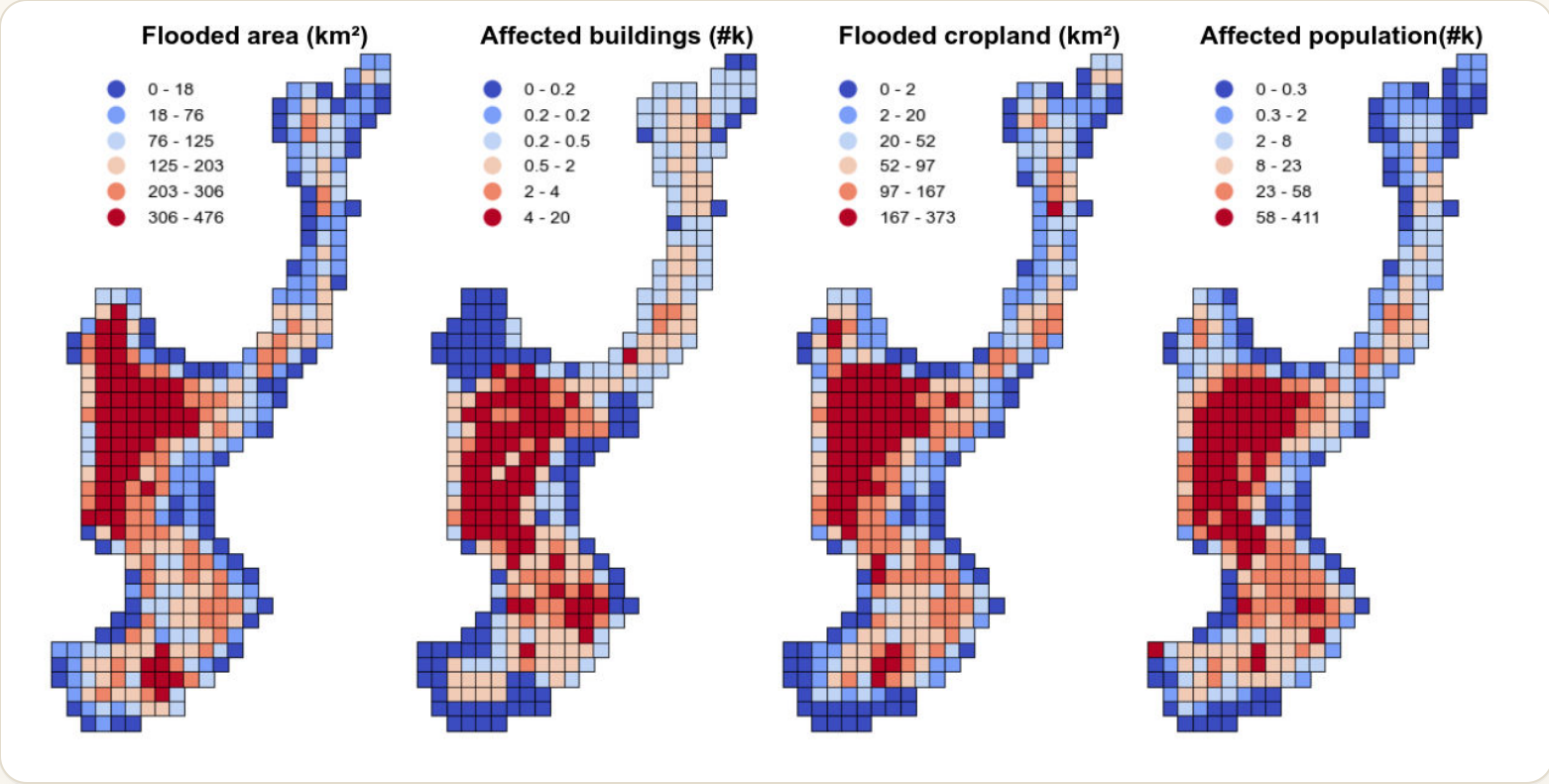


Lismore post-flood analysis

- Cloud-aware model detects **flood traces** (post-peak SWIR signatures)
- Relaxed threshold → 3-class map: land / water / flood traces
- Operational value when there is **no imagery at peak flood**

Damage quantification at grid scale

Per-tile statistics from the intersection with ancillary products · Pakistan 2022



Aggregated 25 × 25 km² tiles

54 300 km²
WATER

28 700 km²
CROPLAND

10.8 M
PEOPLE

656 k
BUILDINGS

Summary of today

- **Optical** data is rich in spectral information: even simple **thresholds** on water indices map floods with decent accuracy. Main issue: **clouds**
- **SAR** sees through clouds: calm water is **dark**. Thresholding (Otsu, Edge Otsu, BMax + HAND) is the **workhorse** of operational flood mapping, e.g. the Copernicus EMS
- **WorldFloods**: how to access it, and the principles behind **datasets for deep learning** on satellite images: ML-ready labels, tiles, split by event, no leakage
- **ml4floods**: how to **train**, **evaluate** and **run** deep learning models for flood segmentation

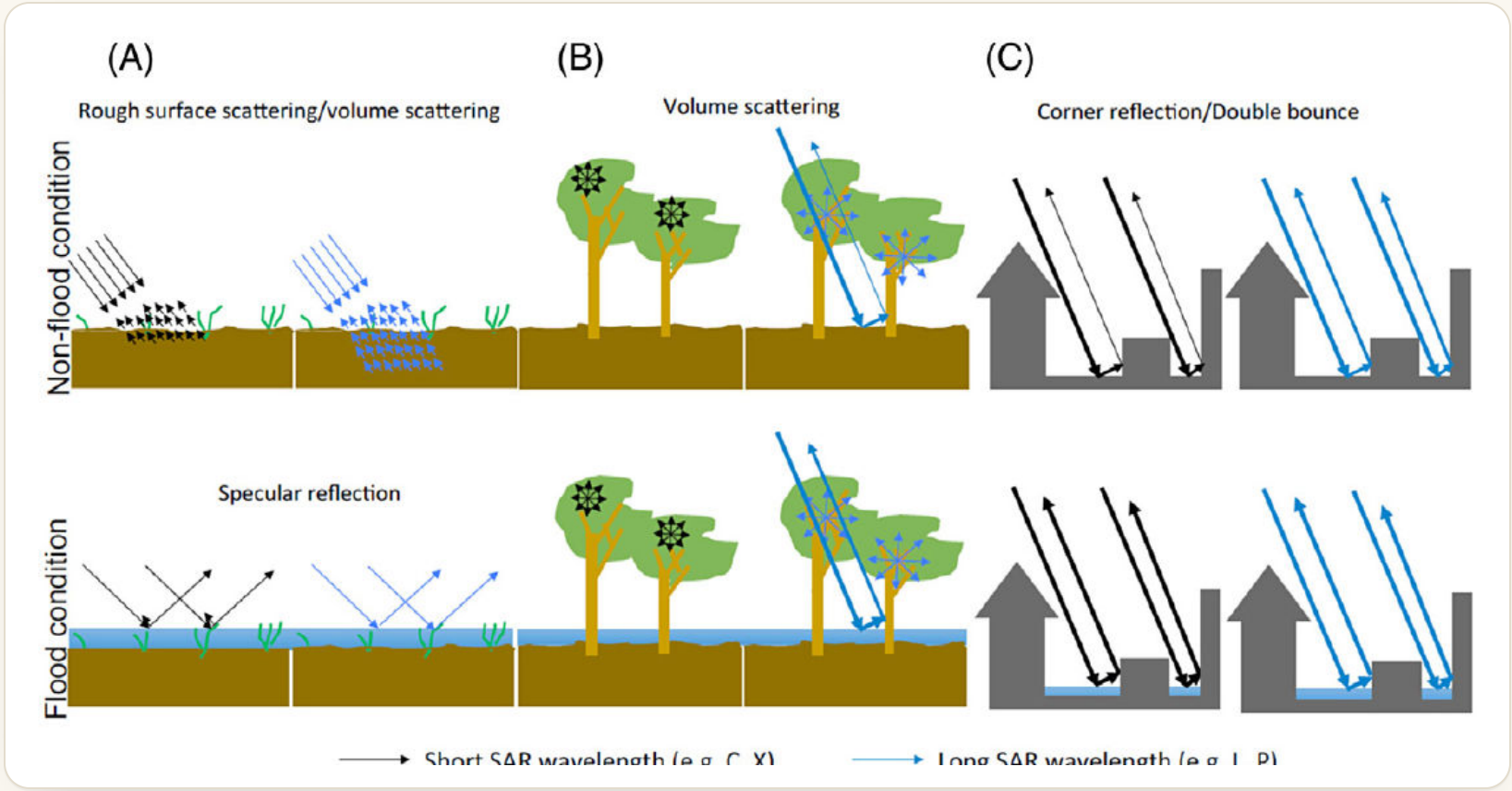
Resources

- ml4floods — github.com/spaceml-org/ml4floods
- WorldFloods v2 — huggingface.co/datasets/isp-uv-es/WorldFloodsv2
- georeader — github.com/spaceml-org/georeader
- NASA ARSET flood trainings — appliedsciences.nasa.gov/arset
- The SAR Handbook (Flores-Anderson et al., 2019)

Thank you!

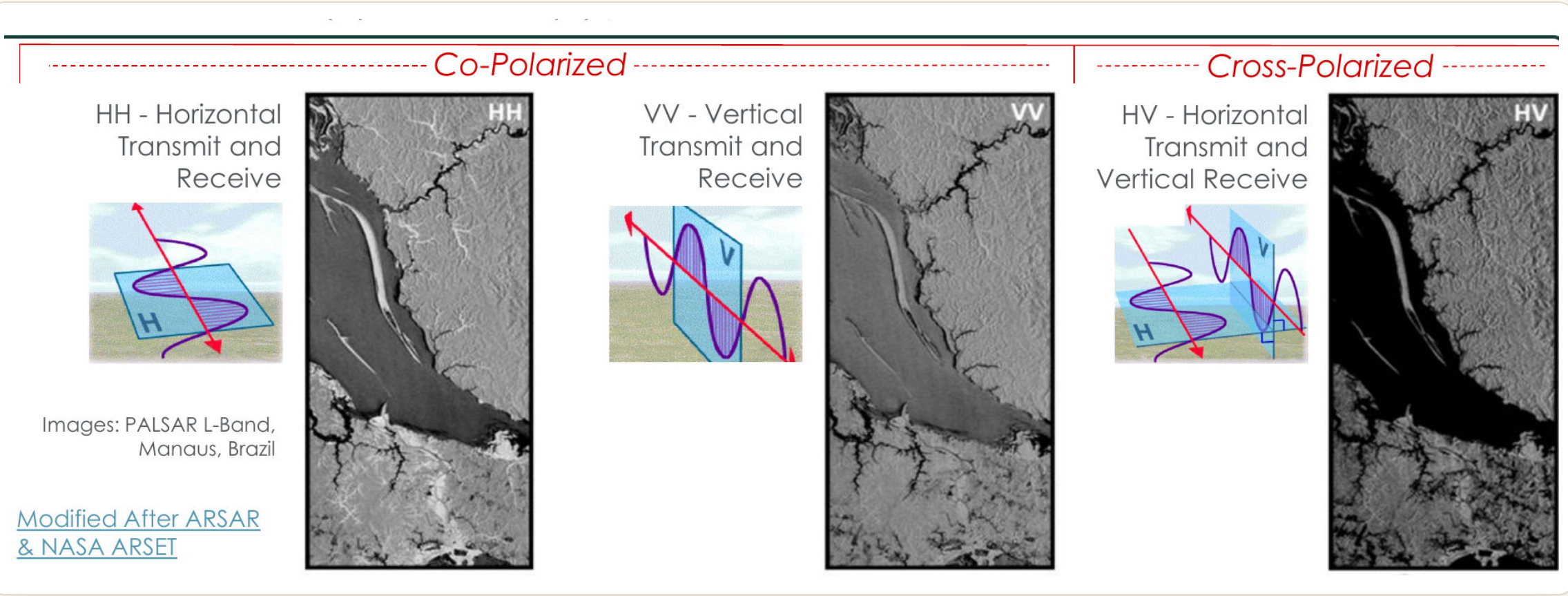
Questions?

Scattering by wavelength (backup)



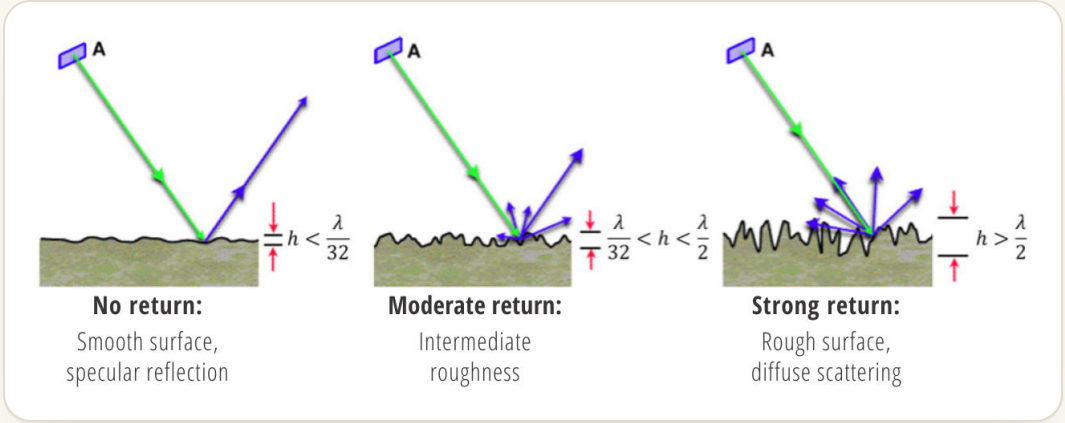
Non-flooded vs flooded, short (C, X) vs long (L, P) wavelengths. Schumann (2021), Earth Observation for Flood Applications, Fig. 14.5, from Schumann & Moller (2015)

Polarisation (backup)

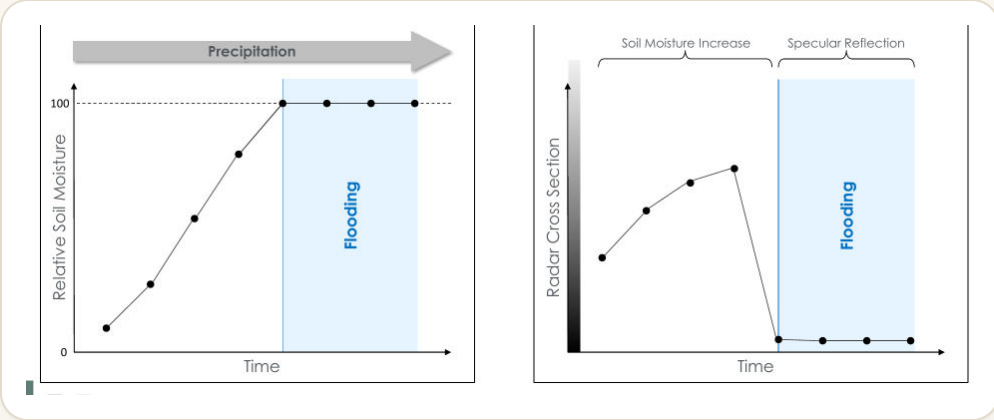


NASA ARSET (2023), SAR for Floods, Part 3 (after ARSAR)

How water looks in SAR (backup)

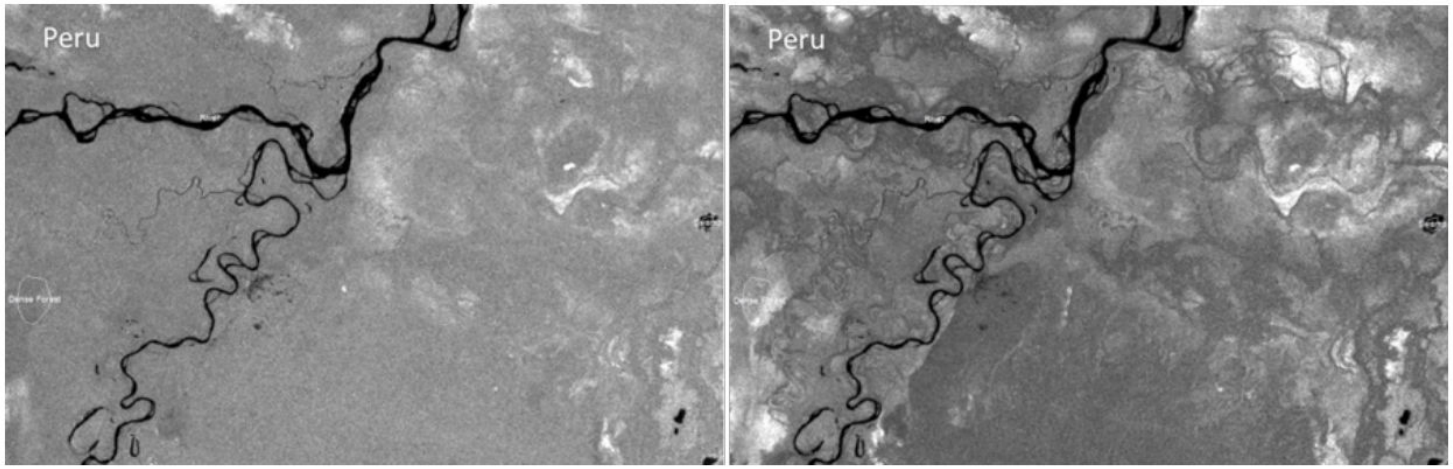


Calm water = smooth surface = specular reflection. Flores-Anderson et al. (2019), The SAR Handbook, Fig. 2.8

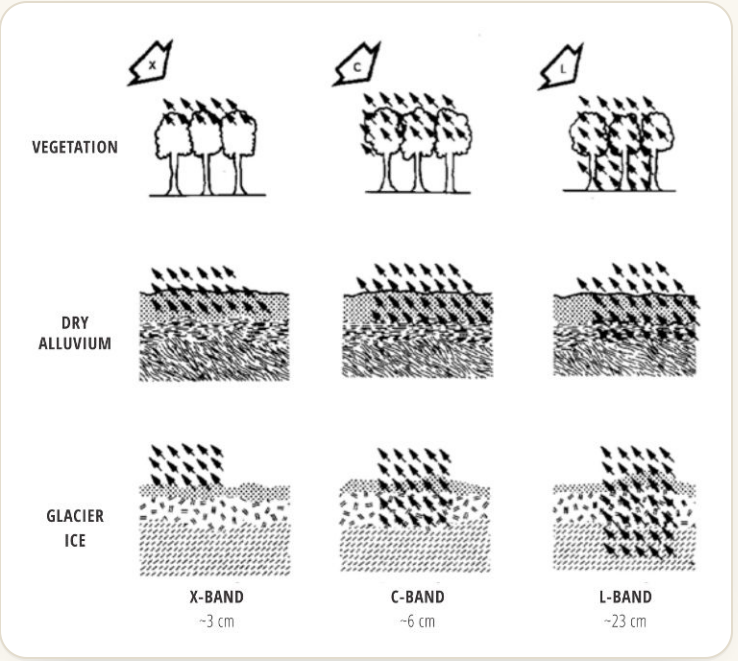


Flooding under vegetation: double bounce. NASA ARSET (2023)

Flooding under forest: a real example (backup)

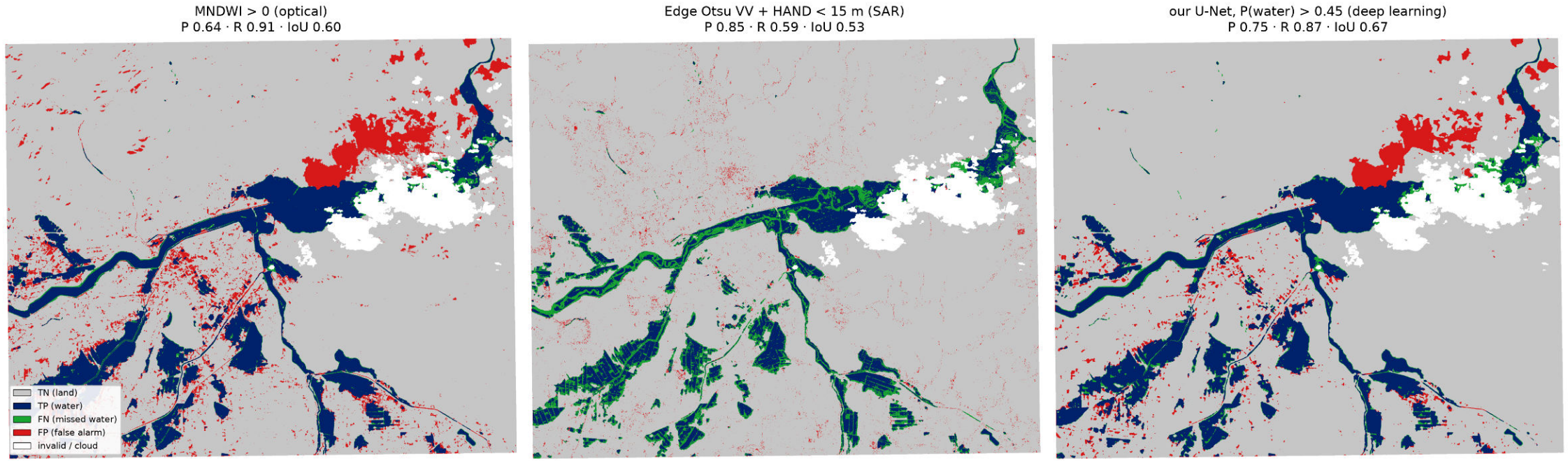


ALOS-1 L-HH, Peru: low- vs high-water season. Flores-Anderson et al. (2019), The SAR Handbook, Fig. 3.6

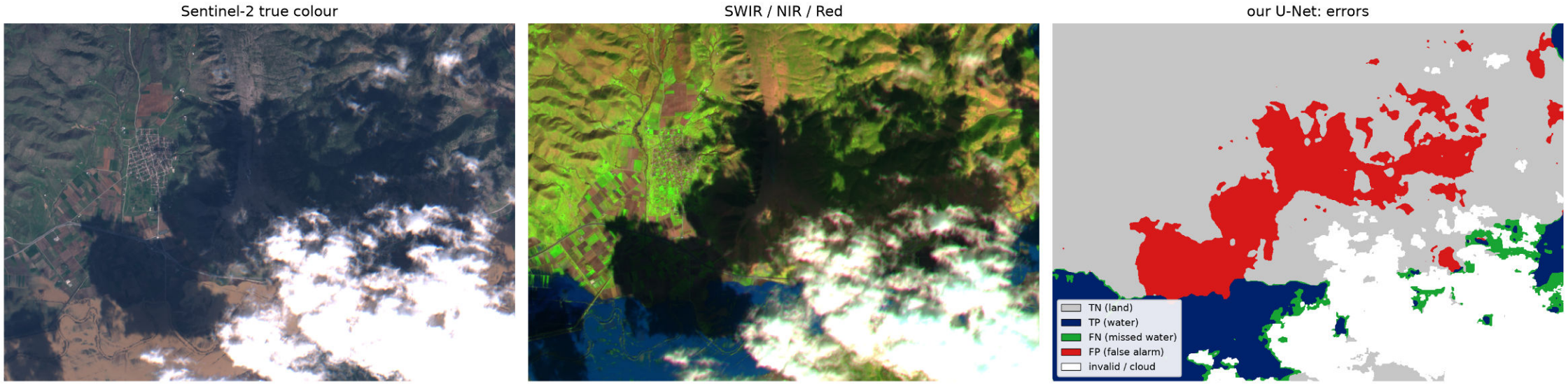


Penetration by wavelength. Flores-Anderson et al. (2019), The SAR Handbook, Fig. 2.6

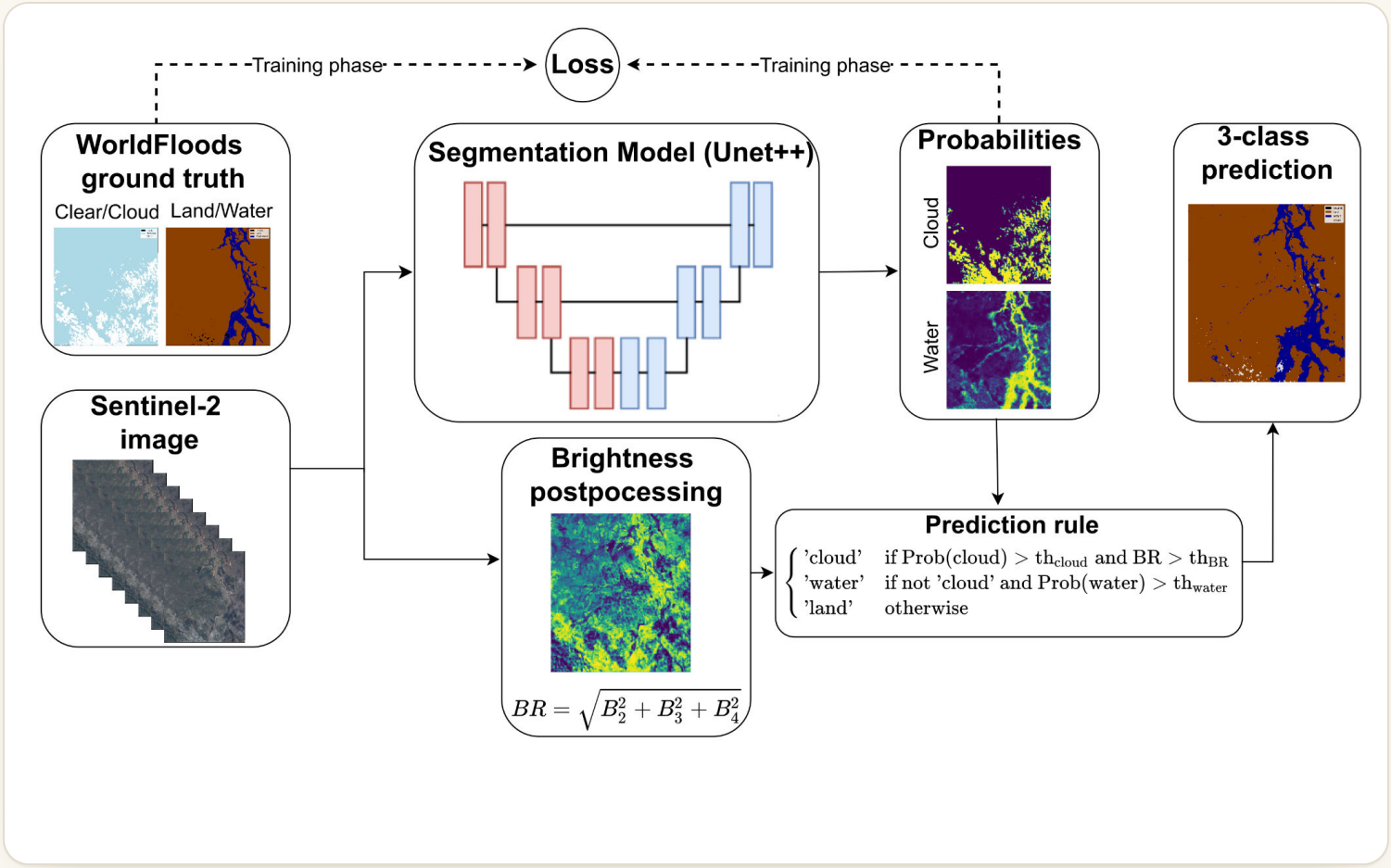
What did we find? Our model vs the rules (backup)



Where does our model fail? (backup)

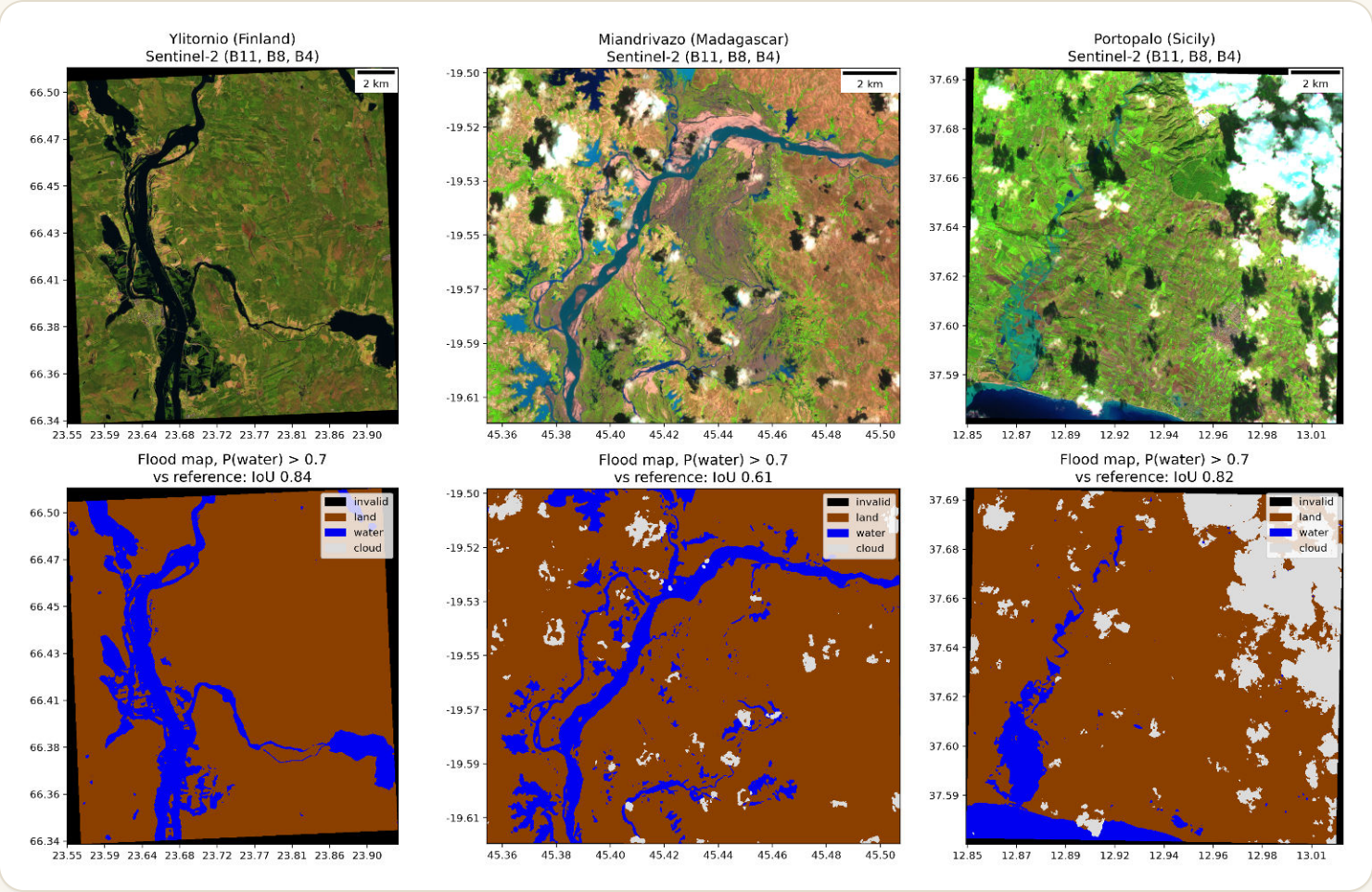


Cloud-aware multioutput framework (backup)

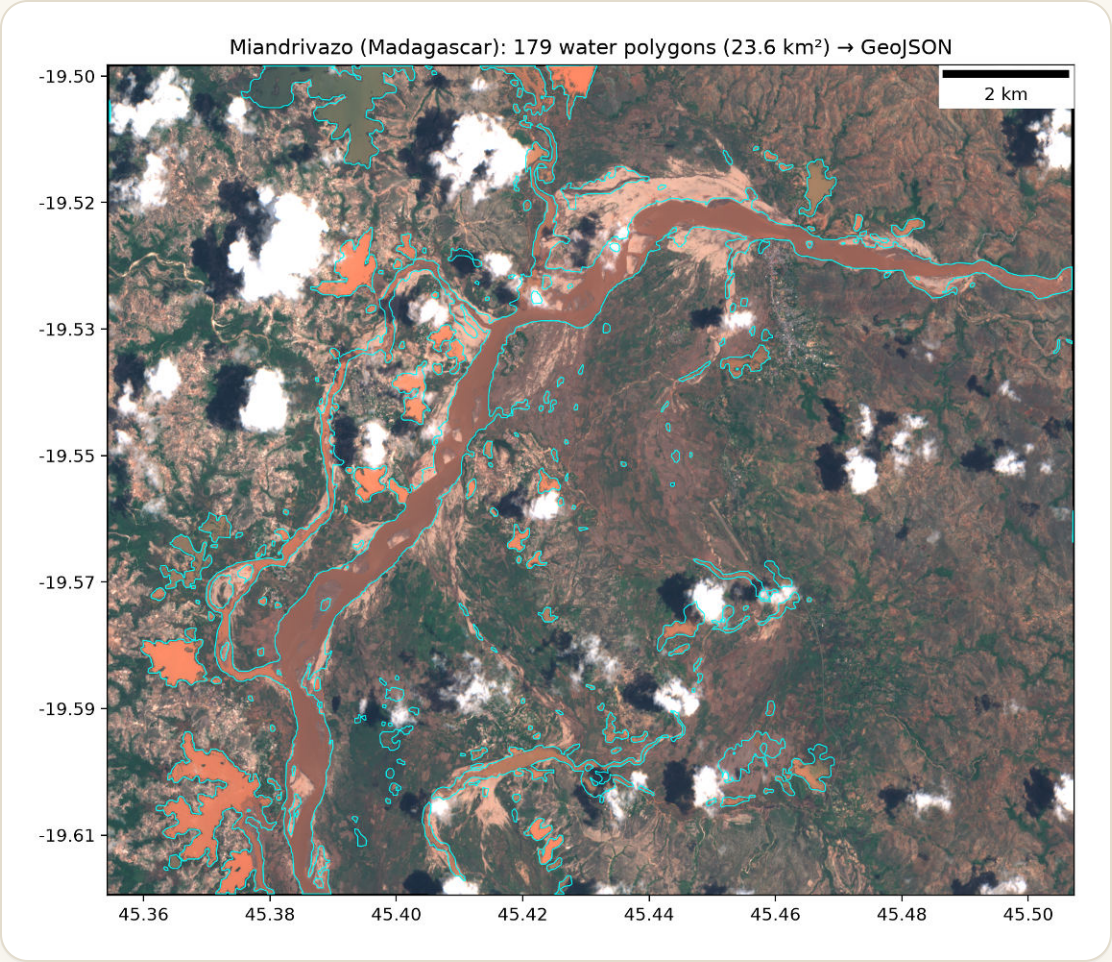


WorldFloods v2 model: two outputs, P(cloud) and P(water) → water can be seen under thin clouds. Portalés-Julià et al. (2023), Scientific Reports

A global model on new floods (backup)

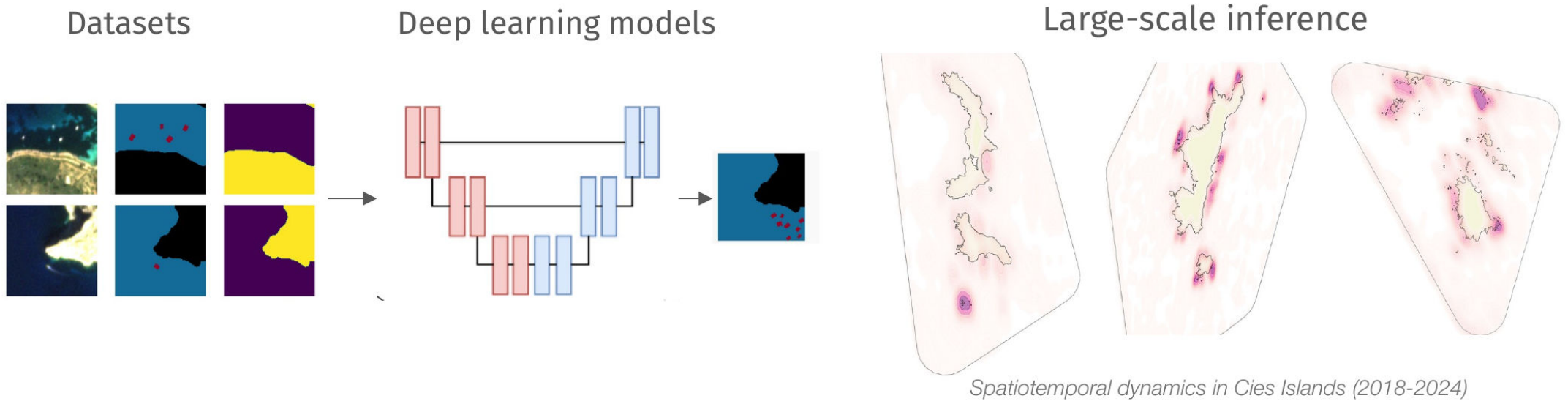


From raster to polygons (backup)



TECMAR project

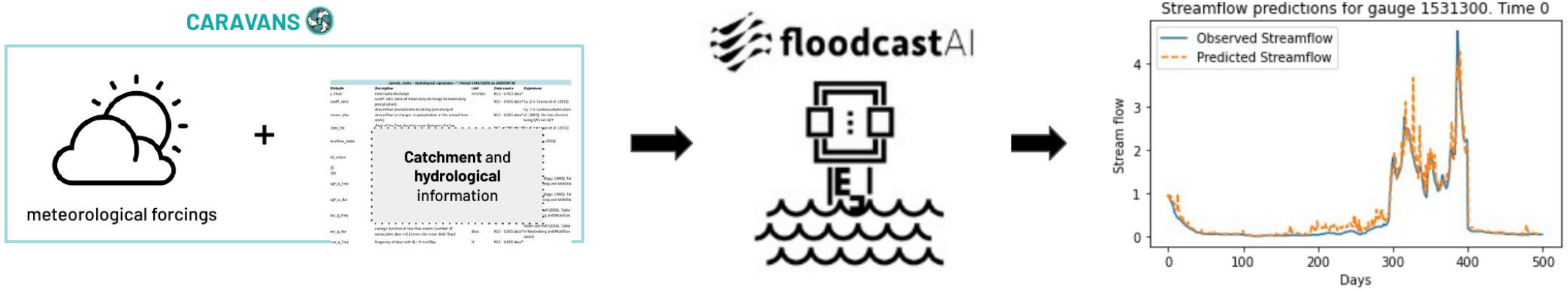
The data-centric methodology of my thesis applied to **small vessel detection** in Marine Protected Areas



Portalés-Julià, E., Menéndez-Blázquez, J., Mateo-García, G., March, D. and Gómez-Chova, L. Satellite detection of small vessels to monitor their potential impact on marine protected areas. ESA Living Planet Symposium 2025

FDL 2022 research sprint

Data-driven **flood prediction** from catchment, hydrological and meteorological data



Bouri, I., Lahariya, M., Nivron, O., Portalés-Julià, E., Backes, D., Bilinski, P. and Schumann, G. (2022). ML framework for global river flood predictions based on the Caravan dataset. arXiv:2212.06719

IFUP project

Moving remote sensing workflows to the cloud with the **Information Factory Urban Pathfinder** (IFUP)

DATA PROVIDERS

- Copernicus
- Landsat 1 to 9
- PlanetScope
- ICEYE
- RADARSAT
- Non-EO data

→

ALGORITHMS

- Cloud masking
- **Flood mapping**
- Flood risk
- Urban mapping
- LULC analysis

→

USERS

- Non-technical EO users
- **Emergency responders**
- Organizations
- Scientists

INFORMATION FACTORY